

1 **Cover Letter**

2 **Rethinking Student Housing in Berlin**

3 Lazareva Nataliia, Urban Design M.Sc., Technical University of Berlin

4 Dear Editors-in-Chief,

5
6 please find the enclosed manuscript "Rethinking Student Housing in Berlin" which we are submitting for exclusive
7 consideration for publication in Computers & Geosciences. We confirm that the submission follows all the require-
8 ments and includes all the items of the submission checklist.

9
10 The manuscript presents:

11
12 We provide the source codes in a public repository with details listed in the section "Code availability".

13
14 Thanks for your consideration.

15
16 Sincerely,

17
18 Lazareva Nataliia, Urban Design M.Sc., Technical University of Berlin

19

Rethinking Student Housing in Berlin

20 Lazareva Nataliia^a, Urban Design M.Sc.^b and Technical University of Berlin^c

21

22 **ABSTRACT**

23 Placeholder text is often used in design to indicate where content will eventually be placed. It

24 serves as a visual guide for designers and developers, enabling them to lay out pages and adjust

25 typography, spacing, and color schemes before the final content is inserted. An example of

26 filler text is provided below for the purpose of demonstrating layout elements in a generic layout

27 scenario. This filler content is intentionally devoid of specific meaning, meaning it contains

28 no hidden or actionable messages, and is instead built from commonly used words and phrases

29 to provide a textured baseline for creative exploration. It can also aid in streamlining editorial

30 workflows, especially when translation or localization teams must preview how copy will appear

31 in a finished design. Typical placeholder narratives may include references to common design

32 practices, UI components, grammar conventions, and visual hierarchy principles all brought

33 together in a fluid tone

34

35

36 **1. 0. Preliminary Pages**

ORCID(s):

37 **1.1. 0.1 Title Page**
38 ***1.1.1. Technical University of Berlin***
39 ***Rethinking Student Housing in Berlin: DataDriven Approach* Author:**
40 Nataliia Lazareva
41 **Supervisors:**
42 Prof. Dr. Lech Suwala
43 Dr. Ing. Andreas Brück
44 Master Thesis
45 2026

46	1.2. 0.2 Table of Contents
47	<i>1.2.1. 0. Preliminary Pages</i>
48	0.1 Title Page
49	0.2 Table Of Contens
50	0.3 List Of Figures
51	0.4 List Of Tables
52	<i>1.2.2. 1. Introduction</i>
53	1.1 European Student Accommodation Crisis
54	1.2 Berlin Housing Crisis History
55	1.3 Theoretical Framework
56	1.4 Rethinking Student Housing
57	<i>1.2.3. 2. Capacities, Demand & Spatial Distribution</i>
58	2.1 Overview
59	2.2 Urban Data Mining
60	2.3 Descriptive Market Statistics
61	2.4 Global Spatial Autocorrelation
62	<i>1.2.4. 3. Student Housing Market Imbalance</i>
63	3.1 Overview
64	3.2 Central Place Analysis
65	3.3 Market Pressure Analysis
66	<i>1.2.5. 4. Students Behind the Numbers</i>
67	4.1 Overview
68	4.2 Students Needs And Preferences
69	4.3 Students Experiences
70	<i>1.2.6. 5. Data-Driven Decisions</i>
71	5.1 Conclusion
72	5.2 Reflection
73	<i>1.2.7. 6. Appendix</i>
74	6.1 References

1.3. 0.3 List of Figures

Figure	Title	Page
1	Publication Year Time Series	1
2	Publication Type Distribution	1
3	Method of Study Distribution	1
4	Data Source Distribution	1
5	Spatial Distribution of Case Studies Across Europe	1
6	Spatial Distribution of Case Studies Across Germany	1
7	Time Overlay Network Visualisa- tion	1
8	Visual Comparison Initial Unique Student Accommodation Addresses with Final Centroids	1
9	Visual Comparison Initial Unique Higher Education Universities Ad- dresses with Final Centroids	1
10	Student Accommodation Capacity Distribution	1
11	Student Accommodation Capacity Normal Distribution Histogram	1
12	Student Accommodation Price Dis- tribution	1
13	Student Accommodation Living Area Distribution	1
14	Student Accommodation Capacity Distribution by Ownership Model	1
15	Accommodation Spatial Distribu- tion by Ownership Model	1
16	State and Private Owned Student Accommodation Normal Capacity Distribution Comparison	1
17	Higher Education Universities Stu-	1

Table	Title	Page
1	Literature Review Cities	1
2	Manual Data Collection Sources for Student Accommodation	1
3	Manual Data Collection Sources for Higher Education Universities	1
4	Student Accommodation Numeric Fields Summary	1
5	Key Student Accommodation Ownership Companies	1
6	Student Accommodation Ownership Model Numeric Fields Summary	1
7	Higher Education Universities Numeric Fields Summary	1
8	Higher Education Universities Ownership Model Numeric Fields Summary	1
9	Student Accommodation Capacity Moran's I index and P-value for Different Number of Neighbors (k)	1
10	Higher Education Universities International Students Demand Moran's I index and P-value for Different Number of Neighbors (k)	1
11	Central Higher Education Universities.	1
12	Travel Time Summary Statistics	1
13	Market Areas	1
14	Centrality Central Place Metrics	1
15	Service Range Adequacy Central Place Metrics	1
	Hierarchy Rank Central Place Metrics	1

Research aimed to explore the topic, identifying major trends, challenges, and research gaps, in order to form scientific objectives and relevance of research for Berlin Case Study.

2.1. 1.1 European Student Accommodation Crisis

In order to better understand the current state of research on European Student Accommodation Crisis topic, a comprehensive literature research was conducted. *Research aimed to explore the topic, identifying major trends, challenges, and research gaps, in order to form scientific objectives and relevance of research for Berlin Case Study.*

Type of Literature Research: Bibliographical Mapping

Tools: Zotero, Python, VOSviewer

Topic: European Student Accommodation Crisis

Period of the search: 2010-2025

Area of search: European Union

Academic Database: Scopus, Emerald, Google Scholar, TU Berlin Library

Language: English

2.1.1. Literature Review Type

Bibliometric analysis is defined as “a part of scientometrics for utilising mathematical and statistical methods to analyse scientific activities in a research field” (Louarn et al., 2025). Thus, bibliographical mapping can be seen as a structured, data-driven approach for organising existing knowledge regarding the selected topic. It can be applied to vast interdisciplinary literature, developing structured networks determined by keywords, co-citations, bibliometrics coupling, and direct citation analysis. This kind of approach matches with the overall aim to introduce innovative and structured analytical methods in urban design research. The methodology followed in review type comprises four stages: data collection, eligibility analysis, general descriptive and content analysis. (Louarn et al., 2025).

2.1.2. Limitations

In the bibliometric analyses, certain limitations were set, which should be taken into account when interpreting the results. Firstly, document type was limited to ‘journal article’ and ‘book section’ to avoid the double-collecting of similar research papers. However, during the collecting process ‘reports’ were added as a grey literature sources. Secondly, the search was limited to articles written in English. Thirdly, the search area was initially limited to Germany, which was considered an especially narrow approach. Subsequently, the search area was extended to the European Union, which provided a more comprehensive view of the problem. Last but not least, the period of search was limited to 2010-2025 years, during which the governmental student accommodation availability rate in Germany was no more than 10% (Student Housing in Germany Statistical Overview, 2023). Those limitations assist in focusing on research

110 that is immediately connected with the Student Housing Crisis in Berlin and retain only relevant information.

111 **2.1.3. Set Up**

112 Import Libraries

113 `import json`

114 `import pandas as pd`

115 `import geopandas as gpd`

116 `import matplotlib.pyplot as plt`

117 `import numpy as np`

118 `from shapely.geometry import Point`

119 **2.1.4. Data Mining**

120 The literature collection was conducted in June 2025 using three academic databases: Scopus, Emerald, Google
121 Scholar and TU Berlin Library as a grey literature source. The search terms was organised in five preliminary clusters:

- 122 1. Student Housing Market Terms: `student housing OR student accommodation OR dormitory OR Wohnheim`
123 `OR affordable housing`
- 124 2. Policy: `housing crisis OR rental market OR real estate investment OR financialization of`
125 `housing`
- 126 3. Urban Design Terms: `urban displacement OR studentification OR gentrification OR social`
127 `housing OR urban periphery`
- 128 4. Geographic Terms: `Berlin OR Germany OR EU OR European Union`
- 129 5. Demographic Terms: `international students OR Erasmus students OR student mobility`

130 The search resulted in a total of 50 articles from the four data sources, which serve as an initial orientation and
131 provide a conceptual foundation for European Student Accommodation Crisis topic.

132 `file_path = '../inputs/1.1_european_student_accommodation_crisis/literature_review_list.json'`

133

134 `# Load the JSON data`

135 `with open(file_path) as file:`

136 `data = json.load(file)`

137

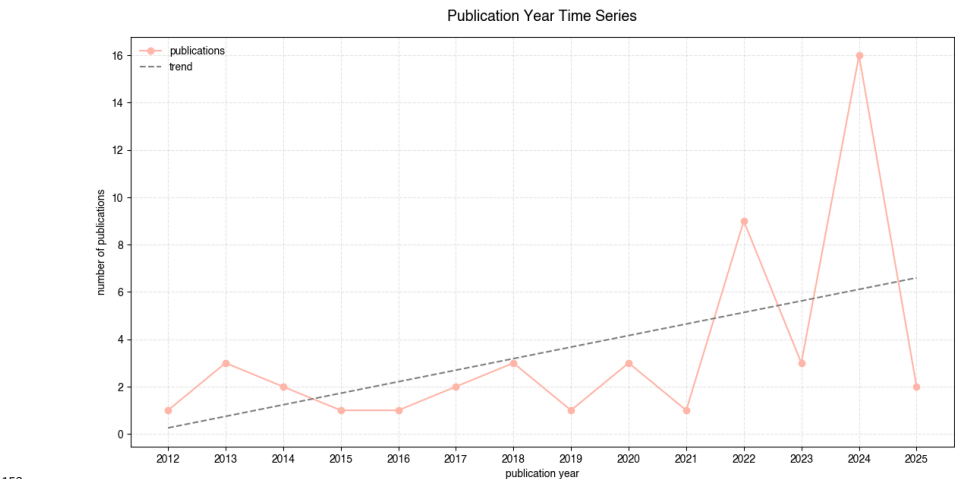
138 `# Extract items (journal articles and attachments) from the JSON`

```
139 items = data.get('items', [])
140
141 # Convert the items into a DataFrame
142 df = pd.json_normalize(items)
```

143 **2.1.5. General Descriptive Analysis**

144 **Time Series**

145 The Study Period (2010-2025) has seen steady growth in publication trends (Figure 1). Between 2012 and 2021,
146 the average annual number of publications was 2. In comparison, between 2021 and 2025 average publications number
147 of per ear was 4 times higher - approximately 8 publications per year. In 2024, the number of annual publications was
148 eight times as many as those published in 2014. The regular growth in the number of publications could be explained by
149 the increasing acknowledgment of the importance of student housing crisis. Moreover, the number of articles related
150 to student housing market regulations increased over the past 5 years. This could be seen as a part of the solution to
151 the student housing crisis. Thus, providing new approaches to student housing, regulating the market, and recognising
152 the right of students to social housing be taken into account in forming a theoretical framework.

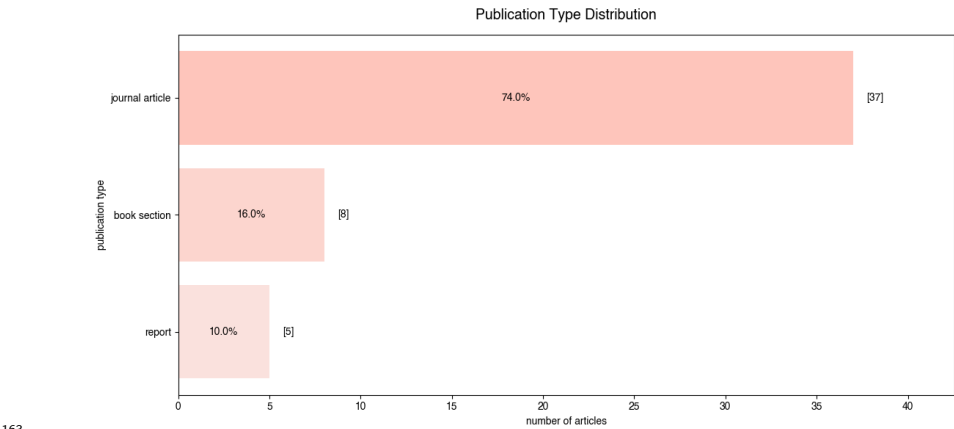


154 *Figure 1. Publication Year Time Series*

155 **Publication Type Distribution**

156 Publication Type analysis (Figure 2) depicts the dominance of peer-reviewed journal articles (74%) and book chap-
157 ters (16%) with the strong academic foundation of the research field on student housing market and urban design. In
158 contrast, the reports (10%), representing grey literature, offer valuable insights from policy institutions and housing or-
159 ganisations. Although non-academic, these sources provide empirical evidence and real-world perspectives that bridge
160 the gap between theory and practice. The combination of academic and grey literature thus ensures a balanced and
161 multidimensional understanding of the student housing crisis, where data-driven analysis is informed both by academic

162 research and applied policy contexts.



163

164 *Figure 2. Publication Type Distribution*

165 **Method of Study Distribution**

166 Methods of Study (Figure 3) refer to the techniques and tools used in research to collect, analyse, and interpret data.

167 Such methods can be broadly categorised into qualitative, quantitative, and mixed-method approaches. Qualitative

168 study methods (29.2%) have a strong emphasis on understanding the social dynamics and subjective experiences,

169 highlighting topics such as cultural diversity, moving abroad, and community living. Quantitative studies (33.3%)

170 indicate a desire for evidence-based insights related to housing costs, student satisfaction, or demographic trends.

171 However, the majority of studies are using mixed-method approaches with an emphasis on understanding the social

172 processes within the data-driven framework. Thus, merging qualitative methods with quantitative ones is the most

173 relevant approach to access the Student Hosing Crisis topic.

Method of Study Distribution

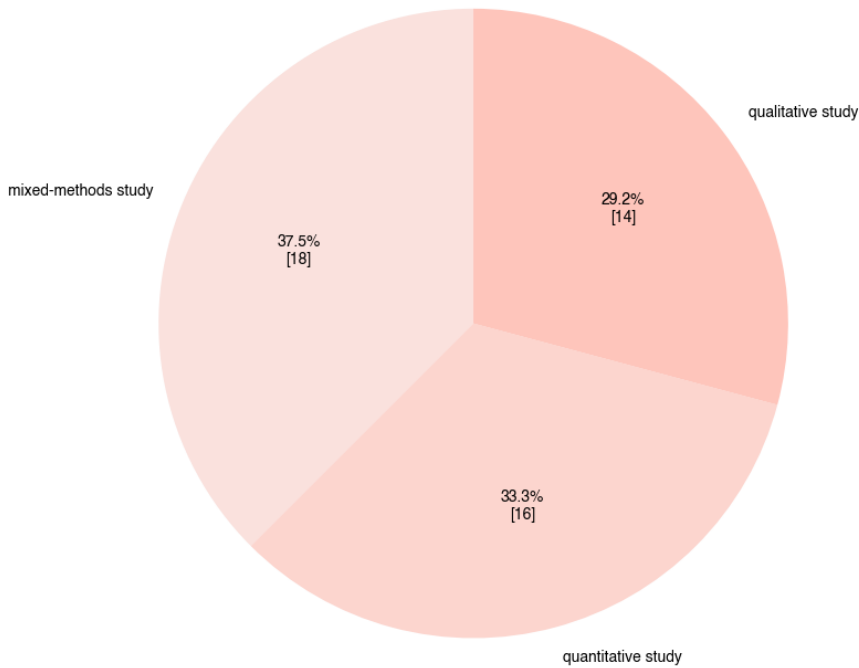


Figure 3. Method of Study Distribution

Data Source Distribution

During the data source distribution analysis (Figure 4), four primary data source categories were identified: numerical data (30.3%), social data (27.5%), literature review (25.7%), and other data (16.5%). The balance between four categories reflects the interdisciplinary nature of the Student Crisis topic. Statistics, institutional records, market data provide measurable insights into housing, while social data (surveys, interviews, case studies, street observations) capture students' lived experiences and perceptions. The strong representation of literature sources ensures theoretical grounding in urban design, economics and sociology.

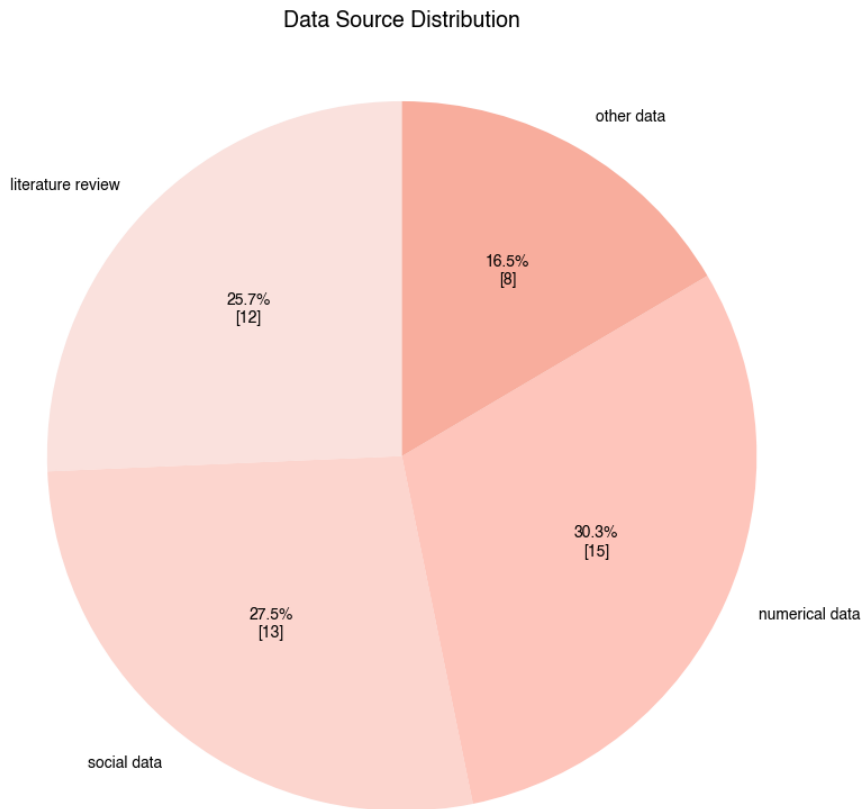


Figure 4. Data Source Distribution

Studies Areas

In selected literature, 46 cities were mentioned as case studies. The most frequently mentioned cities are Berlin, Hamburg, Stockholm and Vienna (Table 1), all of which have populations of over 1.5 million. This can be explained by the fact that the more people live in a city, the easier it is to obtain a representative sample for social research. Large cities also tend to have more open statistics and data.

```
city.head(4)
```

191

	city_name	count	longitude	latitude	geometry
0	Berlin	24	13.404954	52.520008	POINT (391776.158 5820073.118)
1	Hamburg	6	9.993682	53.551086	POINT (168441.945 5945241.585)
2	Stockholm	3	18.068581	59.329323	POINT (674570.668 6580745.518)
3	Vienna	3	16.373819	48.208174	POINT (602066.671 5340350.73)

192

Table 1. Literature Review Cities

193

194

195

196

197

198

Spatial Distribution of Case Studies (Figure 5) shows geographical origin of the academic and empirical datasets used in the literature, highlighting that topic of European Student Accommodation Crisis is researched mainly in Western Europe (e.g., Germany, Austria, Netherlands) and Northern Europe (e.g., the UK, Scandinavia), while East Europe (e.g., Poland, Slovakia, Hungary) remains relatively under-researched. It is important to note that, since geographic search terms include “Germany”, the highest data density correlates with the frequency of mentions of Germany in search terms.

199

200

201

202

In the context of the student housing crisis, Germany serves not only as a subject of analysis, but also as a model for comparative European studies. For example, Berlin is compared with Vienna (Holm, Andrej et al., 2024), Prague, Stockholm, and Amsterdam (Wijburg and Aalbers, 2017), as well as other German cities such as Munich, Hamburg, Stuttgart, Hanover, Wiesbaden, and Saarbrücken (Mitze et al., 2015).

listed real estate investment trusts (Wijburg & Aalbers, 2017). The authors describe three waves of financialisation: from the failed liberalisation of housing loans to the mass transfer of municipal housing stock to private equity funds and the subsequent concentration of stock in the hands of large private corporations (Vonovia, Deutsche Wohnen, etc.) (Wijburg & Aalbers, 2017). As a result, the German market has become the next frontline for global investment.

These trends are particularly noticeable in Berlin. According to the IBB Wohnungsmarktbericht 2024, the average cold rent offered in the city in 2024 reached EUR 15/m², which is more than double the comparative rent according to Mietspiegel (EUR 7.21/m²) and represents a gap of around 118% — the largest among major German cities (Investitionsbank Berlin, 2024). At the same time, the rate of new housing construction is falling behind population growth, and the shortage is particularly severe in central areas.

In this context, students, young adults and migrants are the least protected. Student halls of residence cover only about 8% of students in Germany (Deutsches Studierendenwerk, 2024). European-wide data shows that one in four students spends more than 40% of their budget on housing (EUROSTUDENT, 2024).

Thus, international students are most vulnerable to the housing crisis: they are being pushed out of central areas, concentrated in the most marginal segments of the market (shared flats with 6-8 people per room, short-term contracts, illegal or semi-legal rentals) and have a disproportionately high rental burden.

2.1.7. Challenges

Despite notable progress, the literature faces a number of limitations.

First, there is a lack of data and weak integration of sources. Many studies rely on local or one-off cases (Miessner, 2021; von Wittenhorst zu Sonsfeld & Beusker, 2024). Nationwide and Europe-wide studies (EUROSTUDENT, 2024) provide a valuable overview (e.g., the proportion of students spending >40% of their budget on housing), but rarely differentiate data by type of housing (dormitories vs. WGs vs. private rentals) and status (international vs. local students).

Secondly, a significant portion of the studies provide qualitative assessments without quantitative verification. For example, the housing affordability crisis is described through an array of questionnaire complaints and narratives (stress, fraud, language barriers), but is rarely accompanied by spatial modelling or econometric estimates of the impact on mobility participation. Even where quantitative analysis is available, patterns are not always systematised. Miessner (2021) and Avogo et al. (2022) identify important behavioural patterns (private landlords' focus on students), but do not link them to urban structure or patterns of student housing distribution.

Finally, most studies highlight the role of private actors (investors, landlords, platforms), but provide much less analysis of how the state and universities reproduce or moderate inequality through their decisions.

2.1.8. *Gaps*

Based on the literature reviewed, several key gaps can be identified:

Lack of a comprehensive approach to universities and student housing. Universities and Studentenwerke (student unions) are often viewed merely as background players (providers of dormitories) rather than active actors in the housing field. For example, the IBB report analyses the Berlin market in detail, but hardly highlights universities as a separate policy area (Investitionsbank Berlin, 2024).

Insufficient consideration of the role of the public sector in the current housing crisis for students and migrants. Although some authors emphasise that the abandonment of housing subsidies and the privatisation of municipal housing were key triggers of the crisis (Wijburg & Aalbers, 2017), there are still few studies that directly link the reduction in student accommodation/state programmes with the increased vulnerability of the student group in particular. Paradoxically, even in the discussion of social commons (Stoll, 2022), student housing is hardly mentioned, although logically it is a classic candidate for public ownership and democratic governance.

Most studies are either macro-level (national/European surveys) or micro-cases of individual student cities and campuses. There are no studies that systematically examine the spatial distribution of student and migrant housing in large cities (e.g., Berlin) at the planning district level.

These gaps justify the need for research that examines student housing in conjunction with university structures and urban housing policy, uses spatial analysis (at the city and district level), and assesses how different management models (state/municipal, university, private, cooperative) affect the daily lives of students in different ways.

2.2. 1.2 Berlin Housing Crisis History

The history of housing in Berlin is a story of constant crisis and adaptation. Throughout its history, the city has undergone a series of cycles of destruction, reconstruction and transformation. This chapter tells the recent history of Berlin from the post-war period to the present day.

2.2.1. 1945-1989. Post-World War II Developments

When World War II ended, Berlin was heavily damaged and destroyed. All political parties perceived the housing shortage as the most pressing problem of the time and supported state investment into housing and strong regulation of the market. In this respect, estates built under socialism in East Berlin and under capitalism in West Berlin share many commonalities, despite the political differences of their respective regimes during the division (1949–1990) and particularly during the times of the Berlin Wall (1961–1989)(Urban, 2018). During this period, the authorities in East and West Berlin viewed housing construction as a key instrument of social policy.

West Berlin Housing

In the West, most multifamily buildings—both small and large estates—were built by state-sponsored institutions, owned by municipalities or sometimes other public bodies such as trade unions (Urban, 2018). Between 1952 and 1970, over 85% of new housing construction was publicly subsidized, and frequently constructed by state-owned housing companies operating under the principle of common public interest (Fields & Uffer, 2016)

Interventionist housing policy, which funneled large amounts of public disproportionate funding into housing, was set up in the early 1950s, and at the time approved by the West German Federal Government, ultimately aimed at sustaining the city's population levels (Lindenthal and Eichholtz, 2012). Despite government support, by the 1970s these neighbourhoods were criticised for their social isolation and standardised architecture, becoming a symbol of the bureaucratic approach to housing.

East Berlin Housing

In East Germany, the paradigm change towards large estates was far more comprehensive (Urban, 2018). After the Building Conference in 1955, the GDR began the 'industrialisation of the construction industry', which led to the mass construction of houses from pre-fabricated reinforced concrete panels (Plattenbau). Private landownership was almost entirely abandoned and state-led housing associations provided mass housing, becoming the most conspicuous urban form (Fields and Uffer, 2016). Although the new mass construction embodied the idea of socialist equality, over time it became associated with uniformity and social control by the state.

2.2.2. 1989/1991. Political and Spatial Reunion

The fall of the Berlin Wall meant not only political but also spatial unification of two separate housing systems. The unification of the city provided for a massive shock to housing supply.

311 After the opening up of the East, households could live in East Berlin or in surrounding Brandenburg instead
312 of West Berlin. This increased competition is hypothesized to lower prices in peripheral locations of West Berlin.
313 In the same time, the central quarters became “more central”, since they are now in the middle of an overall larger
314 metropolitan area (Lindenthal and Eichholtz, 2012).

315 Municipal policy towards Berlin’s housing estates was somewhat contradictory. On the one hand, in the early 1990s,
316 both the local authority and the national government poured generous subsidies into the housing stock supporting
317 renovation. On the other hand, along with increasing international popularity of market-centred ideas, market was
318 turning against social housing, and thus against the every institution from which these estates derived (Urban, 2018).

319 **2.2.3. 1992-2012. Privatisation, Financialisation and Gentrification**

320 Berlin’s fiscal instability motivated privatisation of its state-owned housing stock and abandonment of social hous-
321 ing subsidies. In 1991, the city owned approximately 480,000 housing units (28% of the entire housing stock), but
322 by the early 2000s, most of this housing had been sold to private and institutional investors (Fields and Uffer, 2016).
323 While privatisation created expectations that private investors would contribute their financial resources to modernize
324 the housing stock, Berlin’s housing demand was low.

325 Between 2000 and 2006, Berlin sold 100,000 dwelling units to international investors, thus sacrificing a system
326 working with long-term success for short-term profit (Fields and Uffer, 2016). Private investors were speculating on a
327 rising market seeing an opportunity to increase rents through modernisation and luxurious upgrading (Urban, 2018).

328 Research by Fields and Uffer (2016) shows that private investment has exacerbated inequality in access to housing,
329 and global financialisation mechanisms have come to determine urban policy more strongly than social priorities. High-
330 yield investors have renovated residential buildings, increased rents and removed flats from the social rental system.
331 This process has been accompanied by the gentrification of central areas and the social displacement of low-income
332 residents to the periphery.

333 **2.2.4. 2012 - 2024. New Actors, Old Problems**

334 Since the early 2010s, Berlin has been experiencing a new wave of housing crisis.

335 Compared to other major German cities, Berlin still has potential for land use both in the city and in the surrounding
336 area (Berlin Strategy, 2018). Which makes it unique for urban planning research.

337 At the same time, a variety of challenges are already looming that the city will have to overcome in the coming
338 years. Above all, these include tasks associated with urban growth. According to the Senate Department for Urban
339 Development, Building and Housing (2022), Berlin already reached a population of 3.75 million. The population
340 forecast assumes that the number of residents will rise to around 3.93 million by 2030. The growth is primarily driven
341 by immigration from abroad (CBRE GmbH, 2025). Such residents accounted for 24.4 percent of Berlin’s population

342 in 2024. The number of newcomers who moved to Berlin from abroad was twice as high as the number of people who
343 moved to the capital from other parts of Germany.

344 Opportunities accompanying this growth include the potential for more affordable housing. However, these devel-
345 opments often come with specific spatial requirements and property policies (Berlin Strategy, 2018). Moreover, the
346 application of planning law become particularly challenging in light of rising costs, inflation, and high interest rates,
347 which carry the risk of social inequality, spatial polarisation, and housing crisis.

348 **2.2.5. Conclusion**

349 Berlin Housing History is a story of recurring crises and compromises. The contradiction between housing as a
350 social right and housing as a market commodity remains unresolved.

2.3. 1.3 Theoretical Framework

Theoretical framework can be seen as a point of view of to research question. In order to make research interdisciplinary it is a need to consider urban and social theories. While there are many theories that could help to solve the student housing crisis in Berlin, this chapter examines only three main theories that respond to literature gaps and influenced the forming of research interest.

2.3.1. *Urban Theories*

Central Place Theory

Central Place Theory helps to construct spatial research at the level of a large city such as Berlin, as well as at the level of its planning districts, defining urban centrality:

For many economic activities located in a city it is important that they be central to some larger market area that extends beyond the boundaries of the urban place itself. This is especially so for activities that serve the community surrounding the urban place (King, 1986).

This theory can be adapted to the student housing crisis by considering universities as central points that establish spatial economic relations in the student housing market.

Central Place Theory, considering universities as central places, helps to unite the existing Berlin student real estate market and the universities that form the demand for this market. Depending on the ownership of student accommodation, the theory helps to define the roles of both the private and public construction sectors.

Thus, the economic interrelationships between higher education universities and student accommodation come sharply into focus when, for example, policy issues concerning metropolitan and regional government reorganisation are raised.

2.3.2. *Social Theories*

Right to the City Theory

Right to the City Theory is a necessary response to the financialisation that has taken place in Berlin over the past decades.

The concept of Right to the City, developed by French philosopher and sociologist Henri Lefebvre, is a critical approach to understanding social relations in the modern city (Lefebvre, 1996). Lefebvre argues that the city is not just a physical space, but a product of social relations where residents have an unconditional collective right to urban life. Lefebvre's concept allows us to rethink the student housing crisis from the perspective of social justice, accessibility, and commercialisation.

The future of the city must be shaped by its inhabitants, not by the interests of capital. (Lefebvre, 1968).

Berlin's urban environment must take into account the needs of all inhabitants, not just the economically privileged groups. Students are an important part of the urban population, providing a driver of change in the city. Every student has the right to social justice. Factors influencing students' housing choices should be taken into account in Berlin's long-term urban planning.

The city is an oeuvre, closer to a work of art than a simple material product. (Lefebvre, 1968).

The housing market in Berlin should be shaped by social demands and not by commercial interests. The city is not an object of speculation. Prioritising private development over the creation of new affordable housing contradicts the student's right to the city.

In Berlin, this theory is particularly relevant and has actually been written into the city constitution:

Everyone has the right to adequate housing. The state promotes the creation and provision of adequate housing, especially for people on low incomes, and the cultivation of condominium ownership – Article 28 (1) of the Berlin Constitution.

This theory aims to protect students' rights to affordable housing, counteract the displacement of students from the central urban fabric, and halt the privatisation of social housing.

Affordability Theory Affordability Theory complements spatial and social analysis with an economic dimension of housing affordability. The main problem with student housing in Berlin is not only a shortage of supply, but also the fact that prices are out of line with students' actual financial capabilities.

The Affordability Theory in Housing, developed by Michael Stone (2006), defines affordability beyond rental prices, incorporating income, basic needs expenses, transport costs, and structural limitations of the housing market. Stone's theory allows us to define affordable housing in a broader economic and social context.

Housing affordability is not just about the cost of housing, but about the balance between housing costs and the ability to meet other basic needs. (Stone, 2006).

The assessment of housing affordability for students should include not only the cost of housing itself and its proportion in relation to the student's income but also the cost of studying at Berlin's universities, as well as basic financial needs such as transport, student insurance, and groceries. In addition to basic financial needs, the term affordability also includes basic social needs such as the quality of housing, the state of the infrastructure, and the possibility for students to participate in urban life. Student housing affordability is the ability to cover housing costs without compromising basic financial and social needs.

Affordability Theory serves as a conceptual basis for analysing student housing needs and preferences and requires a behavioural and qualitative approach. Thus, it complements the quantitative spatial research of Central Place Theory with a qualitative assessment of the actual living conditions of students, revealing the gap between formal housing

414 supply and actual affordability.

415 Together, Central Place Theory, Right to the City Theory, and Affordability Theory form a comprehensive analyt-
416 ical framework that allows us to view the student housing crisis in Berlin as a spatial, social, and economic problem
417 simultaneously. These theories provide a theoretical basis for data-driven analysis aimed at reducing the gap between
418 the opportunities offered by the student housing market and the real needs of students in the context of the ongoing
419 housing crisis.

2.4. 1.4 Rethinking Student Housing in Berlin

2.4.1. Problem Statement

Housing is a basic human need and a fundamental right, comparable in importance to access to water or clean air (Lefebvre, 1968). Stable, affordable and safe housing is a necessary condition for a dignified life, social integration and full participation in urban and educational spaces. In this context, housing should be viewed not only as physical infrastructure, but also as a socio-political tool that shapes access to opportunities, resources and rights.

As Roberta Burghardt (2025) notes, architects, urban planners and designers can influence not only the form of living spaces, but also models of ownership, participation schemes, and mechanisms for managing and allocating housing. Thus, the issue of student housing goes beyond the design of buildings and becomes part of a broader discussion about urban justice and governance.

2.4.2. Context

Historically, in the post-war period, students in Berlin mainly lived in shared rental accommodation. However, over the last few decades, the structure of student housing has changed significantly. Purpose-Built Student Accommodations (PBSA) — specially built and managed residential complexes for students — have begun to play a special role, which is particularly noticeable in the international student segment.

Nevertheless, despite the growth in the number of such facilities, only about 8% of students in Berlin live in PBSA (Deutsches Studierendenwerk, 2024). Most students are still forced to look for accommodation on the general rental market — in shared flats, sublets or corporate models — due to the general rental pattern of the German housing market. As a result, students compete with other population groups for a limited number of available flats, leading to rising prices, housing instability and housing vulnerability.

The process of financialisation of student housing reinforces this imbalance: PBSAs are viewed by investors as stable investment products with high returns, due to the lack of sufficient alternatives in the public sector. As a result, the social function of student housing is taking a step back, making way for market logic (Annas Ejaz, Anika Gercke, 2025).

2.4.3. Limitations

At the same time, a variety of challenges are already looming that the city will have to overcome in the coming years. Above all, these include tasks associated with urban growth. According to the Senate Department for Urban Development, Building and Housing (2022), Berlin already reached a population of 3.75 million. The population forecast assumes that the number of residents will rise to around 3.93 million by 2030. The growth is primarily driven by immigration from abroad (Housing Market Report, 2024). Such residents accounted for 24.4 percent of Berlin's population in 2024. The number of newcomers who moved to Berlin from abroad was twice as high as the number of

451 people who moved to the capital from other parts of Germany.

452 Berlin's housing policy stipulates that at least 30% of new housing must be social housing, with the aim of pre-
453 serving social diversity and housing affordability in all areas of the city (Senate Department for Urban Development,
454 Building and Housing, 2023). However, this target has not yet been sufficiently achieved in the area of student housing.

455 As ESN and ESU (2023) emphasise, expanding public student housing should be a priority across Europe, espe-
456 cially in countries and cities where it is underdeveloped. Such housing plays a key role not only in ensuring affordability,
457 but also in supporting the internationalisation and mobility of students. Higher education institutions, as important
458 urban actors, should engage more actively with local authorities and the private sector, as many of the challenges faced
459 by students are not only market-related but also involve contractual and administrative barriers, where the public sector
460 can play a decisive role.

461 The study focuses exclusively on international students and purpose-built student accommodation, which allows
462 for a deeper understanding of the current structural vulnerabilities of this model, but limits the ability to generalise the
463 results to the entire student population.

464 **2.4.4. Research Question**

465 **How can data-driven decision-making reshape Berlin's student housing, to bridge the gaps between student**
466 **housing market capacities and students demands, addressing basic financial and social needs, during the city's**
467 **ongoing housing crisis?**

468 **2.4.5. Scientific Objectives**

469 In accordance with the question posed, the study has the following scientific objectives:

470 1. Student Housing Market: Capacities, Demand, and Spatial Distribution **Where are student accommodations**
471 **and universities located?**

472 **How are they spatially distributed across the city?**

473 resume Student Housing Market Imbalance **How do student university demand compare to student housing mar-**
474 **ket housing capacities?**

475 **Where do spatial patterns of demand and supply generate elevated market pressure?**

476 resume Students Behind the Numbers **Who are the users of student accommodations?**

477 **What experience they have in ongoing housing crisis? What are common needs, preferences, difficulties and**
478 **problems?**

479 The study aims to combine spatial, quantitative and qualitative analysis to go beyond abstract market indicators. The
480 use of geodata and statistics allows us to identify areas of increased pressure and structural imbalances, while surveys
481 and interviews reveal the everyday experiences of students that are hidden behind aggregated data. This combined
482 approach not only identifies housing shortages, but also understands how they translate into stress, unstable housing
483 strategies, discriminatory practices and social vulnerability, especially among international students. Ultimately, the
484 goal of the study is to form a basis for more equitable and informed decisions in the field of student housing, taking
485 into account the real needs of users and the constraints of the urban context.

3. 2. Capacities, Demand & Spatial Distribution

3.1. 2.1 Overview

Chapter Capacities, Demand & Spatial Distribution gives an overview of Berlin's student housing market. It focuses on evaluation of initial market capacities and demand and how they spread across the city.

3.1.1. What? Student Housing Market: Capacities, Demand, and Spatial Distribution

The student housing market can be understood as the intersection between demand — represented by the number of international students and their housing needs — and capacity, defined by the number of available purpose build student accommodations.

By definition given in MLP Studentenwohnreport (2024):

Demand - mainly depends on the number of students, which often changes by location. Students also compete with young workers, trainees, or commuters, so student numbers only partly show demand. For this research, the target group is international students enrolled in Berlin's higher education, at any level, regardless of current residency.

Capacity- is determined by the number of advertised apartments in the corresponding segments. Alternative housing options, such as student union dormitories, are also available. During the research process, all categories will be merged in a broader term - purpose-built student accommodation, categorised by type of ownership.

For the simplicity of data collection, student housing market will be understood as the intersection between demand — represented by the number of international students and their housing needs — and capacity, defined by the number of available purpose build student accommodations.

In a broader perspective, both demand and capacity are dynamic parameters influenced by university services, urban policy and social drivers. A more detailed discussion of their interrelation will be presented in the Discussion Chapter.

3.1.2. Why? Initial Student Housing Market Description

The aim of this chapter is to describe the current state of Berlin's student housing market by answering the following research questions:

Where are student accommodations and universities located?

How are they spatially distributed across the city?

3.1.3. How? Spatial Data Analysis and Spatial Econometrics

The descriptive process combines Urban Data Mining, Statistical Analysis, and Spatial Autocorrelation to examine market characteristics and explore spatial relationships.

Urban Data Mining

Urban Data Mining is a data collection method that aims to gain new insights by identifying patterns, exceptions, and

517 anomalies in the data (Giseke et al., 2021). It involves selecting, exploring, cleaning, interpreting, and visualising
518 datasets. Data Mining also involves transforming data from non-spatial to spatial. Therefore, it can be seen as the first
519 step of Spatial Data Analysis (SDA). Due to the need to work with large amounts of data. Data Mining is primarily
520 achieved through software, with Python Programming and Artificial Intelligence methods used to further develop urban
521 environments and improve residents' daily lives (Giseke et al., 2021).

522 Urban Data Mining is used - one of the most time-consuming stages. Since there is no pre-made data set that
523 combines data on all student dormitories and educational institutions in Berlin. Therefore, the data had to be collected
524 in two stages: first manually, then automatical preparation. The quality of manual collection depends directly on the
525 time spent on it and carries the risk of incomplete data, as different data sources indicate different sets of attributes.
526 Meanwhile, automatic data preparation allows you to identify inaccuracies in the data and fill in missing attributes.

527 **Descriptive Statistics**

528 Descriptive Statistics is a set of statistical procedures that summarize the essential characteristics of a distribution
529 through calculating and plotting. Statistical methods are employed to analyse spatial data, quantify a spatial process,
530 discover hidden patterns or unexpected trends and model these data in a geographic context(Giseke et al., 2021).

531 Descriptive statistics in the context of this work is considered as the initial stage of data analysis, which allows
532 a quantitative assessment of the characteristics of student housing and higher educational institutions, as well as the
533 identification of initial quantitative patterns.

534 **Spatial Autocorrelation**

535 Spatial Autocorrelation is the degree of spatial dependency, association, or correlation between the value of an obser-
536 vation of a spatial entity and the values of neighboring observations of the same variable. It is unlikely that the spatial
537 distribution of Student Accommodations and Higher Education Universities will be homogeneous in the entire Berlin
538 due to the nonuniformity and noncontinuity of space. The magnitude of spatial autocorrelation may vary from one
539 neighborhood to another due to spatial heterogeneity. To estimate spatial autocorrelation at the local level, there is a
540 need to use local measures of spatial autocorrelation, like Local Moran's I index (Grekousis, 2020).

541 Unlike Urban Data Mining and Descriptive Statistics, Spatial Autocorrelation works directly with spatial dimen-
542 sions and allows you to identify patterns in the spatial distribution of student accommodations and higher education
543 institutions, as well as evaluate the spatial differences between them.

544 Through the combination of quantitative, spatial, and computational approaches, this chapter establishes a founda-
545 tion for the empirical analysis of Berlin's student housing system. It links theoretical concepts of demand and capacity
546 with practical spatial evidence, creating a bridge between urban data science and housing policy research.

547 **3.2. 2.2 Urban Data Mining**

548 Urban Data Mining, or the process of collecting and preparing data for analysis, consisted of two stages, which
549 can be divided into manual collection and automated data preparation. The purpose of the manual stage was to collect
550 data and convert it into tabular form, while the automated process was aimed to convert the tabular data into geospatial
551 data.

552 **3.3. Manual Data Collection**

553 The main task of manual data collection is searching purpose-built student accommodation and higher education
554 universities in Berlin and collecting capacity and demand data, respectively.

555 **3.3.1. Student Accommodation**

556 The Studentenwerk (StW, 2025) website was used as the initial source of data for the purpose of building student
557 accommodation. StW is a public, non-profit organisation that provides essential services to students in Germany, in-
558 cluding affordable housing, operating student dormitories in collaboration with universities. The StW website provides
559 not only detailed information about addresses, prices, accommodation types, and living space in student accommoda-
560 tion, but also references to other publicly funded companies that provide social housing for students, such as Howoge,
561 Gewobag, Berlino, and others. Data on the capacity of public student dormitories was collected from the Study in
562 Germany website, which is supported by the Federal Ministry of Education and Research (BMBF, 2025) and is part
563 of the ‘Germany – Land of Ideas’ campaign.

564 The missing addresses of private companies were collected from online platforms of companies specialising in
565 student housing and property rental in Berlin. While public companies publish data openly, private companies have
566 less transparent policy, which limits the research process and complicates the manual data mining process.

567 A complete list of online sources used for manual data collection on student accommodation is presented in Table
568 2.

	source	comment ¹	link
0	Study in Germany	An official informational platform aimed at in...	https://www.study-in-germany.com/de/
1	HOWOGE Wohnungsbaugesellschaft mbH	Municipal housing company with dedicated stude...	https://www.howo.de/immobiliensuche
2	GESOBAU AG	Public housing provider offering projects like...	https://www.gesobau.de/wohnungsbauprojekte
3	GEWOBAG	New-build and modernized housing projects, som...	https://www.gewobag.de/
4	WBM	Municipal housing firm offering apartments in ...	https://www.wbm.de/neubau-berlin/
5	Berlinovo	Offers student apartments across Berlin. Filte...	https://www.berlinovo.de/de/apartments
6	BGG Berlin	Private housing group with student-suitable ap...	https://www.bgg-berlin.com/de
7	STW Berlin	Official student dorm provider in Berlin; list...	https://www.stw.berlin/en/
8	Studentendorf Berlin	Community-oriented residences in Adlershof and...	https://www.studentendorf.berlin/
9	The FIZZ	International student housing brand; premium h...	https://www.the-fizz.com/en/
10	Neon Wood	Private student housing provider; fully furnis...	https://neonwood.com/de/
11	Home in Berlin	Central Berlin housing platform targeting stud...	https://home-in-berlin.de/
12	Insider Apartments	Furnished rental apartments for students and P...	https://insiderapartments.com/project
13	Studentenhotel Hubertusallee	Hotel-style short- and long-term housing optio...	https://studentenhotel-hubertusallee.de/
14	Studentenwohnheim	External platform listing	http://www.studentenwohnheim.ch/berlin

570 *Table 2. Manual Data Collection Sources for Student Accommodation*

571 The address is a critical parameter for conducting spatial econometric analysis. However, many student accom-
572 modations have more than one building. For example, WH Spandauer Damm comprises five buildings located at
573 Spandauer Damm 134–142 (Studierendenwerk Berlin, 2025). During manual data collection, one of the tasks was
574 to collect unique addresses related to the student housing market in order to calculate centroids afterwards, during
575 automated data preparation.

576 For student housing, the data collected included the address (street, house_nummer, plz, city), the name of
577 the building complex (name), the ownership company name (ownership), the ownership model (ownership_model),
578 number of residential units (!units), their capacity (!students) and type (units_type), maximum and minimum
579 price (price_min, price_max), maximum and minimum living space (area_min, area_max), link to the source
580 (link).

581 As a result of manual data collection, data was collected on 94 proposed student accommodation buildings located
582 at 171 unique addresses.

```
583 fp = r"../inputs/2.2_urban_data_mining/input_pbsa.xlsx"
584
585 pbsa = pd.read_excel(fp, sheet_name='SA', dtype={'street': str, 'house_nummer': str, 'plz': str})
586 pbsa.info()
587 <class 'pandas.core.frame.DataFrame'>
588 RangeIndex: 171 entries, 0 to 170
589 Data columns (total 16 columns):
590 #    Column                Non -Null Count  Dtype
591 -  -  -  -  -  -  -  -  -  -  -  -  -  -  -  -  -
592 0    name                  171 non -null    object
593 1    street                171 non -null    object
594 2    house_nummer          171 non -null    object
595 3    plz                   171 non -null    object
596 4    city                  171 non -null    object
597 5    model                 171 non -null    object
598 6    ownership              171 non -null    object
599 7    ownership_model       171 non -null    object
600 8    !units                93 non -null     float64
```

```

601 9  !students      148 non -null    float64
602 10 units type    168 non -null    object
603 11 price_min     148 non -null    float64
604 12 price_max     96 non -null    float64
605 13 area_min      154 non -null    float64
606 14 area_max      144 non -null    float64
607 15 link          168 non -null    object
608 dtypes: float64(6), object(10)
609 memory usage: 21.5+ KB

```

610 **3.3.2. Higher Education Universities**

611 In order to indicate Berlin's student demand, it was necessary to collect data on all of Berlin's higher education
612 institutions and the number of students enrolled in them. Although the initial plan was to limit the search to public
613 educational institutions only. Nevertheless during the data collection process it became evident that the list and statistics
614 for private universities were also publicly available.

615 The dataset provided by the Statistisches Bundesamt (DESTATIS, 2025) serves as the most comprehensive source,
616 capturing detailed information on student enrollments, origin, and gender at public and private German higher educa-
617 tion institutions. The data provided was so detailed that it included information about branches of Berlin universities
618 in other regions of the country. For example, Media Design School Berlin also has a campus in Munich. These
619 universities were excluded from the statistics as they do not put pressure on the Berlin Student Housing Market.

620 After cleaning, the Higher Education Institution list from the Federal Statistical Office was double-checked with
621 the official Berlin city directory of higher education institutions (Berlin City Portal, 2025) and the official registry
622 maintained by the German Rectors' Conference (Hochschul Kompass, 2025).

623 A complete list of online sources used for manual data collection on higher education institutions is provided in
624 Table 3.

```

625 sources[[
626     "source",
627     "commentz",
628     "link",
629 ]] .iloc[23:]

```


	source	comment ¹	link
23	STW	Official public map of university-related faci...	https://www.stw.berlin
24	Hochschul Kompass	Official registry of higher education institut...	https://www.hochschulkompass.de/ho
25	Berlin City Portal	Official Berlin city direc- tory of higher educa...	https://www.berlin.de/sen/wissensch
26	Federal Statistical Office of Germany (Destatis)	Statistik der Studenten (Code: 21311-0002) is ...	https://www.openstreetmap.org/#map

Table 3. Manual Data Collection Sources for Higher Education Universities

After secondary verification of the list of educational institutions, the next task was to collect unique addresses, as large universities have more than one campus. For example, Charité - Universitätsmedizin Berlin has four campuses in Berlin, which are located at a total of 41 unique addresses. The manual search for addresses was carried out using two services: Open Street Map and Google Maps.

For Higher Education Universities, the data collected included the code according to the Federal Statistical Office, the abbreviation (abr), the name (name) and the full name (fname), the address (adr), the ownership model (own), type of educational institution (stud), number of German students (!s_de), number of foreign students (!s_aus) and total number of students (!s_all).

As a result of manual data collection, data was collected on 70 higher education institutions in Berlin, located at 357 unique addresses.

```
op = r"../inputs/2.2_urban_data_mining/input_uni.xlsx"

uni = pd.read_excel(op, sheet_name='Uni', dtype={'adr': str})

uni.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 357 entries, 0 to 356
Data columns (total 11 columns):
#   Column  Non -Null Count  Dtype
-  -  -  -  -  -  -  -  -  -  -  -  -  -  -  -  -  -  -  -  -  -  -
0   code    357 non-null    object
```

```

652 1   abr      70 non -null    object
653 2   name     70 non -null    object
654 3   fname    70 non -null    object
655 4   adr      357 non -null   object
656 5   own      70 non -null    object
657 6   stud     71 non -null    object
658 7   !s_de    53 non -null    float64
659 8   !s_aus   52 non -null    float64
660 9   !s_all   54 non -null    float64
661 10  url      42 non -null    object
662 dtypes: float64(3), object(8)
663 memory usage: 30.8+ KB

```

664 3.4. Automated Data Preparation

665 During Manual Data Collection, one of the biggest challenges of data-driven research - finding and selecting data
666 - was successfully overcome.

667 The next challenge within Urban Data Mining is data cleaning, formatting, and standardising. Data preparation using
668 the Python programming language not only allows you to identify errors in manual input data, but also automates
669 problem-solving during data import, cleaning, and export, as well as during geocoding.

670 3.4.1. Set Up

671 Import Libraries

```

672 import pandas as pd
673 import numpy as np
674 from tqdm import tqdm
675 import geopandas as gpd
676 from geopy.geocoders import Nominatim, Photon
677 from geopy.extra.rate_limiter import RateLimiter
678 import matplotlib.pyplot as plt
679 from matplotlib.colors import LinearSegmentedColormap
680 from shapely.geometry import MultiPoint
681 from sklearn.cluster import DBSCAN
682 import time

```

Scripts

In order to access spatial analysis and spatial economics methods and Tools, it is necessary to convert addresses to geographical coordinates, which consist of longitude and latitude. Geocoding converts addresses into geographical coordinates to be placed on a map.

Two services, Nominatim and Photon, were used for geocoding. Despite the use of several services, about 10% of addresses were not recognised or were misinterpreted, indicating errors in the manually collected data.

To improve the accuracy of the coordinate search, the names of the city (Berlin) and country (Germany) were added to each address, which made it possible to uniquely identify objects in the German capital. In addition, to account for the particularities of the German language and possible spelling errors, an alternative spelling mechanism was implemented: special characters such as ' β ' were automatically replaced with 'ss', which increased the likelihood of successfully matching the address with the geographic database.

Since the addresses for student accommodation and higher education universities were collected in different formats, two different scripts were also used for geocoding, which differed from each other in their query (df['query']). At the same time, it would be advisable to avoid writing two scripts by collecting data on student accommodation and higher education universities in a single format (for example, df['query'] = df['street'] + ' ' + df['house_nummer'] + ', ' + df['plz'] + ', ' + df['city'] + ', Germany').

```
def geocode_pbsa(df):  
    # Initialize geocoders  
    nom = RateLimiter(Nominatim(user_agent="berlin_housing", timeout=10).geocode,  
                      min_delay_seconds=1)  
    photon = RateLimiter(Photon(user_agent="berlin_housing", timeout=10).geocode,  
                        min_delay_seconds=0.5)  
  
    # Create address strings  
    #df['query'] = df['adr'] + ', Germany'  
    df['query'] = df['street'] + ' ' + df['house_nummer'] + ', ' + df['plz'] + ', ' + df['city']  
    df['query_alt'] = df['query'].str.replace(' \beta', 'ss') # Alternative spelling  
  
    # Geocoding function with retries  
    def geocode(address, geocoder, retries=2):  
        for _ in range(retries):  
            try:
```

```

715         loc = geocoder(address)
716         return (loc.latitude, loc.longitude) if loc else (None, None)
717     except:
718         time.sleep(1)
719     return (None, None)
720
721 # Primary geocoding (Nominatim)
722 tqdm.pandas(desc="Primary geocoding", disable=True)
723 df[['lat', 'lon']] = df['query'].progress_apply(
724     lambda x: pd.Series(geocode(x, nom))
725 )
726
727 # Fallback (Photon for failed addresses)
728 missing = df[df['lat'].isna()]
729 if not missing.empty:
730     tqdm.pandas(desc="Fallback geocoding", disable=True)
731     missing[['lat', 'lon']] = missing['query_alt'].progress_apply(
732         lambda x: pd.Series(geocode(x, photon)))
733     df.update(missing)
734
735 # Convert to GeoDataFrame (Berlin CRS)
736 return gpd.GeoDataFrame(
737     df.drop(['query', 'query_alt'], axis=1, errors='ignore'),
738     geometry=gpd.points_from_xy(df['lon'], df['lat']),
739     crs="EPSG:4326"
740 ).to_crs("EPSG:25833")
741
742 def geocode_uni(df):
743     # Initialize geocoders
744     nom = RateLimiter(Nominatim(user_agent="berlin_housing", timeout=10).geocode,
745                       min_delay_seconds=1)
746     photon = RateLimiter(Photon(user_agent="berlin_housing", timeout=10).geocode,
747                           min_delay_seconds=0.5)

```

```

747
748 # Create address strings
749 df['query'] = df['adr'] + ', Germany'
750 df['query_alt'] = df['query'].str.replace(' \beta', 'ss') # Alternative spelling
751
752 # Geocoding function with retries
753 def geocode(address, geocoder, retries=2):
754     for _ in range(retries):
755         try:
756             loc = geocoder(address)
757             return (loc.latitude, loc.longitude) if loc else (None, None)
758         except:
759             time.sleep(1)
760     return (None, None)
761
762 # Primary geocoding (Nominatim)
763 tqdm.pandas(desc="Primary geocoding", disable=True)
764 df[['lat', 'lon']] = df['query'].progress_apply(
765     lambda x: pd.Series(geocode(x, nom))
766 )
767
768 # Fallback (Photon for failed addresses)
769 missing = df[df['lat'].isna()]
770 if not missing.empty:
771     tqdm.pandas(desc="Fallback geocoding", disable=True)
772     missing[['lat', 'lon']] = missing['query_alt'].progress_apply(
773         lambda x: pd.Series(geocode(x, photon)))
774     df.update(missing)
775
776 # Convert to GeoDataFrame (Berlin CRS)
777 return gpd.GeoDataFrame(
778     df.drop(['query', 'query_alt'], axis=1, errors='ignore'),

```

```

779         geometry=gpd.points_from_xy(df['lon'], df['lat']),
780         crs="EPSG:4326"
781     ).to_crs("EPSG:25833")

782     Berlin Spatial Boundaries

783 pf = r"../inputs/2.2_urban_data_mining/lor_pgr.gpkg"
784 pgr = gpd.read_file(pf)
785 pgr.replace({'-': np.nan, 0: np.nan, '': np.nan}, inplace=True)
786
787 bf = r"../inputs/2.2_urban_data_mining/adm_bz.gpkg"
788 bz = gpd.read_file(bf)
789 bz.replace({'-': np.nan, 0: np.nan, '': np.nan}, inplace=True)
790
791 of = r"../inputs/2.2_urban_data_mining/adm_out.gpkg"
792 out = gpd.read_file(of)
793 out.replace({'-': np.nan, 0: np.nan, '': np.nan}, inplace=True)

```

794 **3.4.2. Data Import**

795 The data obtained as a result of manual data collection served as input data for the automated stage.

```

796 fp = r"../inputs/2.2_urban_data_mining/input_pbsa.xlsx"
797
798 pbsa = pd.read_excel(fp, sheet_name='SA', dtype={'street': str, 'house_nummer': str, 'plz': str})
799
800 op = r"../inputs/2.2_urban_data_mining/input_uni.xlsx"
801
802 uni = pd.read_excel(op, sheet_name='Uni', dtype={'adr': str})

```

803 **3.4.3. Geocoding**

804 As part of the automated data preparation stage, the addresses of student accommodation and higher educa-
805 tion institutions were geocoded to obtain accurate geographical coordinates for each unique address. The data was
806 then converted into a geospatial format (GeoDataFrame) and reprojected into the official Berlin coordinate system
807 (EPSG:25833).

808 **Student Accommodation**

```
809 pbsa = geocode_pbsa(pbsa)

810 pbsa.info()

811 <class 'geopandas.geodataframe.GeoDataFrame'>
812 RangeIndex: 171 entries, 0 to 170
813 Data columns (total 19 columns):
814 #   Column                Non -Null Count  Dtype
815 - - - - -
816 0   name                  171 non -null    object
817 1   street               171 non -null    object
818 2   house_nummer         171 non -null    object
819 3   plz                  171 non -null    object
820 4   city                 171 non -null    object
821 5   model                171 non -null    object
822 6   ownership            171 non -null    object
823 7   ownership_model      171 non -null    object
824 8   !units              93 non -null     float64
825 9   !students           148 non -null    float64
826 10  units type           168 non -null    object
827 11  price_min            148 non -null    float64
828 12  price_max            96 non -null     float64
829 13  area_min             154 non -null    float64
830 14  area_max             144 non -null    float64
831 15  link                 168 non -null    object
832 16  lat                  171 non -null    float64
833 17  lon                  171 non -null    float64
834 18  geometry             171 non -null    geometry
835 dtypes: float64(8), geometry(1), object(10)
836 memory usage: 25.5+ KB
```

837 **Higher Education Universities**

```
838 uni = geocode_uni(uni)
```

```
839 uni.info()

840 <class 'geopandas.geodataframe.GeoDataFrame'>

841 RangeIndex: 357 entries, 0 to 356

842 Data columns (total 14 columns):
843  #   Column      Non -Null Count  Dtype
844  --  --
845  0    code        357 non -null    object
846  1    abr          70 non -null    object
847  2    name        70 non -null    object
848  3    fname       70 non -null    object
849  4    adr         357 non -null    object
850  5    own         70 non -null    object
851  6    stud        71 non -null    object
852  7    !s_de       53 non -null    float64
853  8    !s_aus      52 non -null    float64
854  9    !s_all      54 non -null    float64
855  10   url         42 non -null    object
856  11   lat         357 non -null    float64
857  12   lon         357 non -null    float64
858  13   geometry    357 non -null    geometry
859 dtypes: float64(5), geometry(1), object(8)
860 memory usage: 39.2+ KB
```

861 This results in a shift from tabular data to geospatial data with a new attribute geometry.

862 **3.4.4. Data Cleaning**

863 **Student Accommodation**

864 Student Housing capacity is calculated as a single capacity indicator (!cap) according to the following rule: if the
865 number of students living in student accommodation (students) is known, it is taken; if not, the number of housing
866 units (units) is taken, while purposed build student accommodation, where both values are unknown, is excluded.

```
867 pbsa['!cap'] = pbsa['!students'].where(
868     pbsa['!students'].notnull(),
```



```

869     pbsa['!units']
870 )
871
872 pbsa = pbsa.dropna(subset=['!cap'])

```

873 After that, there is a transition from unique addresses to centroids or geographical centers (Figure XX), for which
874 grouping by name is performed (`pbsa.groupby("name")`). Thus, student dormitories with more than one building
875 are assigned a single point, and the capacity of all buildings is summed (!cap': "sum").

```

876 pbsa_c = (
877     pbsa.groupby('name')
878     .agg({
879         'geometry': lambda x: MultiPoint([pt for pt in x]).centroid,
880         'plz': 'first',
881         'city': 'first',
882         'model': 'first',
883         'ownership': 'first',
884         'ownership_model': 'first',
885         '!cap': 'sum',
886         'units type': 'first',
887         'price_min': 'first',
888         'price_max': 'first',
889         'area_min': 'first',
890         'area_max': 'first',
891         'link': 'first'
892     })
893     .reset_index()
894 )
895
896 pbsa_c = gpd.GeoDataFrame(pbsa_c, geometry='geometry', crs=pbsa.crs)

```

897 This results in a transition from 168 points to 88 centroids, which simplifies and speeds up further use of data in
898 the data analysis (Figure 8).

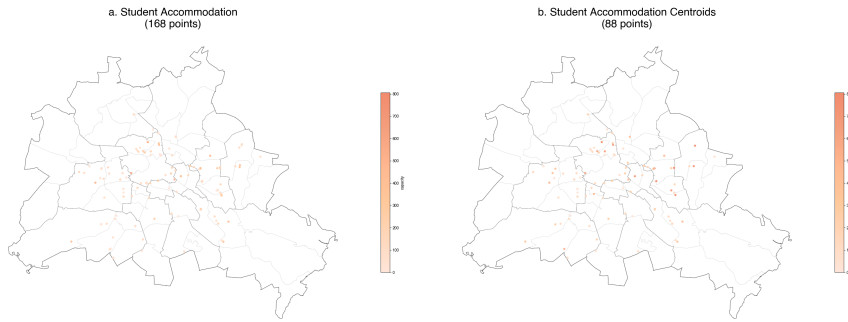


Figure 8. Visual Comparison Initial Unique Student Accommodation Addresses with Final Centroids

It should be noted that unique addresses were collected specifically for the correct calculation of centroids. Meanwhile, for student communities, the calculation of centroids is relatively simple due to the unique names for each community and the proximity of its buildings. For universities, however, the task is not so straightforward, as grouping by name will not give the desired result, since a university with the same name may be located in different parts of the city, for example, Humboldt-Universität, which has three campuses: Campus Mitte, Campus Adlershof, and Campus Nord.

Higher Education Universities

Higher Education Universities' demand for each educational institution is calculated as the number of international students enrolled in it. Universities for which there is no data on students are excluded.

```
uni = uni.dropna(subset=['!s_all'])
```

Afterwards, there is a transition from unique addresses to centroids (Figure XX). For this purpose, grouping by code (`uni['code']`) is performed, while demand indicators: the number of German students (`!s_de`), the number of foreign students (`!s_aus`) and the total number of students (`!s_all`) are distributed evenly among unique addresses — buildings — with a single unique code (`for code in uni['code'].unique()`). This allows us to obtain an 'average' representation of the demand indicators for each individual building, while these data for different buildings are initially combined.

```
copy_cols = ['abr', 'name', 'fname', 'own', 'stud', 'url']
```

```
split_cols = ['!s_de', '!s_aus', '!s_all']
```

```
# Loop over unique codes
```

```
for code in uni['code'].unique():
```

```
    mask = uni['code'] == code
```

```
    group = uni[mask]
```

924

925 # Copy values from first row in group

926 for col in copy_cols:

927 uni.loc[mask, col] = group.iloc[0][col]

928

929 # Distribute numeric values

930 for col in split_cols:

931 if pd.notnull(group.iloc[0][col]):

932 uni.loc[mask, col] = round(group.iloc[0][col] / len(group), 2)

933 else:

934 uni.loc[mask, col] = pd.NA

935 To further transition from unique addresses to a single address for a complex of university buildings, the data
 936 were grouped based on a distance threshold. The threshold parameter was responsible for separating unique addresses
 937 belonging to the same university campus based on the spatial factor. The threshold value was obtained through multiple
 938 manual tests. At the same time, the demand values for each unique address were summed up.

939 distance_threshold = 650 # in meters

940 records = []

941

942 for code, group in uni.groupby("code"):

943 coords = np.array([(geom.x, geom.y) for geom in group.geometry])

944 if len(coords) == 0:

945 continue

946 # Cluster points that are close to each other

947 clustering = DBSCAN(eps=distance_threshold, min_samples=1).fit(coords)

948 group = group.copy()

949 group['cluster'] = clustering.labels_

950 for cluster_id, cluster_points in group.groupby('cluster'):

951 mp = MultiPoint(cluster_points.geometry.tolist())

952 # Prepare a record with all columns

953 record = cluster_points.iloc[0].to_dict()

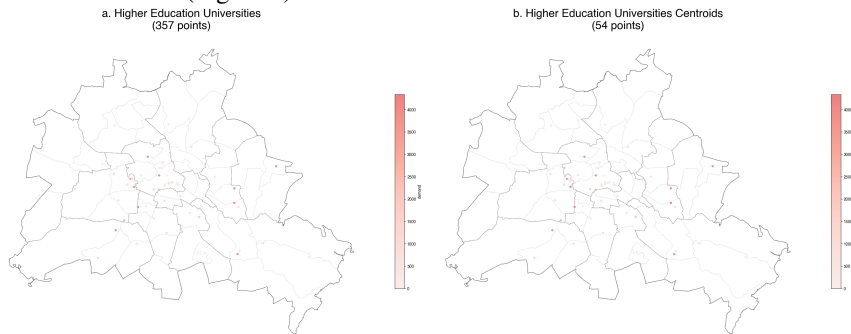
954 record['geometry'] = mp.centroid

```

955     # Sum the specified columns
956     for col in ['!s_de', '!s_aus', '!s_all']:
957         record[col] = cluster_points[col].sum()
958     record['cluster'] = f"{code}_{cluster_id+1}"
959     records.append(record)
960
961 # Create the GeoDataFrame
962 uni_c = gpd.GeoDataFrame(records, geometry='geometry', crs=uni.crs)

```

963 Thus, unique addresses for individual buildings were grouped not only by code but also by university campus
964 affiliation. Next, geometric centers (centroids) were calculated for each group. This resulted in a transition from 357
965 points to 54 centroids (Figure 9).



966
967 *Figure 9. Visual Comparison Initial Unique Higher Education Universities Addresses with Final Centroids*

968 The verification method was a manual visual comparison of the final centroids with the original unique addresses
969 in QGIS using the Measure Tool.

970 **3.4.5. Data Export**

971 The processed data for analysing the capacity and demand for student housing is stored in geographic packages
972 (.gpkg) — an open format for storing geospatial data, which differs from the tabular format (.exel) in that it contains
973 coordinates.

```

974 ef = r"../outputs/2.2_urban_data_mining/pbsa.gpkg"
975 pbsa.to_file(ef, driver="GPKG")
976
977 ef = r"../outputs/2.2_urban_data_mining/pbsa_c.gpkg"
978 pbsa_c.to_file(ef, driver="GPKG")
979

```

```
980 ef = r"../outputs/2.2_urban_data_mining/uni.gpkg"
981 uni_copy.to_file(ef, driver="GPKG")
982
983 cf = r"../outputs/2.2_urban_data_mining/uni_c.gpkg"
984 uni_c.to_file(cf, driver="GPKG")
```

985 3.5. 2.3 Descriptive Market Statistics

986 In the 2024/25 winter semester, a total of 49,801 students with non-German citizenship were enrolled. This is more
987 than in the previous winter semester (+1,439). The proportion of international students thus increased from 24.1% to
988 24.8% (Statistik Berlin Brandenburg, 2025). According to data collected in the Urban Data Mining chapter, the total
989 capacity of planned student accommodation is 25,817 students. In other words, only 51.8% of international students
990 can be accommodated in halls of residence if there is no competition from local German students.

991 While describing a dataset is usually the first task in any analysis, in order to rethink the student housing crisis,
992 it is necessary to use descriptive statistics (also called ‘summary statistics’). Descriptive statistics quickly provide an
993 understanding of data variability and allow for the identification of possible patterns (Grekousis, 2020).

994 3.5.1. Set Up

995 Import Libraries

```
996 import os
997 import numpy as np
998 import pandas as pd
999 import matplotlib.pyplot as plt
1000 import seaborn as sns
1001 import matplotlib.path_effects as path_effects
1002 import geopandas as gpd
1003 from shapely.geometry import Point, MultiPoint
1004 from pyproj import CRS
1005 from scipy.stats import norm

1006 Berlin Spatial Boundaries

1007 pf = r"../inputs/2.3_descriptive_market_statistics/lor_pgr.gpkg"
1008 pgr = gpd.read_file(pf)
1009 pgr.replace({' -': np.nan, 0: np.nan, '' : np.nan}, inplace=True)
1010
1011 bf = r"../inputs/2.3_descriptive_market_statistics/adm_bz.gpkg"
1012 bz = gpd.read_file(bf)
1013 bz.replace({' -': np.nan, 0: np.nan, '' : np.nan}, inplace=True)
1014
```

```
1015 of = r"../inputs/2.3_descriptive_market_statistics/adm_out.gpkg"
1016 out = gpd.read_file(of)
1017 out.replace({'-': np.nan, 0: np.nan, '' : np.nan}, inplace=True)
```

1018 **3.6. Student Accommodation Capacity**

1019 The geospatial student accommodation data, obtained as a result of Urban Data Mining, served as a base for
1020 statistical research.

```
1021 ef = r"../inputs/2.3_descriptive_market_statistics/pbsa_c.gpkg"
1022 pbsa_c = gpd.read_file(ef)
```

1023 **3.6.1. Common Statictics**

1024 The numeric fields are the basis for collecting statistics, for which it is possible to calculate the sum (sum), mean
1025 (mean), median (median) and standard deviation (std), minimum (min) and maximum (max). In the case of Stu-
1026 dent Accommodation, such data includes: capacity (!cap) maximum and minimum price (price_min, price_max),
1027 maximum and minimum living space (area_min, area_max).

```
1028 cols_numeric = ['!cap', 'price_min', 'price_max', 'area_min', 'area_max']
```

1029 Summary Statistics (Table 4) shows considerable diversity in student accommodation data, as evidenced by sig-
1030 nificant deviations from the mean value for the main numeric fields (std).

```
1031 summary = pbsa_c[cols_numeric].agg(['mean', 'median', 'std', 'min', 'max'])
1032 summary = summary.T.round(2)
```

1033
1034

```
1035 summary
```

	mean	median	std	min	max
!cap	293.38	200.0	248.18	24.0	1062.0
price_min	447.33	400.0	136.85	256.0	720.0
price_max	615.94	469.5	310.19	352.0	1499.0
area_min	15.11	16.0	4.05	9.0	25.0
area_max	30.79	27.0	17.08	10.0	149.0

1037 *Table 4. Student Accommodation Numeric Fields Summary*

Capacity indicators (!cap) show significant dispersion (Figure 10), where the median value is equal to the standard deviation (mean $\approx$ std), reflecting the polarised nature of housing supply — from small private dormitories, for example, Reichsstrasse Dormitory designed for 61 students, to large public-owned residential complexes, such as WH Siegmunds Hof from Studierendenwerk Berlin with more than 500 places.

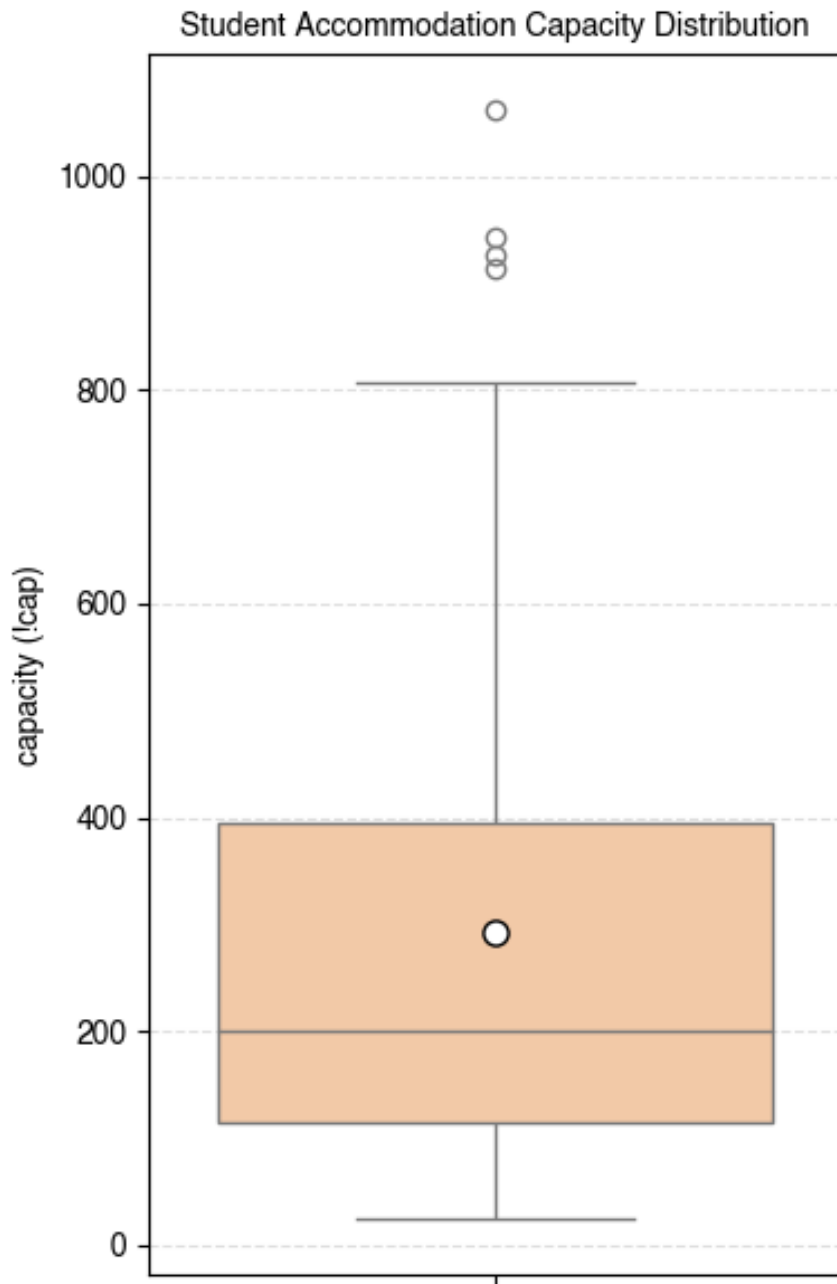
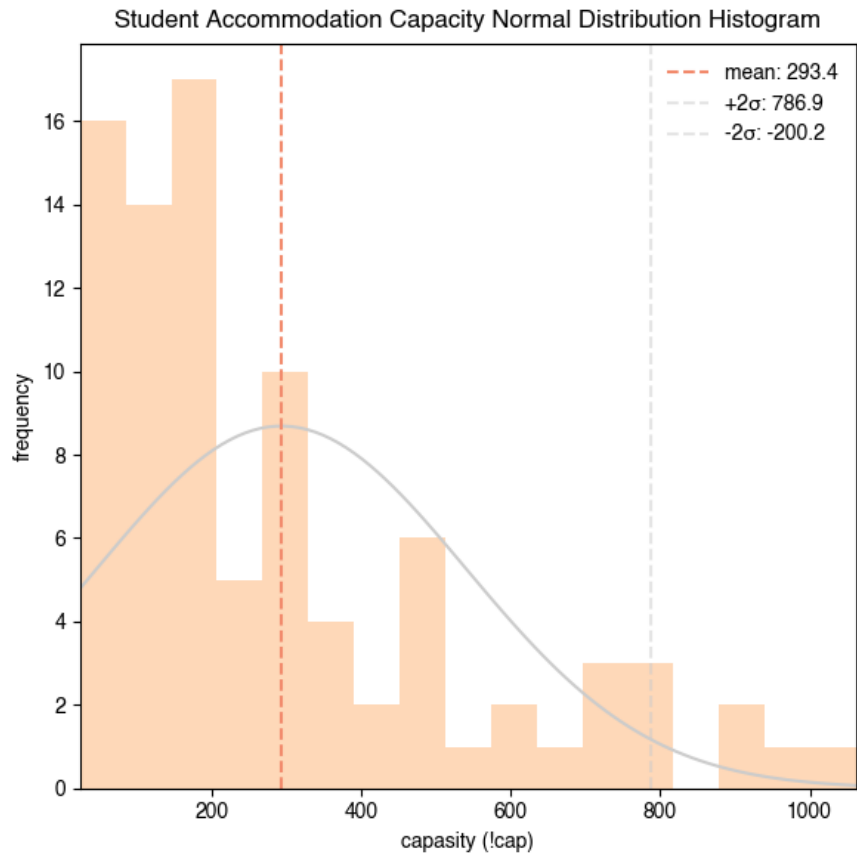


Figure 10. Student Accommodation Capacity Distribution

The Student Accommodation Capacity Normal Distribution Histogram (Figure 11) shows that student accommodation consists of small halls of residence with fewer than 300 students, indicating fragmentation and a small scale of

supply within Berlin’s existing urban structure.

Only a few large complexes exceed 600–800 places, reflecting isolated high-density developments, often located on the periphery or on redeveloped sites.



1049

1050 *Figure 11. Student Accommodation Capacity Normal Distribution Histogram*

1051 Price Distribution (Figure 12) shows a clear disparity between minimum and maximum student accommodation
1052 prices in Berlin, illustrating both baseline affordability constraints and significant upper-end price volatility. Minimum
1053 prices on average are 447 ± 136 , which aligns with Berlin’s Mietspiegel, suggesting a relatively narrow—but still
1054 high—cost floor for most students. In contrast, maximum prices have multiple high-value outliers, reaching above
1055 1,400, indicating the presence of high-cost or highly commodified units within the market.

1056 When discussing student housing prices, it is important to remember that housing is a human right. (Lefebvre,
1057 1968), not an object of trade. The Berlin Constitution clearly state: Everyone has the right to adequate
1058 housing. The state promotes the creation and provision of adequate housing, especially for
1059 people on low incomes, and the cultivation of condominium ownership. Students can be seen as peo-
1060 ple of low income, who can not afford such high-cost options.

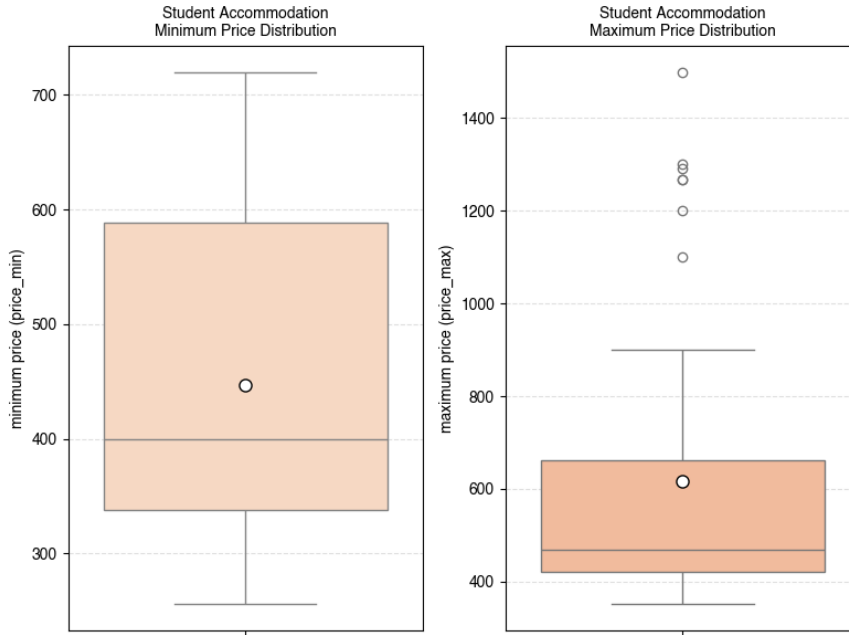


Figure 12. Student Accommodation Price Distribution

Spatial characteristics ($area_{min}$, $area_{max}$) demonstrate additional heterogeneity — living spaces range from compact studios ($<12 \text{ m}^2$, $\sigma=\pm 4.7$) to spacious shared flats ($>40 \text{ m}^2$, $\sigma=\pm 11.2$), illustrating differences in the morphology of student housing (Figure 13).

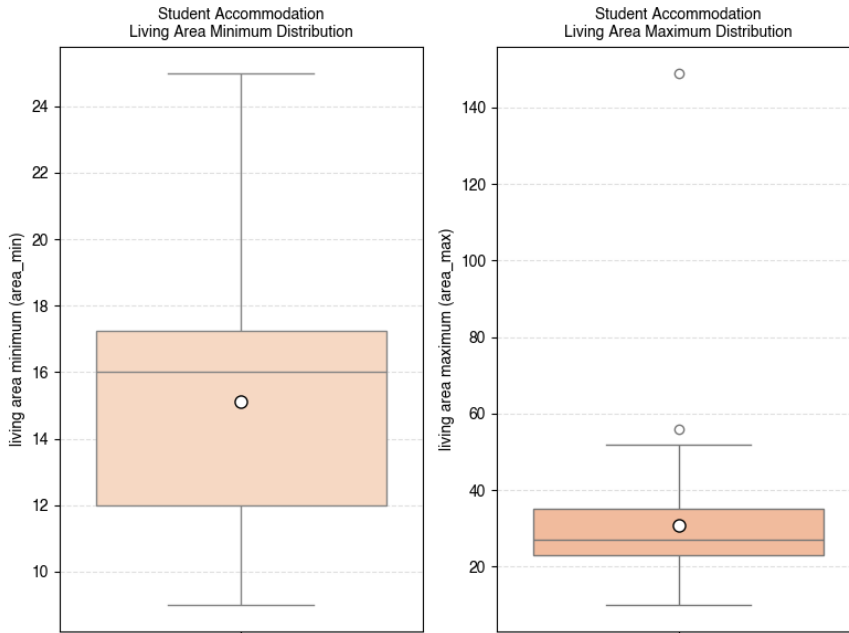


Figure 13. Student Accommodation Living Area Distribution

The significant variability (Table XX) identified at the beginning of the chapter requires additional statistical processing. The heterogeneity essentially reflects the fragmented housing market in Berlin, where supply imbalances lead

1070 to differences in housing typology across different urban areas.

1071 For a more detailed analysis of the housing market and evaluation of numeric data, student accommodations were
1072 divided into groups according to ownership model (own).

1073 **3.6.2. Ownership Model**

1074 Key Student Accommodation Ownership Companies (Table 5) are the state-owned companies Studierendenwerk
1075 Berlin, Berlinovo, Degewo and the private companies Home in Berlin GmbH and Campus Viva GmbH. Together,
1076 these five companies own 68.9% of the student accommodation market.

```
1077 summary_ownership = (pbsa_c.groupby('ownership', dropna=False)[cols_numeric].agg(['sum', 'mean',  
1078  
1079 summary_ownership.sort_values(by=('!cap', 'sum'), ascending=False, inplace=True)  
1080  
1081 summary_ownership.head(5)
```

	!cap				price_min				price_max				area_min				area_max			
	sum	mean	median	std	sum	mean	median	std	sum	mean	median	std	sum	mean	median	std	sum	mean	median	std
ownership																				
Studierendenwerk Berlin	9942.9	1010.6	883.0	285.9	1010.1	1030.5	832.2	204.3	441355.2	407.1	422.0	59.8	394.0	12.7	112.0	3.63	903.0	29.1	328.0	8.03
Berlinovo	3787.0	444.2	232.0	282.1	940.0	37.7	110.0	78.7	0.0	NaN	NaN	NaN	145.0	16.1	116.0	0.60	162.0	20.2	520.0	2.76
Home in Berlin GmbH	890.0	115.0	78.5	15.1	5715.6	19.1	659.5	96.5	6779.0	779.0	779.0	NaN	91.0	15.1	713.0	4.22	102.0	25.5	26.0	5.74
Campus Viva GmbH	108.6	54.0	54.0	359.2	190.6	95.0	695.0	176.7	1499.0	499.0	499.0	NaN	38.0	19.0	19.0	1.41	72.0	36.0	36.0	14.14
Degewo	1065.0	32.5	32.5	557.9	0.0	NaN	NaN	NaN	0.0	NaN	NaN	NaN	39.0	19.5	19.5	3.54	61.0	30.5	30.5	7.78

1083 *Table 5. Key Student Accommodation Ownership Companies*

1084 When dividing student accommodation into state and privately owned (Table 6), the gap between the minimum
1085 and maximum prices (price_min and price_max) for monthly rent for different ownership models become evident.
1086 Table XX analyses the numerical fields for student accommodation according to different ownership types.

1087 The capacity of state and private accommodation differs slightly. Nevertheless, prices for state-owned accommo-
1088 dation vary from 368 to 439, while privately owned accommodation costs significantly more: from 590 to 968.

1089 At the same time, the minimum income for students set by the Federal Foreign Office in 2025 for foreign students
1090 to obtain a visa is 992 per month. This makes private accommodation unaffordable for most international students.

1091 Reducing the price gap between state and private accommodation through price regulation could be part of the
1092 solution to the student housing crisis.

```
1093 summary_ownership_model = (  
1094     pbsa_c  
1095     .groupby('ownership_model', dropna=False)[cols_numeric]  
1096     .agg(['mean', 'median', 'std']).round(2)  
1097     .sort_index()  
1098 )  
1099  
1100 summary_ownership_model
```

	!cap			price_min			price_max			area_min			area_max			
	mean	median	std	mean	median	std	mean	median	std	mean	median	std	mean	median	std	
	ownership_model															
1101	Privat-	295.50	245.0	205.93	590.52	600.0	96.00	968.94	887.5	307.59	16.00	17.0	3.18	29.60	27.0	8.46
	Owned															
	State-	292.28	162.5	269.13	368.43	352.0	79.89	439.44	422.0	60.25	14.72	15.0	4.36	31.29	27.0	19.65
	Owned															

1102 Table 6. Student Accommodation Ownership Model Numeric Fields Summary

1103 Despite their inaccessibility in terms of cost, private student accommodation accounts for 34.3% of the market
1104 supply (Figure 14). Private accommodation is not uncommon – it houses about one-fifth of all international students.
1105 This suggests that the private market should be regulated systematically, rather than on a case-by-case basis.

Student Accommodation Capacity Distribution by Ownership Model

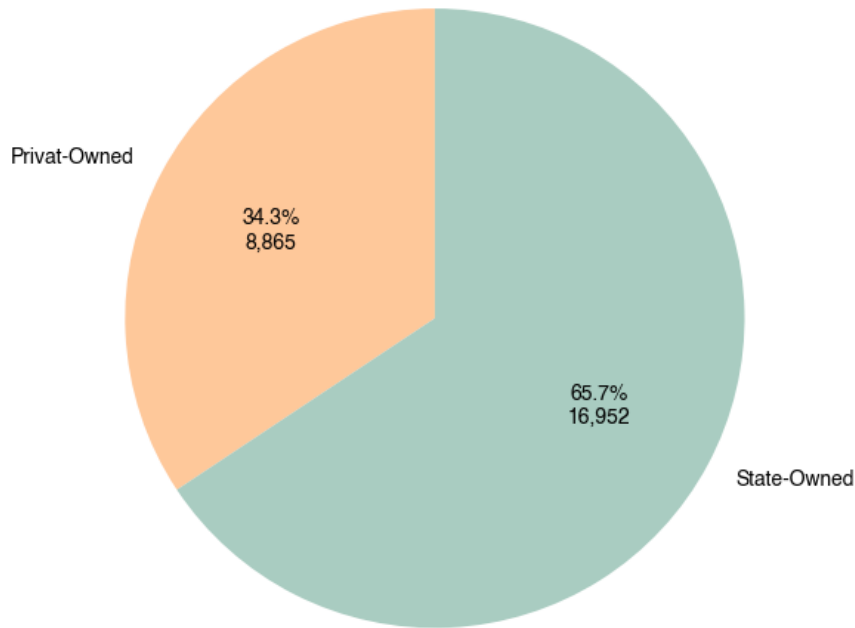


Figure 14. Student Accommodation Capacity Distribution by Ownership Model

The map (Figure 15) illustrates the spatial distribution of student housing in Berlin by ownership model, revealing clear structural patterns that contribute to the ongoing student housing crisis in the city.

Public residences are more evenly distributed across the periphery, reflecting their role as affordable, welfare-oriented infrastructure, but their limited density indicates insufficient public capacity relative to demand.

Private facilities are more concentrated in central areas and high-demand areas, particularly in Mitte, Friedrichshain-Kreuzberg and Charlottenburg-Wilmersdorf, reflecting market logic that prioritises profitable locations with high rents over equal access.

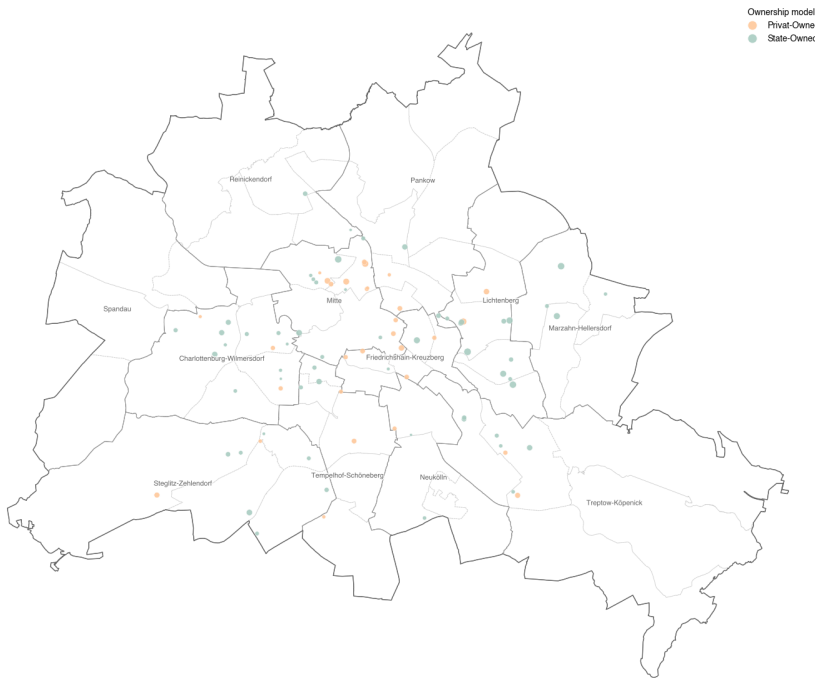


Figure 15. Student Accommodation Spatial Distribution by Ownership Model

Two histograms (Figure 16) show that public and private student residences in Berlin have almost identical average capacities (≈ 290 – 300 places), but their distribution differs in aspects that are relevant to the housing crisis.

Public facilities have a wider range, including several large complexes with more than 900 places (e.g. WH Coppistraße, Iranische Straße, Ludwig-Renn-Straße Hellersdorf, WH Aristotelessteig), reflecting their role in providing significant long-term public capacity.

In private student accommodation, by contrast, small and medium-sized facilities predominate, with fewer large complexes, indicating a market that tends to prioritise scattered, small investments over large-scale projects.

The public sector is at least attempting to implement large projects, while private supply remains fragmented.

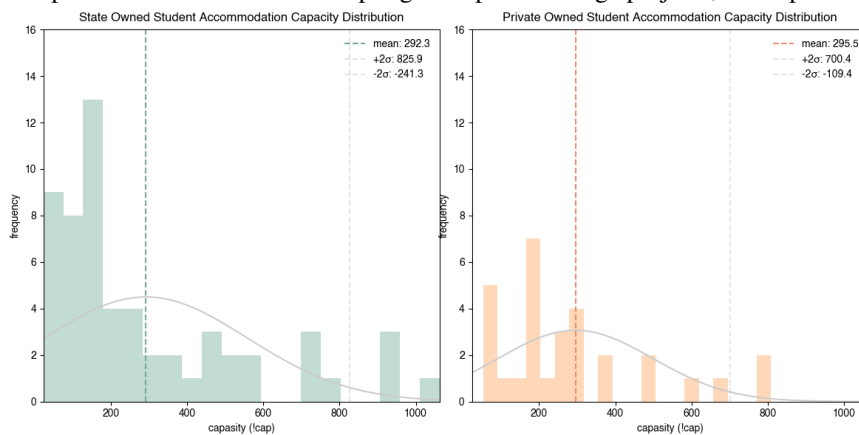


Figure 16. State and Private Owned Student Accommodation Normal Capacity Distribution Comparison

This division highlights a fundamental problem: public student housing is too infrequent and located on the outskirts to offset affordability pressures, while private housing is concentrated in the most expensive areas. As a result, students face geographical and financial mismatches, reinforcing the need for data-driven planning strategies that re-balance supply, expand capacity, and redirect development to underserved areas.

Full descriptive statistics on student accommodation are available for review in the Materials & Data chapter.

```
output_path = r"../outputs/2.3_descriptive_market_statistics/student_acomodation_statisticcs.xlsx"
```

```
# Write DataFrames to separate sheets in one Excel file
```

```
with pd.ExcelWriter(output_path, engine='openpyxl') as writer:
```

```
    summary.to_excel(writer, sheet_name='common_statistics')
```

```
    summary_ownership.to_excel(writer, sheet_name='by_ownership')
```

```
    summary_ownership_model.to_excel(writer, sheet_name='by_ownership_model')
```

```
    summary_unit_type.to_excel(writer, sheet_name='by_unit_type')
```

3.7. Higher Education Universities Students Demand

The higher education universities' geospatial data, obtained as a result of Urban Data Mining, served as a base for statistical research.

```
cf = r"../inputs/2.3_descriptive_market_statistics/uni_c.gpkg"
```

```
uni_c = gpd.read_file(cf)
```

3.7.1. Common Statictics

In the case of Higher Education Universities, the numerical data are the number of German students (!s_de), the number of foreign students (!s_aus) and the total number of students (!s_all).

```
cols_numeric = ['!s_all', '!s_de', '!s_aus']
```

Higher Education Universities Data (Table 7) shows a sharp difference in student numbers between universities in Berlin. The average university enrolment is 3,720 students, but a typical university (median 842 places) is 4.4 times smaller. This means that several large universities take on the main part of the load (such as Free University, Humboldt University and Technical University of Berlin).

Foreign students make up 24.1% of the total, meaning that one in five students in Berlin is a foreigner. Comparing the standard deviations for German and foreign students, one can see segmentation based on nationality, which

1155 manifests itself in a 3.3-times higher dispersion of German students than foreign students, revealing a cluster model
1156 of foreign student placement in large state universities. It is noteworthy that Free University (8,280 foreign students),
1157 Humboldt University (6,415 foreign students) and Technical University of Berlin (10,262 foreign students) account for
1158 24,957 foreign students, or 51.5% of total international student demand.

```
1159 summary = uni_c[cols_numeric].agg(['mean', 'median', 'std', 'sum'])  
1160 summary = summary.T.round(2)  
1161  
1162 summary
```

	mean	median	std	sum
1163 !s_all	3720.15	842.0	8498.32	200888.0
!s_de	2822.80	489.0	6604.27	152431.0
!s_aus	897.35	172.5	1999.27	48457.0

1164 *Table 7. Higher Education Universities Numeric Fields Summary*

1165 Free University, Humboldt University and Technical University Berlin are exceptions to the rule (or so-called
1166 ‘statistical outliers’). Most universities have up to 600 students. Thus, Higher Education Universities Students Demand
1167 Distribution (Figure 17) is far from normal.

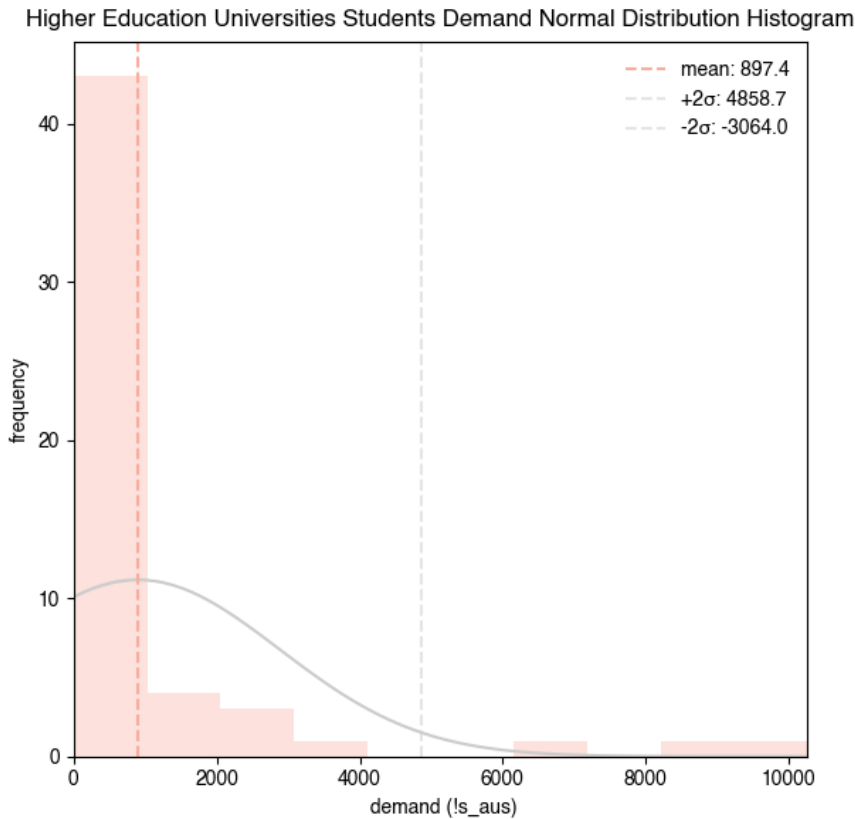


Figure 17. Higher Education Universities Students Demand Normal Distribution Histogram

In order to gain a deeper understanding of the imbalance in supply, by analogy with student accommodations, higher education universities can be divided into two groups based on their ownership model (own): Public and Private Universities.

3.7.2. Ownership Model

Higher Education Universities Ownership Model Numeric Fields Summary (Table 8) shows a dramatic structural difference between public and private higher education institutions in Berlin. Public universities dominate the academic landscape, accepting an average of more than 9,000 students per institution, while private universities accept fewer than 1,000 students. This gap persists among both German and foreign students: public universities enrol the majority of German students (an average of $\approx 7,100$) and attract significantly more foreign students (an average of $\approx 2,000$) than private institutions.

The much higher overall figures and variability in the public sector highlight the concentrated demand pressure on public universities, which are the main drivers of the student housing market.

```
summary_ownership_model = (
    uni_c
```

```
1184     .groupby('own', dropna=False)[cols_numeric]
1185     .agg(['mean', 'median', 'std', 'sum']).round(2)
1186     .sort_index()
1187 )
1188
1189 summary_ownership_model
```

	!s_all				!s_de				!s_aus			
	mean	median	std	sum	mean	median	std	sum	mean	median	std	sum
own												
Privat	804.89	597.0	727.67	28171.0	498.49	267.0	584.14	17447.0	306.40	102.0	493.02	10724.0
Uni-												
ver-												
sity												
Public	9090.37	4226.0	12832.96	172717.07	104.42	2805.0	9895.76	134984.01	985.95	400.0	3062.69	37733.0
Uni-												
ver-												
sity												

1191 *Table 8. Higher Education Universities Ownership Model Numeric Fields Summary*

1192 Private universities contribute less significantly to the total number of students (Figure 18). Nevertheless, 22.1%

1193 of foreign students choose private universities for their education. Of all students at private universities, 38% are

1194 foreigners (vs. 22% at state universities).

Higher Education Universities Demand Distribution by Ownership Model

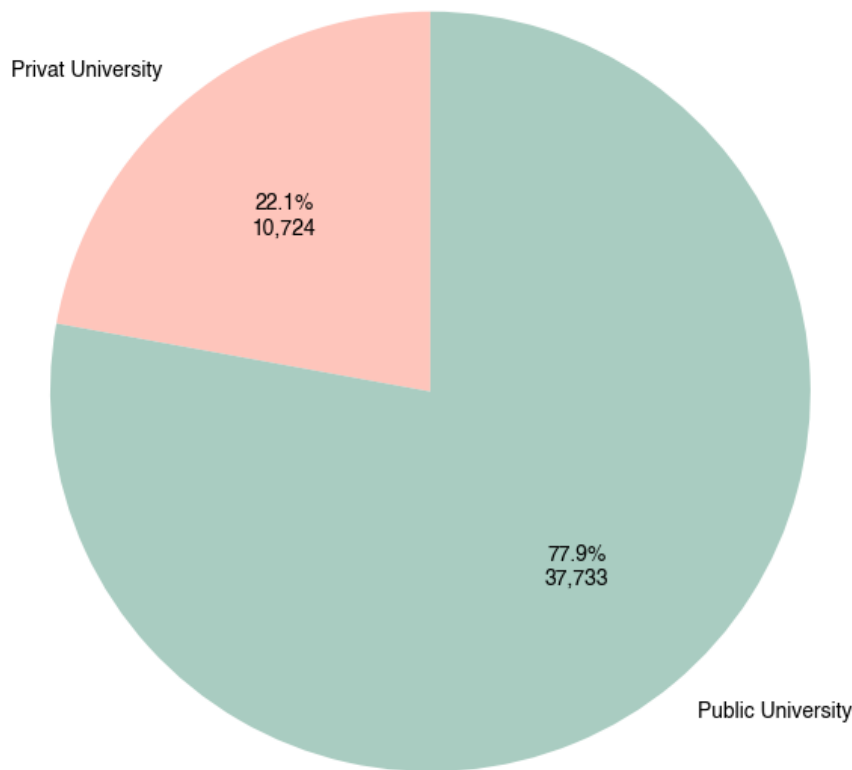


Figure 18. Higher Education Universities Demand Distribution by Ownership Model

The spatial distribution of higher education institutions in Berlin (Figure 19) shows a strong central concentration of private universities, in contrast to the more dispersed location of public universities throughout the city.

Private universities are mainly concentrated in Mitte, Charlottenburg-Wilmersdorf and Friedrichshain-Kreuzberg – areas that are already characterised by high demand for housing – while public universities occupy both central areas and more peripheral areas such as Lichtenberg, Steglitz-Zehlendorf and Treptow-Köpenick. This uneven geography increases spatial inequality in the demand for student housing: large public universities create high pressure on the housing market across the city, while private universities add local spikes in demand in already highly densely populated central areas.

For large public universities, state-owned dormitories remain a priority, while the private sector requires integration into the urban housing ecosystem and lower prices as a market driver that is developing faster than the public sector.

The map highlights a broader structural problem: the spatial logic of Berlin's university system does not correspond to the distribution of student housing, exacerbating the city's ongoing student housing crisis.

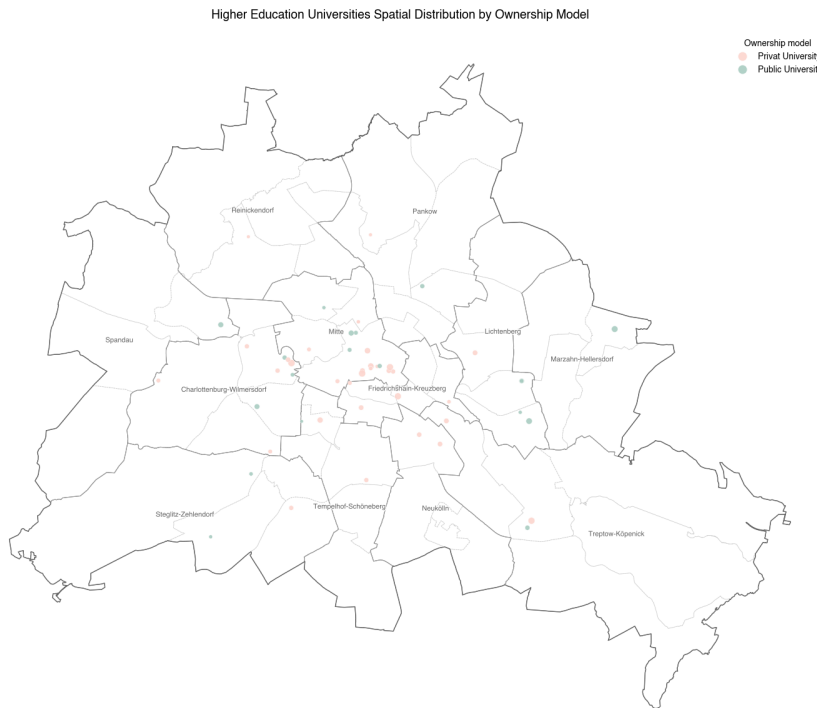


Figure 19. Higher Education Universities Spatial Distribution by Ownership Model

Two histograms (Figure 20) show that public and private universities in Berlin have different demand indicators: 1,986 for public universities and 306 for private universities.

Public universities have a wider range, including several large complexes with more than 8,000 students. (e.g., Technical University of Berlin and Free University of Berlin), indicating a high concentration of international demand in a limited number of campuses.

In private halls of residence, on the other hand, the distribution is more compact, but still spread out. The vast majority of private universities have low or medium demand, while large complexes have up to 2,500 international students.

The contrast between the two distributions confirms that the housing demand of international students in Berlin is unevenly distributed, creating local areas of market overload.

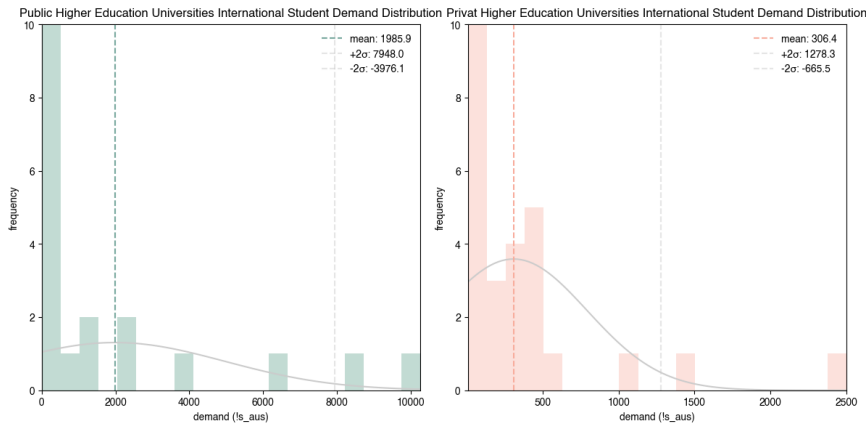


Figure 20. Public and Private Owned Higher Education Universities International Student Demand Distribution Comparison

Full descriptive statistics on student accommodation are available for review in the Materials and Data Chapter.

```
output_path = r"../outputs/2.3_descriptive_market_statistics/higher_education_universities.xlsx"
```

```
# Write DataFrames to separate sheets in one Excel file
```

```
with pd.ExcelWriter(output_path, engine='openpyxl') as writer:
```

```
    summary.to_excel(writer, sheet_name='common_statistics')
```

```
    summary_ownership_model.to_excel(writer, sheet_name='by_ownership_model')
```

```
    summary_uni_type.to_excel(writer, sheet_name='by_uni_type')
```

3.8. 2.4 Global Spatial Autocorrelation

In the Descriptive Market Statistics Chapter, a descriptive analysis of two datasets was conducted: student accommodations and higher education universities. However, in order to access spatial distribution across cities, it is necessary to measure spatial dependency.

Spatial autocorrelation measures spatial dependency. In other words, how much the value of a variable in a specific location is related to the values of the same variable at neighboring locations. Spatial autocorrelation (either positive or negative) is a key concept in geographic analysis. Spatial autocorrelation can be explained by the first law of geography. According to the law, objects in a neighborhood tend to have more similarities and interactions than those lying further away (Grekousis, 2020).

Set Up Import Libraries

```
import os

import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

import matplotlib as mpl

import seaborn as sns

from matplotlib.colors import LinearSegmentedColormap

import matplotlib.patches as mpatches

import matplotlib.path_effects as path_effects

import geopandas as gpd

from shapely.geometry import Point, MultiPoint

from pyproj import CRS

from scipy.stats import norm

import libpysal

from esda import Moran

from libpysal.weights.spatial_lag import lag_spatial

from libpysal.weights import KNN

from libpysal.weights import weights

Berlin Spatial Boundaries

pf = r"../inputs/2.4_global_spatial_autocorrelation/lor_pgr.gpkg"
```

```

1262 pgr = gpd.read_file(pf)
1263 pgr.replace({' -': np.nan, 0: np.nan, '' : np.nan}, inplace=True)
1264
1265 bf = r"../inputs/2.4_global_spatial_autocorrelation/adm_bz.gpkg"
1266 bz = gpd.read_file(bf)
1267 bz.replace({' -': np.nan, 0: np.nan, '' : np.nan}, inplace=True)
1268
1269 of = r"../inputs/2.4_global_spatial_autocorrelation/adm_out.gpkg"
1270 out = gpd.read_file(of)
1271 out.replace({' -': np.nan, 0: np.nan, '' : np.nan}, inplace=True)
1272
1273 mf = r"../inputs/2.4_global_spatial_autocorrelation/berlin_mauer.gpkg"
1274 mauer = gpd.read_file(mf)
1275 mauer.replace({' -': np.nan, 0: np.nan, '' : np.nan}, inplace=True)

```

1276 **3.9. Student Accommodation Capacity**

1277 **3.9.1. Data Import**

1278 The data obtained as a result of Urban Data Mining was used as input data for the analysis (since no changes were
 1279 made to the data during Descriptive Market Statistics).

```

1280 pf = r"../inputs/2.4_global_spatial_autocorrelation/pbsa_c.gpkg"
1281 pbsa_c = gpd.read_file(pf)

```

1282 Before calculating global spatial autocorrelation, it is also necessary to perform tests: Moran's I Scatterplots Test
 1283 and K-Nearest Neighbor Test, as well as calculate the Significance Threshold test.

1284 Since the research focused on market parameters, global autocorrelation will focus on capacity, which measures
 1285 the market supply.

1286 **3.9.2. Standardisation**

1287 Data standardisation is used to assess the correlation of neighbouring values relative to the overall average and to
 1288 eliminate the outliers identified in the previous chapter. Standardisation converts data into dimensionless values, which
 1289 allows for accurate comparison of clustering for different student accommodations. Positive standardised capacity
 1290 values (!cap_std) indicate above-average objects, while negative values indicate below-average objects. This will
 1291 help to clearly identify student accommodation clusters in the future.

1292 pbsa_c["!cap_std"] = pbsa_c["!cap"] - pbsa_c["!cap"].mean()

1293 3.9.3. *Moran's I Scatterplots Test & K-Nearest Neighbor Test*

1294 Several diagnostic measures can be used to identify spatial autocorrelation. Those that estimate spatial autocorrela-
1295 tion by a single value for the entire study area are named global spatial autocorrelation measures. The most commonly
1296 used is Moran's I (Grekousis, 2020).

1297 **Moran's I** computes global spatial autocorrelation by taking into account
1298 feature locations and attribute values (of a single attribute) simultaneously. It is calculated by the following formula:

$$I = \frac{n}{\sum_i \sum_j w_{ij}} \cdot \frac{\sum_i \sum_j w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\sum_i (x_i - \bar{x})^2} \quad (1)$$

1299 where:

1300 n — number of spatial features

1301 x — attribute value of feature i (a variable in a spatial analysis context)

1302 x — attribute value of feature j

1303 $\bar{x}$ — mean of this attribute

1304 w — spatial weight between feature i and j

1305 w — aggregation of all spatial weights

1306 In order to calculate spatial weights between features, K-nearest neighbor weights are used. K-nearest neighbor
1307 weights are distance-based weights defines the neighbor set of a particular observation as containing its nearest obser-
1308 vations, where the user specifies the number of nearest neighbors (k). If we do not test k, Moran's I Index might have
1309 values beyond the [-1,1] boundaries (Grekousis, 2020).

1310 To understand the intuition behind its math, it is useful to begin with a graphical interpretation: the Moran Plot
1311 for different numbers of neighbors (Figure XX). The Moran Plot is a way of visualizing a spatial dataset to explore the
1312 nature and strength of spatial autocorrelation. It is essentially a traditional scatter plot in which the variable of interest
1313 is displayed against its spatial lag (Rey et al., 2023).

1314 The Moran's I scatterplots (Figure XX) across increasing k-values (`k_values = list(range(1, 21))`) reveal
1315 that student accommodation capacity in Berlin exhibits a weak but consistently positive spatial autocorrelation, which
1316 strengthens as the neighborhood definition expands. Each point represents a student accommodation and its relation-
1317 ship to the capacities of its surrounding neighbors. The four quadrants provide insight into the spatial logic of student
1318 housing distribution:

1319 **Q1 – High–High (HH)**

1320 These locations have high student accommodation capacity and are surrounded by other areas with similarly high
1321 capacities. These clusters signal where the market has already developed a strong supply.

1322 **Q2 – Low–High (LH)**

1323 These student accommodations have low capacity themselves, but are neighboring high-capacity ones. They often
1324 represent transition zones—places that have the potential to absorb future student housing development because they
1325 border well-served districts. These locations highlight opportunity areas where relatively small additions could signif-
1326 icantly improve spatial balance.

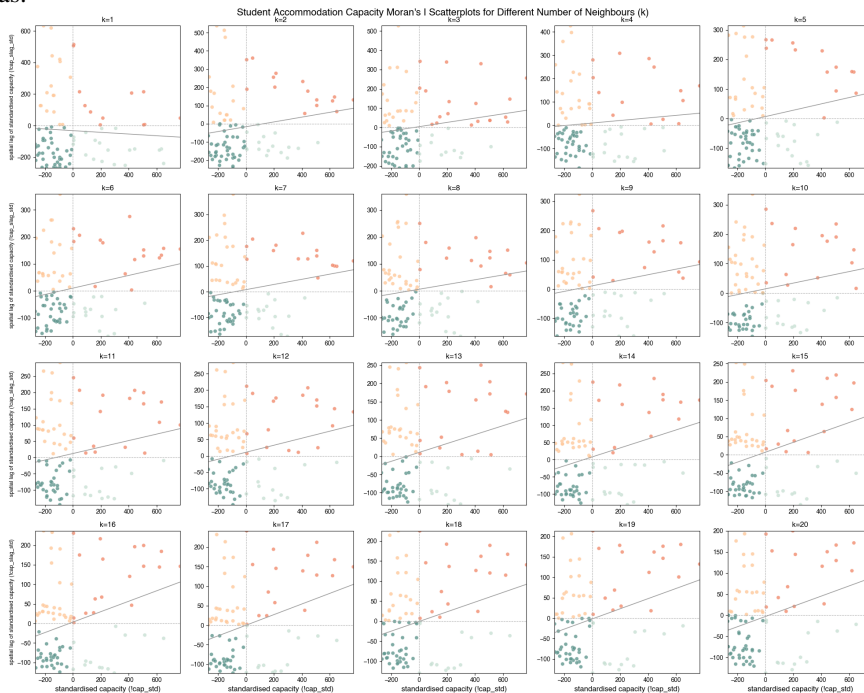
1327 **Q3 – Low–Low (LL)**

1328 These locations exhibit low capacity and are surrounded by other low-capacity areas, marking systemic undersupply
1329 zones. For policy-making, these locations indicate structurally disadvantaged parts of the city, where students face
1330 limited options.

1331 **Q4 – High–Low (HL)**

1332 These student accommodations have high capacity but are surrounded by low-capacity neighbors. Such areas represent
1333 isolated clusters—individual dorms or private student residences located in otherwise undersupplied districts. They
1334 highlight mismatches where supply exists but lacks integration into broader student housing networks, potentially
1335 limiting accessibility.

1336 In conclusion, the Moran's I scatterplots illustrate that Berlin's student housing supply is spatially uneven, featuring
1337 concentrated high-capacity cores, isolated developments, transitional opportunity zones, and persistent undersupply
1338 areas.



1339

Figure 21. Student Accommodation Capacity Moran's I Scatterplots for Different Number of Neighbors (k)

The difference between the scatterplots (Figure 21) must be evaluated based on Moran's I index and P-value. Through these metrics, we assess whether spatial autocorrelation is statistically significant. To do this, we need to calculate the parameters for each value of the number of nearest neighbors (k) (Table 9).

Moran's I index and P-value can be interpreted as follows. A Moran's I index score higher than 0.3 is an indication of relatively strong positive autocorrelation, while a score lower than -0.3 is an indication of relatively strong negative autocorrelation. If the p-value is large (usually $p > 0.05$), the results are not statistically significant, and we cannot reject spatial dependency (Grekousis, 2020).

```
np.random.seed(55)

k_values = [ 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20]

morans_I = []

p_values = []

for k in k_values:

    w = libpysal.weights.KNN.from_dataframe(pbsa_c, k=k)

    w.transform = "R"

    moran = Moran(pbsa_c['!cap'], w)

    morans_I.append(moran.I)

    p_values.append(moran.p_sim)

# Create DataFrame
pbsa_k_test = pd.DataFrame({

    'k': k_values,

    "Moran's I": morans_I,

    'P -value': p_values

})

pbsa_k_test
```

	k	Moran's I	P-value
0	2	0.131920	0.076
1	3	0.110553	0.057
2	4	0.054687	0.163
3	5	0.107291	0.030
4	6	0.117724	0.017
5	7	0.100457	0.018
6	8	0.088330	0.039
7	9	0.095008	0.016
8	10	0.097889	0.012
9	11	0.099631	0.013
10	12	0.106905	0.008
11	13	0.122254	0.005
12	14	0.130466	0.002
13	15	0.135846	0.003
14	16	0.133675	0.001
15	17	0.136209	0.001
16	18	0.120816	0.002
17	19	0.124204	0.001
18	20	0.121062	0.002

Table 9. Student Accommodation Capacity Moran's I index and P-value for Different Number of Neighbors (k)

The task of finding the desired value of the number of nearest neighbors (k) can be solved by plotting a graph of the dependence of Moran's I index and P-value on k (Figure 22). On this graph, it is necessary to find the maximum values of the Moran's I index modulus and then compare the P-value.

The maximum values of Moran's I index are achieved at k from 15 to 17. Since there are only 88 student dormitories, the minimum value of 15 neighbors (k = 15) was chosen. That is, each student dormitory is compared with its 15 closest neighbors in order to identify spatial dependence.

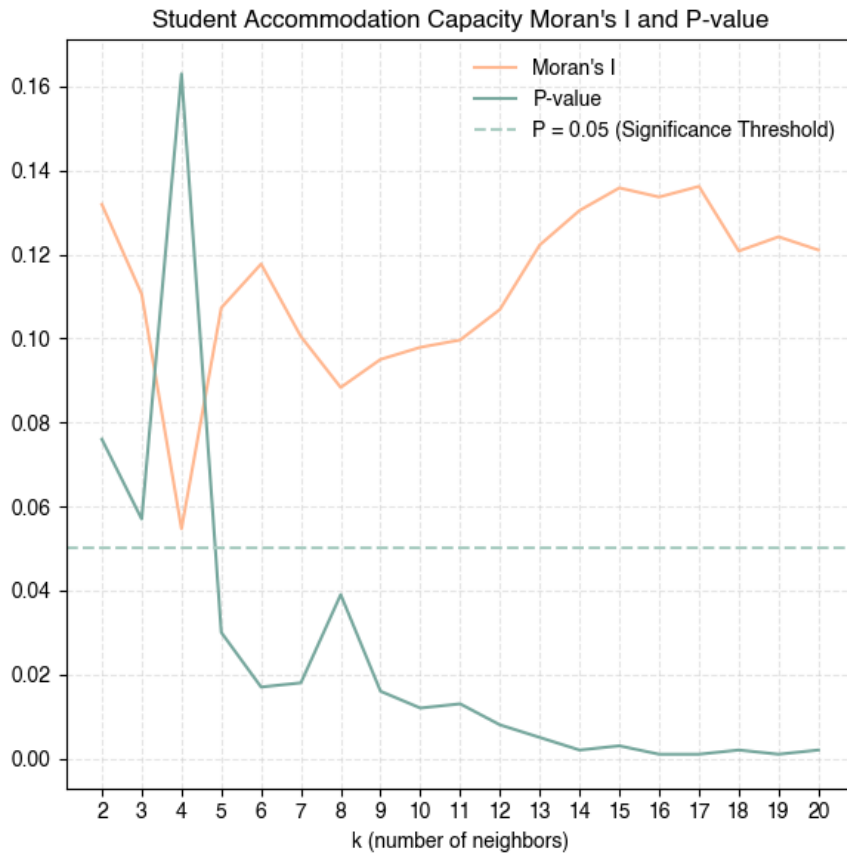


Figure 22. Student Accommodation Capacity Moran's I index and P-value for Different Number of Neighbors (k)

3.9.4. Significance Threshold Test

In the original Moran's I Scatterplots for Different Number of Neighbors (Figure XX), the points are distributed across four quadrants: HH, HL, LH, LL. However, points near zero may accidentally fall into false categories, even though they do not demonstrate real spatial dependence. This can lead to distorted results and the identification of false points of spatial autocorrelation.

The Significance Threshold Test sets a significance threshold (`threshold`) for standardized capacity (`!cap_std`) and spatial lag standardized capacity (`!cap_slag_std`), and filters out not significant points where both variables take neutral values.

Option 1 - Percentile -based threshold: 234.375

(75th percentile of absolute values)

Option 2 - Standard deviation -based threshold: 124.090

($0.5 * \text{standard deviation}$)

Option 3 - Fixed threshold: 50.000

1392 (Fixed value for data range)
1393 Option 4 - Median -based threshold: 78.750
1394 (0.5 * median of absolute values)

1395

1396 Test 1: threshold = 234.375

1397 NS: 66 (75.0%)

1398 HH: 8 (9.1%)

1399 LH: 6 (6.8%)

1400 LL: 6 (6.8%)

1401 HL: 2 (2.3%)

1402

1403 Test 2: threshold = 124.090

1404 NS: 25 (28.4%)

1405 HH: 15 (17.0%)

1406 LH: 10 (11.4%)

1407 LL: 26 (29.5%)

1408 HL: 12 (13.6%)

1409

1410 Test 3: threshold = 50.000

1411 NS: 6 (6.8%)

1412 HH: 16 (18.2%)

1413 LH: 12 (13.6%)

1414 LL: 35 (39.8%)

1415 HL: 19 (21.6%)

1416

1417 Test 4: threshold = 78.750

1418 NS: 11 (12.5%)

1419 HH: 16 (18.2%)

1420 LH: 12 (13.6%)

1421 LL: 35 (39.8%)

1422 HL: 14 (15.9%)

1423 In order to correctly interpret the Significance Threshold Test, it is necessary to pay attention to the distribution of

1424 points across four squares - spatial clusters.

1425 **Option 1 - Percentile-based threshold: 234.375**

1426 With a high threshold value, 75% of the points became neutral, and the global autocorrelation analysis lost its meaning.

1427 **Option 2 - Standard deviation-based threshold: 124.090**

1428 It will eliminate the outliers that were described in the Descriptive Market Statistics chapter, as it is calculated according
1429 to the standard deviation.

1430 **Option 3 - Fixed threshold: 50.000**

1431 A threshold value that is too low results in the percentage of neutral points being too low at 6.8%.

1432 **Option 4 - Median-based threshold: 78.750**

1433 Median-based threshold does not react to abnormally large values. Therefore, this threshold creates a threshold that
1434 reflects the real “central” distribution structure (`threshold = threshold_median`).

1435 Using three tests: Moran’s I Scatterplots Test, K-Nearest Neighbour Test, and Significance Threshold Test, it can
1436 be confirmed that the results of local spatial autocorrelation are stable.

1437 *3.9.5. Spatial Weights & Moran’s I & P-value*

1438 After performing all necessary tests, it is possible to proceed to the direct calculation of Spatial Weights, Moran’s
1439 I, and P-value for the selected value of numbers of nearest neighbors ($k=15$). To do this, the method of spatial weights
1440 standardization must be defined. Row-standardization (`w.transform = ‘R’`) makes Moran’s I and other spatial in-
1441 dices comparable between different areas, even if the number of neighbors for objects is different. For each row of the
1442 weight matrix (each object), the sum of all neighbor weights is normalized to 1.

$$w_{ij}^{(R)} = \frac{w_{ij}}{\sum_j w_{ij}} \quad (2)$$

1443 `w = KNN.from_dataframe(pbsa_c, k=15)`

1444

1445 `w.transform = "R"`

1446

1447 `values = pbsa_c[‘!cap’]`

1448

1449 `moran = Moran(values, w)`

1450 `print(f"Moran’s I: {moran.I:.3f}")`

1451 `print(f"P -value: {moran.p_sim:.3f}")`

1452 Moran's I: 0.136

1453 P-value: 0.001

1454 Moran's I (Moran's I: 0.136) indicates positive spatial autocorrelation. In the context of urban planning, this
1455 indicates that student accommodations have a weak but stable tendency to form clusters.

1456 P-value (P-value: 0.001) indicates that the probability of obtaining such a result by chance is 0.1%. In the
1457 context of urban planning, this shows that the observed spatial clusters actually exist and are not random.

1458 3.9.6. Spatial Lag & Standardised Spatial Lag

1459 After calculating spatial weights, it is necessary to compute the spatial lag. The spatial lag operator is one of the
1460 most common and direct applications of spatial weights matrices (called **W** formally) in spatial analysis. Conceptually,
1461 the spatial lag captures the behavior of a variable in the immediate surroundings of each location; in that respect, it is
1462 akin to a local smoother of a variable (Rey et al., 2023).

1463 We can formally express it in matrix notation as:

$$Y_{sl} = \mathbf{W}Y \quad (3)$$

1464 or, in individual notation as:

$$y_{sl-i} = \sum_j w_{ij} y_j \quad (4)$$

1465 where:

1466 w_{ij} is the cell in **W** on the i -th row and j -th column, thus capturing the spatial relationship between observations
1467 i and j , in other words y_{sl-i} is the weighted average values in a specified neighborhood.

1468 y_{sl-i} thus captures the product of the values and weights of each observation other than i in the dataset.

1469 Spatial lag is a key element of many spatial analysis techniques including spatial autocorrelation (Rey et al., 2023).

1470 `pbsa_c["!cap_slag"] = lag_spatial(w, pbsa_c["!cap"])`

1471 `pbsa_c["!cap_std"] = pbsa_c["!cap"] - pbsa_c["!cap"].mean()`

1472 `pbsa_c["!cap_slag_std"] = lag_spatial(w, pbsa_c["!cap_std"])`

1473 The maps (Figure 23) illustrate the various stages of data preparation for global autocorrelation — from ac-
1474 tual capacity values (!cap) and their spatial relationships (!cap_slag) to their standardized values (!cap_std and

1475 !cap_slag_std).

1476 **a. Student Housing Capacity (!cap)**

1477 The first map shows the absolute capacity of student housing. Despite the presence of large facilities in central areas,
1478 the overall structure remains fragmented, with significant heterogeneity between districts. This map reflects the actual
1479 market and serves as a starting point for analysis.

1480 **b. Student Housing Capacity Influence (!cap_slag)**

1481 The second map displays the spatial lag of capacity — the average capacity in neighboring locations. It captures the
1482 influence of neighboring facilities, smoothing out local peaks and revealing latent clusters that are not visible in the
1483 absolute data. This shows not only where large dormitories are located, but also where their surroundings are saturated
1484 or deficient.

1485 **c. Standardized Capacity (!cap_std)**

1486 The third map shows standardised values (centered around the mean). Here you can see which areas have above- or
1487 below-average capacity for the city, rather than simply “large/small.” This is a key step for Moran’s I, as it allows you
1488 to identify deviations from the overall trend rather than absolute values.

1489 **d. Standardized Capacity Influence (!cap_slag_std)**

1490 The fourth map displays standardized spatial lag—how much neighboring areas are above or below average. In combi-
1491 nation with map (c), it shows where High–High (HH) and Low–Low (LL) clusters form and where anomalies (High–
1492 Low / Low–High) occur. These patterns form the basis for interpreting the Moran Scatterplot.

1493 Together, maps (a–d) demonstrate that the distribution of student housing in Berlin is highly heterogeneous and
1494 structurally dependent on the spatial environment. The transition from absolute values (!cap) to standardized spatial
1495 lags (!cap_slag_std) allows us to identify hidden clusters and areas of shortage — key information for rethinking
1496 student housing placement strategies and developing spatially sensitive solutions.

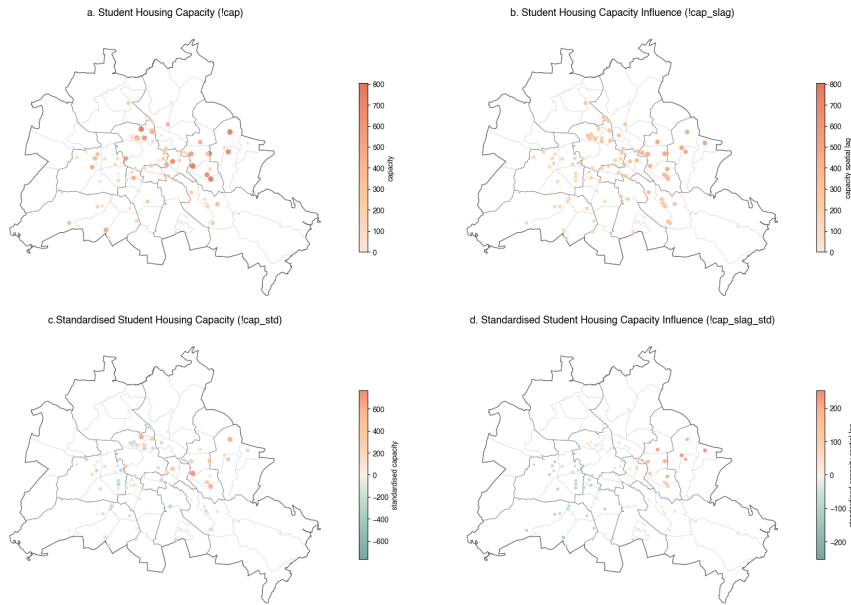


Figure 23. Stages of Preparing Student Accommodation Capacity Data

Moran's graph (Figure 24) shows the spatial relationship between the standardised capacity of student housing (!cap_std) and its spatial lag (!cap_slag_std) — that is, the average capacity of neighbouring locations. Dividing the graph into four quadrants allows us to identify types of spatial patterns:

Q1 – High–High (HH)

These are high-capacity clusters where several large student residences are concentrated close to each other. For example, Studentenwohnheim Alfred-Jung-Straße, with a capacity of 800, is located less than 200 metres from another large student accommodation facility, WH Storkower Straße, with a capacity of 508 students, forming a stable centre for foreign students. At the same time, 12 of the 17 student facilities in the HH quadrant are state-owned. This leads to the conclusion that these centres of attraction are formed by state-owned facilities.

Q2 – Low–High (LH) These are local pockets near developed clusters where a particular facility has low capacity compared to its saturated surroundings. For example, four facilities located in Wed: Norderoog, Studentenwohnheim Paul-Hertz-Haus, Süderoog, WH Augustenburger Platz with a total capacity of 471 students. Such areas are particularly important for analysis, as they may represent missed opportunities for growth.

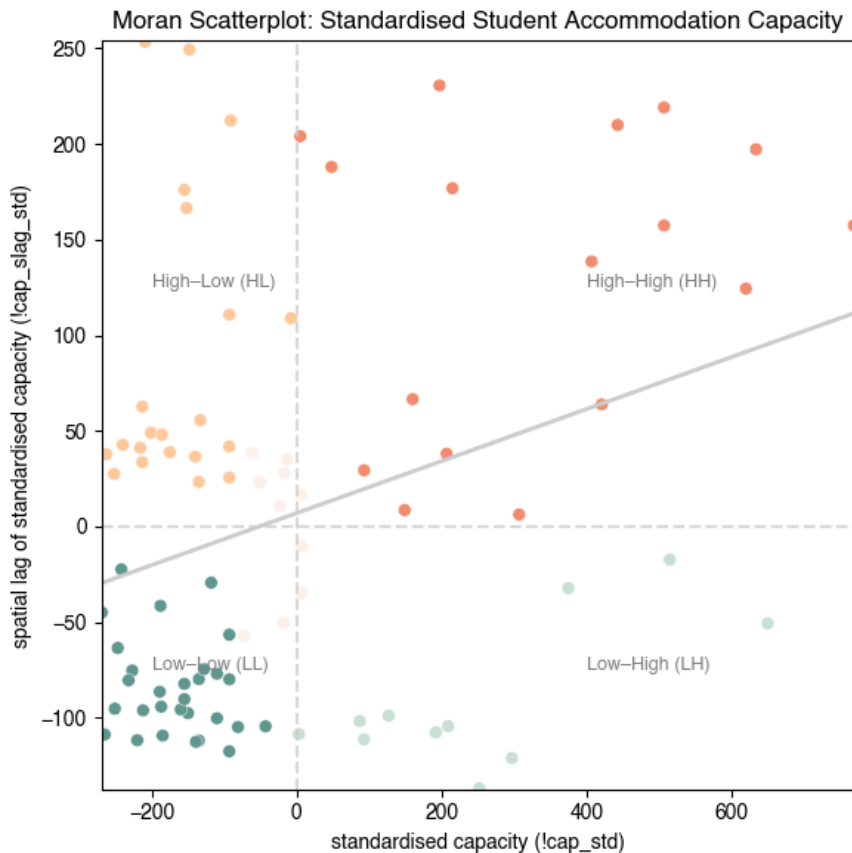
Q3 – Low–Low (LL)

These are supply shortage clusters where both the property and its surroundings have low capacity. For example, student accommodation in the direct vicinity of the Technical University of Berlin (Insider Apartments Leibniz Straße, WH Fraunhoferstraße, WH Hardenbergstraße). Such areas indicate structurally underserved areas – potential territories for development.

Q4 – High–Low (HL)

1518 These are spatial anomalies where one large property appears in an environment where neighbouring areas have low
1519 supply. For example, Student Village Berlin Adlershof, is an early indicator of the development of new student areas.

1520 The graph shows that the distribution of student housing capacity in Berlin includes both stable high-density clus-
1521 ters (HH) and structural gaps (LL), as well as a number of spatial anomalies (HL, LH), indicating uneven market
1522 development. These patterns describe indicators of the student housing crisis.



1523
1524 *Figure 24. Student Accommodation Capacity Moran Scatterplot*

1525 The map (Figure 25) visualises the results of global spatial autocorrelation, which allows spatial clusters of student
1526 housing capacity to be identified.

1527 In addition to identifying point clusters, the map clearly demonstrates the structural difference between East and
1528 West Berlin, which persists decades after the city's reunification. The map reveals clear spatial segregation.

1529 East Berlin shows a high concentration of large student complexes and compact clusters. In the districts of Licht-
1530 enberg, Marzahn-Hellersdorf, and parts of the central districts, large halls of residence (HH) are spatially clustered,
1531 indicating strategically strong market areas. Individual high-capacity facilities (HL) are located in areas with low sup-
1532 ply (e.g., in the south-east of the city). This is partly due to the historical legacy of the former East Berlin, where
1533 student residences were actively developed as part of the social infrastructure.

Western Berlin, on the other hand, is characterised by a predominance of stable areas with low capacity and a shortage of supply. Areas of systemic shortage are concentrated mainly in Charlottenburg-Wilmersdorf, Tempelhof-Schöneberg, Neukölln, and Steglitz-Zehlendorf. Properties with low capacity are surrounded by large student complexes, most often in areas where local conditions or urban planning restrictions hinder expansion, such as in Mitte. This reflects a market that is more oriented towards the private rental sector, where student accommodation has developed more slowly.

The spatial distribution of student housing in Berlin remains uneven and historically conditioned: the east has concentrated large, state-owned housing clusters, while the west faces a systemic supply shortage. This asymmetry increases pressure on rental markets in western districts and serves as a key signal of the ongoing student housing crisis.

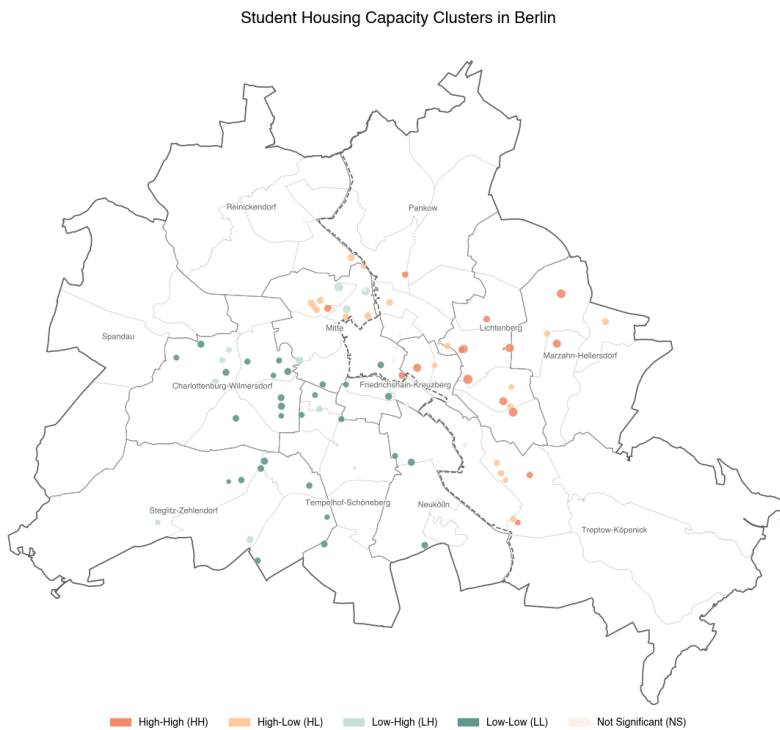


Figure 25. Student Accommodation Capacity Clusters in Berlin

3.9.7. Data Export

The student accommodation data obtained from global spatial autocorrelation was stored for further market analysis.

```
ef = r'../outputs/2.4_global_spatial_autocorrelation/pbsa_capacity_analysis.gpkg'
pbsa_c.to_file(ef, driver="GPKG")
```

1551 **3.10. Higher Education Universities International Demand**

1552 Similarly, an analysis of the spatial distribution of higher education university student demand across the city was
1553 conducted.

1554 **3.10.1. Data Import**

```
1555 uf = r'../inputs/2.4_global_spatial_autocorrelation/uni_c.gpkg'  
1556 uni_c = gpd.read_file(uf)
```

1557 **3.10.2. Standardisation**

1558 Since the research focused on international students, global autocorrelation will focus on number of international
1559 students (!s_au), which measures the market demand.

```
1560 uni_c["!s_au_std"] = uni_c["!s_au"] - uni_c["!s_au"].mean()
```

1561 **3.10.3. Moran's I Scatterplots Test & K-Nearest Neighbor Test**

1562 Moran's I Scatterplots (Figure 26) demonstrate how the spatial structure of international student demand changes as
1563 the number of nearest neighbours increases (`k_values = list(range(1, 21))`) (Table 10). The main conclusion
1564 is that, unlike the student housing market, demand from universities does not form stable spatial clusters. Most points
1565 in all graphs are concentrated near zero on the standardised international demand axis (!s_au_std), indicating that
1566 neighbouring universities do not have a strong influence on demand values.

1567 **Q1 – High–High (HH)**

1568 Unlike student dormitories, there are virtually no high-density clusters because large universities are located far from
1569 each other, and their demand values do not correlate with each other, i.e., universities attract foreign students regardless
1570 of their location.

1571 **Q2 – Low–High (LH)**

1572 These are local pockets near universities with a large number of foreign students, where a particular university has low
1573 demand compared to its saturated surroundings.

1574 **Q3 – Low–Low (LL)**

1575 For most values of the number of nearest neighbours (k), there is a high concentration of universities with a low number
1576 of foreign students surrounded by similar institutions. This indicates a generally uniform low density of international
1577 student demand across the city.

1578 **Q4 – High–Low (HL)**

1579 Several universities with very high numbers of international students are isolated “peaks of demand”, not forming
1580 full-fledged spatial clusters.

In conclusion, Moran's I scatterplots illustrate that the demand generated by international students is structurally dispersed and does not form clusters across the city.

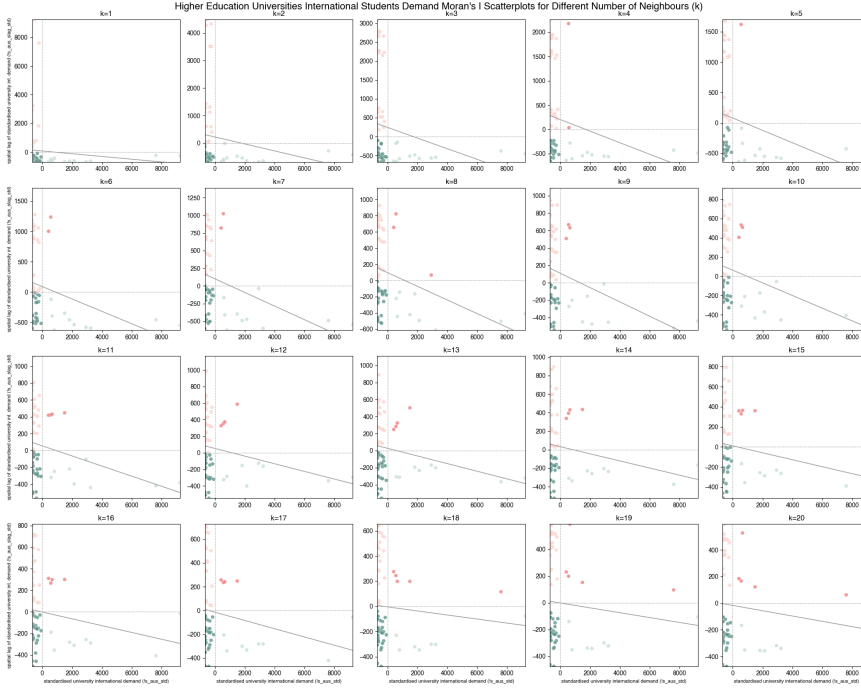


Figure 26. Higher Education Universities International Students Demand Moran's I Scatterplots for Different Number of Neighbors (k)

```
np.random.seed(55)

k_values = [2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20]
morans_I = []
p_values = []

for k in k_values:
    w = libpysal.weights.KNN.from_dataframe(uni_c, k=k)
    w.transform = "R"
    moran = Moran(uni_c['!s_aua'], w)
    morans_I.append(moran.I)
    p_values.append(moran.p_sim)

# Create DataFrame
```

```

1600 uni_k_test = pd.DataFrame({
1601     'k': k_values,
1602     "Moran's I": morans_I,
1603     'P -value': p_values
1604 })
1605
1606 uni_k_test

```

	k	Moran's I	P-value
0	2	-0.126627	0.022
1	3	-0.140827	0.003
2	4	-0.119096	0.003
3	5	-0.101951	0.009
4	6	-0.102231	0.004
5	7	-0.096559	0.004
6	8	-0.084318	0.014
7	9	-0.082463	0.005
8	10	-0.065725	0.039
9	11	-0.060102	0.061
10	12	-0.046249	0.138
11	13	-0.044825	0.134
12	14	-0.038477	0.221
13	15	-0.034638	0.255
14	16	-0.031578	0.280
15	17	-0.034663	0.224
16	18	-0.016068	0.472
17	19	-0.018201	0.477
18	20	-0.019882	0.448

1608 *Table 10. Higher Education Universities International Students Demand Moran's I index and P-value for Different*
1609 *Number of Neighbors (k)*

1610 The Moran's I distribution graph and P-value (Figure 27) show that the P-value takes acceptable values ($P < 0.05$)
1611 when k is between 2 and 10. Since we are talking about global spatial autocorrelation, the purpose of which is to

1612 identify patterns throughout Berlin, it is necessary to choose the maximum value of k. At the same time, the P-value
1613 when choosing 10 nearest neighbours is critically close to the significance threshold. Therefore, the previous value
1614 k=9 was chosen.

1615 *Figure 27. Higher Education Universities International Students Demand Moran's I index and P-value for Different*
1616 *Number of Neighbors (k)*

1617 **3.10.4. Significance Threshold Test**

1618 By analogy with student accommodation capacity, the significance threshold was calculated as half of the median
1619 of the absolute value (median_abs * 0.5).

1620 Option 1 - Percentile -based threshold: 679.990

1621 (75th percentile of absolute values)

1622 Option 2 - Standard deviation -based threshold: 828.251

1623 (0.5 * standard deviation)

1624 Option 3 - Fixed threshold: 550.000

1625 (Fixed value for your data range)

1626 Option 4 - Median -based threshold: 304.745

1627 (0.5 * median of absolute values)

1628

1629 Test 1: threshold = 679.990

1630 NS: 54 (76.1%)

1631 HH: 2 (2.8%)

1632 LH: 6 (8.5%)

1633 LL: 5 (7.0%)

1634 HL: 4 (5.6%)

1635

1636 Test 2: threshold = 828.251

1637 NS: 64 (90.1%)

1638 HH: 2 (2.8%)

1639 LH: 5 (7.0%)

1640 LL: 0 (0.0%)

1641 HL: 0 (0.0%)

1642

```

1643 Test 3: threshold = 550.000
1644     NS: 23 (32.4%)
1645     HH: 4 (5.6%)
1646     LH: 7 (9.9%)
1647     LL: 23 (32.4%)
1648     HL: 14 (19.7%)

```

```

1649

```

```

1650 Test 4: threshold = 304.745
1651     NS: 5 (7.0%)
1652     HH: 5 (7.0%)
1653     LH: 7 (9.9%)
1654     LL: 34 (47.9%)
1655     HL: 20 (28.2%)

```

1656 *3.10.5. Spatial Weights & Moran's I & Spatial Lag*

```

1657 # Generate spatial weights from the GeoDataFrame
1658 w = KNN.from_dataframe(uni_c, k=9)
1659 w.transform = "R" # Row -standardize weights
1660
1661 values = uni_c['!s_aus']
1662
1663 moran = Moran(values, w)
1664 print(f"Moran's I: {moran.I:.3f}")
1665 print(f"P -value: {moran.p_sim:.3f}")

```

```

1666 Moran's I: -0.082
1667 P -value: 0.009

```

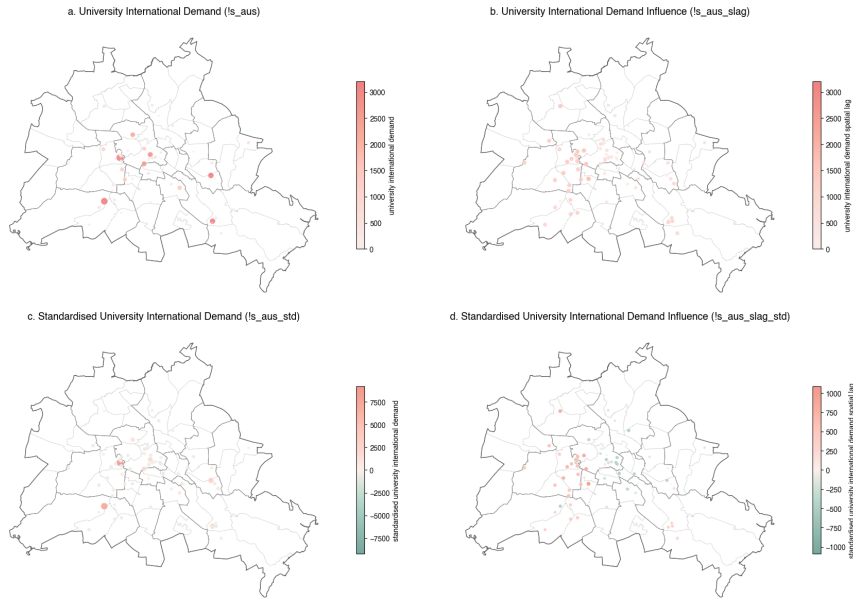
1668 Moran's I (Moran's I: -0.082) indicates a weak negative spatial autocorrelation. In the context of urban plan-
 1669 ning, this indicates that higher education universities do not create spatial clusters; in other words, they are unevenly
 1670 distributed throughout the city.

1671 P-value (P-value: 0.007) indicates that the probability of obtaining such a result by chance is 0.7%, i.e., the
 1672 result obtained is statistically significant.

1673 3.10.6. Spatial Lag & Standardised Spatial Lag

```
1674 # Add spatial lag and standardized columns
1675 uni_c["!s_aus_slag"] = lag_spatial(w, uni_c["!s_aus"])
1676 uni_c["!s_aus_std"] = uni_c["!s_aus"] - uni_c["!s_aus"].mean()
1677 uni_c["!s_aus_slag_std"] = lag_spatial(w, uni_c["!s_aus_std"])
```

1678 The maps (Figure 28) visualise the different stages of preparing international demand data for spatial analysis.



1679

1680 *Figure 28. Stages of Preparing Higher Education Universities International Students Demand Data*

1681 Moran's graph (Figure 29) visualises the spatial relationships between standardised international university de-
1682 mand (!s_aus_std) and its spatial lag (!s_aus_slag_std). The division into four quadrants allows us to identify
1683 characteristic spatial patterns of demand distribution and understand where areas of high competition or, conversely,
1684 a lack of educational attractiveness are forming.

1685 **Q1 – High–High (HH)**

1686 These are clusters of high international demand, where universities with a large proportion of foreign students are
1687 located next to similarly significant neighbours. Such areas form stable 'magnets' for international students, for ex-
1688 ample, the College of Business and Law (HWR – 1,336 foreign students), the University of the Arts Berlin (UKB –
1689 1,244 foreign students), and ESCP Business School Berlin (ESCP – 1,092 foreign students). At the same time, two of
1690 the three universities in the HH quadrant are publicly owned (HWR and UKB) and located in Willensdorf, while the
1691 private ESCP University is located in Charlottenburg. These three universities form a powerful spatial core of global
1692 education in West Berlin.

1693 **Q2 – High–Low (HL)**

Spatial anomalies, where a university with high international demand is located among neighbouring universities with low values. This is the largest cluster among all universities. It mainly consists of individual private universities that have a localised impact and signal the beginning of the formation of new educational centres. Among the universities belonging to the HL cluster but funded by the state are the Technical University of Berlin, the Protestant University of Applied Sciences Berlin, and the Charité – University Medicine Berlin campuses in the south-west of the city.

Q3 – Low–Low (LL)

Clusters of low demand, where both the university itself and its surroundings attract relatively few foreign students. LL zones indicate areas where international educational activity is weak, such as Pankow, Lichtenberg, Friedrichshain-Kreuzberg, and Neukölln. Among the reasons for unrealised educational potential, one can assume a lack of infrastructure, limited academic activity, or weak integration into the city's structure.

Q4 – Low–High (LH)

These are local failures, where a university demonstrates low international demand, but its environment demonstrates high demand. At the same time, seven of the nine universities in the LH quadrant are publicly owned. Universities such as the Free University of Berlin and Humboldt University of Berlin have the potential to participate in existing educational clusters, but are not yet strategically integrated into them.

This distribution indicates competition between universities, the existence of 'hot spots' of international activity, and, at the same time, structural territorial gaps. All these patterns are important for studying the student housing crisis, as international demand directly correlates with increased pressure on the rental market, especially in highly attractive local clusters.

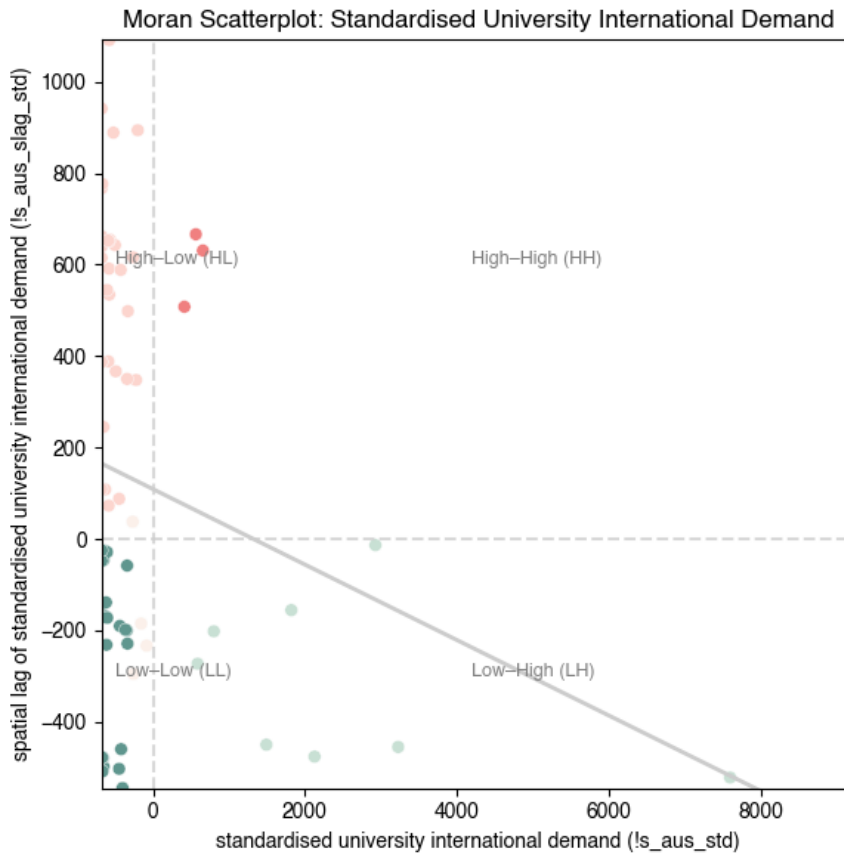


Figure 29. Higher Education Universities International Students Demand Moran Scatterplot

The map showing the spatial distribution of international student demand in Berlin (Figure 30) reveals that, unlike the supply of student housing, demand does not form stable clusters of high concentration. Instead of the expected spatial agglomeration around large public universities, there is a dispersion (scattering) of values. In other words, universities with high numbers of international students are not grouped geographically, but rather distributed in a fragmented manner and often separated from similar neighbours.

The east-west divide highlights Berlin's historical asymmetry: the West traditionally concentrates international academic demand, while the East remains unattractive in an international context.

The demand for international students is distributed more chaotically and does not correlate with the spatial patterns of student housing supply, which exacerbates the structural imbalance between the housing market and the city's academic infrastructure.

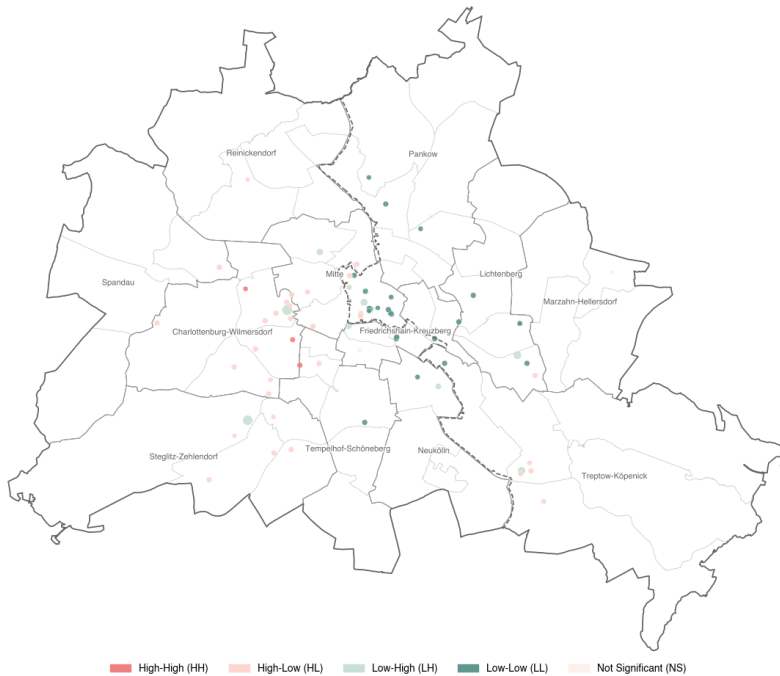


Figure 30. Higher Education Universities International Demand Dispersion in Berlin

3.10.7. Data Export

The data on higher education universities' international students demand obtained as a result of global spatial autocorrelation was collected for further market analysis.

```
hf = r'../outputs/2.4_global_spatial_autocorrelation/uni_int_demand_analysis.gpkg'
uni_c.to_file(hf, driver="GPKG")
```

4. 3. Student Housing Market Imbalance

4.1. 3.1 Overview

Chapter Student Housing Market Imbalance analyses the imbalance between university demand and student housing capacity.

4.1.1. *What? Student Housing Market Imbalance*

Student Housing Market Imbalance can be understood as a set of statistically significant patterns that constitute mismatches between student housing capacity supply and university demand for higher education.

Among these patterns, one of the most important is pressure on the student housing market. Based on the conclusions drawn in the Capacities, Demand & Spatial Distribution Chapter, it can be assumed that pressure on the market is not evenly distributed.

4.1.2. *Why? Assess Spatial Demand–Supply Imbalances*

The aim of this chapter is to assess the student housing market imbalances, identify areas where demand exceeds supply by answering the following research questions:

How do student university demand compare to student housing market housing capacities?

Where do spatial patterns of demand and supply generate elevated market pressure?

The answers to these questions are necessary to understand how and where pressure on the student housing market is formed in the city.

4.1.3. *How? Spatial Data Analysis*

Unlike the Capacities, Demand & Spatial Distribution Chapter, where the analysis of the two systems was conducted sequentially, this chapter aimed to combine the data using spatial join and conduct a comparative analysis of the student housing market according to the Central Place Theory outlined in the Theoretical Framework chapter.

Spatial Join

A spatial join combines represented geographic objects and their associated attributes based on a spatial relationship (University of West Florida and Morgan, 2023). In other words, spatial joins allow you to combine multiple data sets based on spatial characteristics. For example, you can calculate the total student housing capacity and university demand for the Mitte district.

However, such a rough approach is not suitable for a detailed analysis of the student housing market, as Berlin's districts vary greatly in size, so simple spatial intersection and subsequent aggregation distort the real picture of market pressure. In order to increase the accuracy of the spatial join, an isochrone accessibility analysis was performed.

Isochrone Analysis

An isochrone in geography and urban planning is a polygon that depicts the area accessible from a point within a

certain time threshold, i.e., isochrones calculate accessibility in terms of time (OpenStreetMap Wiki, 2025). This allows administrative district boundaries to be replaced with more functional zones - zones of student mobility. Thus, isochrone analysis eliminates the problem of mismatched district areas and focuses on the actual patterns of student movement.

Based on isochrones, it is possible to perform a spatial join between university demand and student housing capacity. However, such isochrones often overlap with each other, which leads to the problem of overlap. One solution to the problem of overlap is rasterisation.

Rasterisation

Rasterisation converts vectors into rasters. In Geospatial Science, it usually means converting vector geometry to raster geometry. It can be used to summarise data from several objects. Since accessibility isochrones intersect with each other, rasterisation — the conversion of geometry into a regular grid — is performed to match spatially heterogeneous supply and demand data.

These research methods allow for a parallel assessment of market areas and show how spatial patterns exacerbate or mitigate the shortage of student housing, forming or neutralising market pressure.

Thus, the chapter demonstrates the geography of the student housing market, identifying the most problematic areas.

4.2. 3.2 Central Place Analysis

The chapter Student Housing Market: Capacities, Demand, and Spatial Distribution provides a consistent analysis of two systems: student accommodations and higher education universities.

To proceed to market analysis, it is necessary to examine how these two systems interact with each other, forming the student accommodation market. Central Place Theory, outlined in the chapter Theoretical Framework, assists in navigating the analysis process.

Set Up Import Libraries

```
import os

import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

import matplotlib as mpl

import geopandas as gpd

from shapely.geometry import Point, MultiPoint

from pyproj import CRS

from scipy.stats import norm

from matplotlib.colors import LinearSegmentedColormap

import matplotlib.path_effects as path_effects

import string

import requests

import time

from shapely.geometry import Polygon

import seaborn as sns

from shapely.geometry import MultiPolygon


Berlin Spatial Boundaries


pf = r"../inputs/3.2_central_place_analysis/lor_pgr.gpkg"

pgr = gpd.read_file(pf)

pgr.replace({'-': np.nan, 0: np.nan, '' : np.nan}, inplace=True)


bf = r"../inputs/3.2_central_place_analysis/adm_bz.gpkg"
```



```

1809 bz = gpd.read_file(bf)
1810 bz.replace({' -': np.nan, 0: np.nan, '' : np.nan}, inplace=True)
1811
1812 of = r"../inputs/3.2_central_place_analysis/adm_out.gpkg"
1813 out = gpd.read_file(of)
1814 out.replace({' -': np.nan, 0: np.nan, '' : np.nan}, inplace=True)

```

1815 The analysis of student accommodation capacity and higher education universities' international student demand
1816 data, obtained as a result of Global Spatial Autocorrelation, served as the basis for student housing market research.

```

1817 ef = r'../inputs/3.2_central_place_analysis/pbsa_capacity_analysis.gpkg'
1818 pbsa_cap = gpd.read_file(ef)
1819
1820 mf = r'../inputs/3.2_central_place_analysis/uni_int_demand_analysis.gpkg'
1821 uni_dem = gpd.read_file(mf, driver="GPKG")

```

1822 **4.3. Central Places Definition**

1823 In Central Place Theory King (1986) argue that the idea that there is a smooth and regular progression in urban
1824 centres from the smallest up to the largest center - is not a very useful one, and that greater insight can be gained into
1825 the nature of an urban pattern by referring to discrete population-size classes.

1826 Taking this into account, central universities should be defined not as the 'largest' in terms of numbers, but as
1827 institutions that perform central functions in the urban system. Their significance is determined not so much by the
1828 scale of student enrolment as by their functional role, their position in the urban and educational hierarchy, and their
1829 ability to serve wider areas of attraction.

1830 In order to determine which universities are considered central, the following concerns must be taken into account.

1831 Firstly, based on spatial economics analysis of university clusters, only High-High (HH), Low-High (LH) and
1832 High-Low (HL) clusters are capable of generating sustained high demand in the housing market, fulfilling the func-
1833 tional roles of centres. Meanwhile, Low-Low (LL) clusters do not form centres of demand.

1834 Secondly, the number of central universities must be limited, as 26 out of 52 universities cannot be central by
1835 definition.

1836 Only by taking this into account can a quantitative test be conducted using the international student demand
1837 (!s_au) threshold.

1838 Option 1 - Percentile -based threshold: 414.000

1839 (75th percentile of absolute values)

1840 Option 2 - Standard deviation -based threshold: 828.251

1841 (0.5 * standard deviation)

1842 Option 3 - Fixed threshold: 550.000

1843 (Fixed value for your data range)

1844 Option 4 - Median -based threshold: 51.000

1845 (0.5 * median of absolute values)

1846

1847 Test 1: threshold = 414.000

1848 Total features: 15

1849 Total demand: 40958.7

1850 Features with cluster = 'NS': 0

1851 Features with cluster = 'HH': 3

1852 Features with cluster = 'LH': 9

1853 Features with cluster = 'HL': 3

1854 Features with cluster = 'LL': 0

1855

1856 Test 2: threshold = 828.251

1857 Total features: 12

1858 Total demand: 39607.76

1859 Features with cluster = 'NS': 0

1860 Features with cluster = 'HH': 3

1861 Features with cluster = 'LH': 9

1862 Features with cluster = 'HL': 0

1863 Features with cluster = 'LL': 0

1864

1865 Test 3: threshold = 550.000

1866 Total features: 12

1867 Total demand: 39607.76

1868 Features with cluster = 'NS': 0

1869 Features with cluster = 'HH': 3

1870 Features with cluster = 'LH': 9

```

1871 Features with cluster = 'HL': 0
1872 Features with cluster = 'LL': 0
1873
1874 Test 4: threshold = 51.000
1875 Total features: 45
1876 Total demand: 46240.259999999995
1877 Features with cluster = 'NS': 0
1878 Features with cluster = 'HH': 3
1879 Features with cluster = 'LH': 9
1880 Features with cluster = 'HL': 19
1881 Features with cluster = 'LL': 14

```

1882 In order to correctly interpret the Threshold Test, it is necessary to pay attention to the number of universities and
1883 their distribution across four spatial clusters.

1884 **Option 1 - Percentile-based threshold: 414,000**

1885 With a percentile-based threshold, the upper quarter of the distribution is selected, taking into account three spatial
1886 clusters - HH, LH and HL. It also significantly reduces the sample from 52 to 15 universities, fixing the central nodes
1887 of the pair. Therefore, this threshold creates a threshold that reflects the real structure of the student housing market.
1888 (theshold = threshold_percentile).

1889 **Option 2 - Standard deviation-based threshold: 828.251**

1890 **Option 3 - Fixed threshold: 550.000**

1891 Standard deviation-based and fixed thresholds exclude HL clusters that are early indicators of the development of new
1892 student attraction zones.

1893 **Option 4 - Median-based threshold: 51.000**

1894 With a median-based threshold, almost all universities become 'central,' which devalues the category itself.

1895 Thus, using the spatial clusters and percentile-based threshold filter, 15 universities (Table XX) were identified as
1896 central.

1897 Thus, using the spatial clusters and percentile-based threshold filter, 15 universities (Table 11) were identified as
1898 central.

```

1899 central_places = uni_dem[
1900     (uni_dem['cluster_aus'].isin(['HL', 'HH', 'LH'])) &
1901     (uni_dem['!s_aus'] >= threshold_percentile)

```

```
1902 ]  
  
1903 central_places[[  
1904     "code",  
1905     "abr",  
1906     "name",  
1907     "own",  
1908     "cluster",  
1909     "!s_aus"  
1910 ]]
```

	code	abr	name	own	cluster	!s_au
2	HS0180	CU	Charite Univer- sitätsmedizin Berlin	- Public University	Uni- LH	1266.50
3	HS0180	CU	Charite Univer- sitätsmedizin Berlin	- Public University	Uni- HL	455.94
6	HS0200	HU	Humboldt- Universität Berlin	Public University	Uni- LH	3608.46
7	HS0200	HU	Humboldt- Universität Berlin	Public University	Uni- LH	2806.58
16	HS1380	FU	Freie Univer- sität Berlin	Public University	Uni- LH	8280.02
21	HS1690	TU	Technische Universität Berlin	Public University	Uni- LH	9914.01
23	HS1700	ESCP	ESCP- Europe Wirtschaft- shochschule Berlin	Privat University	Uni- HH	1092.00
30	HS2290	BIU	Berlin Univ. Applied Sciences	Int. of University	Privat Uni- HL	417.00
36	HS2790	UAB	Universität der Künste Berlin	Public University	Uni- HH	1243.34
39	HS3270	GIU	German International	Privat University	Uni- HL	478.00

4.4. Market Areas Definition

Once the central universities have been identified, the next step is to define market areas that jointly reflect data on higher education universities' international demand and student accommodation capacity. As noted in the Overview chapter, there are several approaches to combining these data. Here are three primary approaches.

The first approach is to use Berlin's existing administrative boundaries, such as Life-Oriented Spaces (LOR - *Lebensweltlich Orientierte Räume*) or their sub-level Neighbourhood Districts (BZR - *Bezirksregionen*). However, this method creates significant gaps: many areas end up "empty" because they have neither universities nor student accommodation. This indicates that neighbourhood districts are too small for analysing the urban supply and demand system. The next level, Prognosis Areas (PGR), on the contrary, is too large and aggregates too heterogeneous areas. Thus, the existing life-oriented spaces do not reflect the real geography of the student housing market.

The second approach is to build buffer zones around central universities, creating new geometries for spatial data integration. However, due to Berlin's heterogeneous morphology, it is impossible to correctly determine a universal buffer distance. In addition, this method results in all market areas being the same size, which contradicts the actual structure of urban accessibility and central place theory.

The third approach is to use accessibility isochrones, i.e., zones formed on the basis of actual travel time. Unlike administrative boundaries or fixed buffers, isochrones reflect the actual spatial relationship and student mobility. This allows for the creation of flexible and functionally relevant market areas in which supply and demand can be matched more accurately.

To construct accessibility isochrones, the study used the Travel Time API service, which allows modelling reachability zones based on real-time travel. A distinctive feature of the Travel Time service is the ability to generate isochrones based on public transport estimates ('type': 'public_transport').

Below is an example of using the Travel Time API to create 15-minute accessibility zones (`travel_time = 900`).

```
# url = 'https://api.traveltimeapp.com/v4/time -map'

# tt_app_id = 'my_app_id'
# tt_api_key = 'my_api_key'

# headers = {
#     'content -type': 'application/json',
#     'x -application -id': tt_app_id,
```

```

1943 #     'x -api -key': tt_api_key
1944 # }
1945
1946 # travel_time = 900 # 15 minutes
1947
1948 # for idx, row in central_places.iterrows():
1949 #     lon = float(row['lon'])
1950 #     lat = float(row['lat'])
1951
1952 #     payload = {
1953 #         "arrival_searches": [
1954 #             {
1955 #                 "id": f"iso_{idx}",
1956 #                 "coords": {"lat": lat, "lng": lon},
1957 #                 "arrival_time": "2025 -08 -13T09:00:00+01:00",
1958 #                 "travel_time": travel_time,
1959 #                 "transportation": {"type": "public_transport", "disable_border_crossing": True
1960 #             }
1961 #         ]
1962 #     }
1963
1964 #     resp = requests.post(url, json=payload, headers=headers)
1965 #     print(f"request {idx}: {resp.status_code} {resp.reason} for location {[lon, lat]}")
1966
1967 #     if resp.ok:
1968 #         data = resp.json()
1969 #         for res in data.get('results', []):
1970 #             search_id = res.get('search_id')
1971 #             for shp in res.get('shapes', []):
1972 #                 shell = shp.get('shell', [])
1973 #                 if not shell:
1974 #                     continue

```

```

1975 #         coords = [(pt['lng'], pt['lat']) for pt in shell]
1976 #         if coords and coords[0] != coords[-1]:
1977 #             coords.append(coords[0])
1978 #         poly = Polygon(coords)
1979 #         g = gpd.GeoDataFrame([{'geometry': poly, 'tt_id': search_id,
1980 #                                'origin_lon': lon, 'origin_lat': lat}],
1981 #                                crs='EPSG:4326')
1982 #         iso_gdf = gpd.GeoDataFrame(pd.concat([iso_gdf, g],
1983 #                                                ignore_index=True,
1984 #                                                sort=False))
1985
1986 #         time.sleep(15)
1987
1988 # print(f"got {len(iso_gdf)} isochrone polygons.")
1989
1990 # iso_gdf["geometry"] = iso_gdf["geometry"].buffer(0)
1991
1992 # iso_gdf.to_crs(epsg=25833)
1993
1994 # ef = r"../outputs/3.2_central_place_analysis/iso_without_data_15_min.gpkg"
1995
1996 # iso_gdf.to_file(ef, driver="GPKG")

```

1994 **4.5. Travel Time Test**

1995 Since isochrones are sensitive to travel time, a travel time test was conducted (Figure 31) to determine market areas
1996 where travel time varied from 10 to 35 minutes with 5-minute intervals (isochrone_times = [10, 15, 20, 25,
1997 30, 35]).

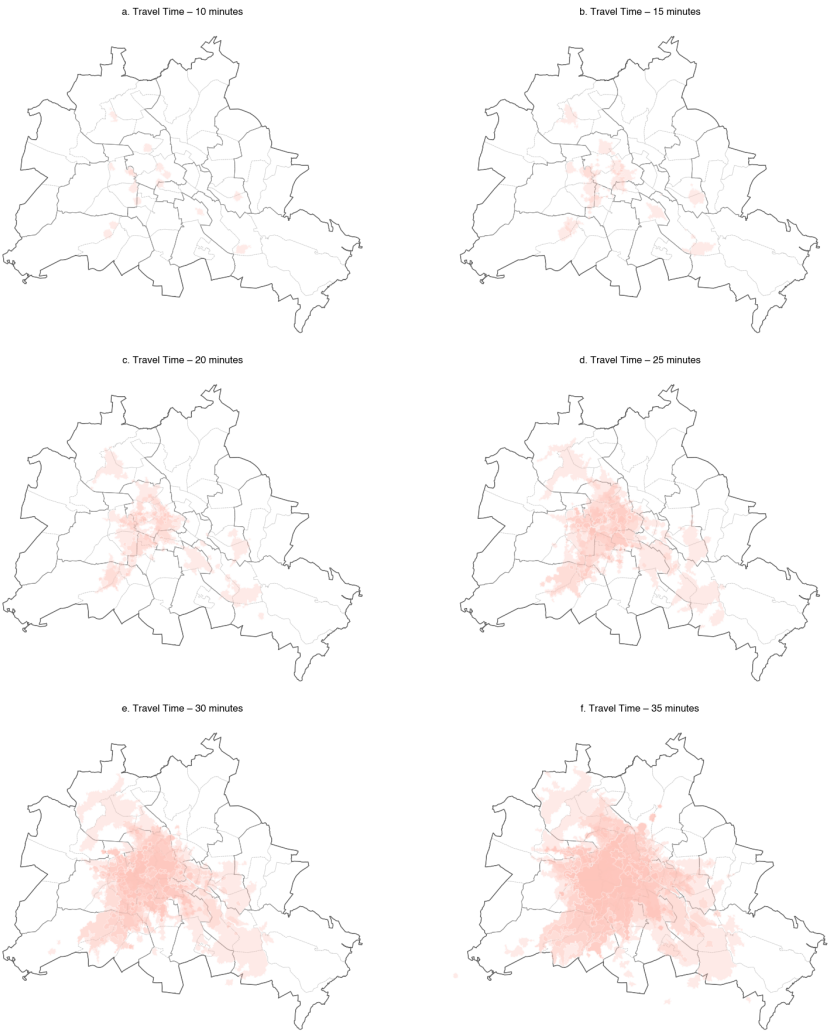
```

1998 # Define isochrone time ranges
1999 isochrone_times = [10, 15, 20, 25, 30, 35]
2000 base_path = "../outputs/3.2_central_place_analysis"
2001
2002 # Initialize results storage
2003 results_summary = []
2004 pbsa_market_summary = None
2005 iso_dem_ready = None

```



```
2006 pbsa_cap_ready = None
2007 summary_df = None
2008 total_capacity_complete = pbsa_cap["!cap"].sum()
2009 total_demand_complete = uni_dem["!s_aus"].sum()
2010
2011 pbsa_cap = pbsa_cap.to_crs("EPSG:25833")
2012 central_places = central_places.to_crs("EPSG:25833")
```



2013
2014 *Figure 31. Travel-Time API Test*

2015 When constructing isochrones, it is important to correctly interpret the allocation of student housing capacity and
2016 higher education universities demand.

2017 The allocation of student accommodations is performed using a spatial join between points and isochrone polygons.
2018 Since one student accommodations facility may be located within the catchment area of several universities, directly

assigning its full capacity to each area would lead to a significant overestimation of supply. To avoid double-counting, each student accommodation facility first receives information about the number of isochrones in which it is located, after which its capacity is divided equally among all relevant areas. The adjusted values are then aggregated at the market area level. This approach allows for the correct distribution of available housing in accordance with the actual spatial structure of accessibility, taking into account the overlap of the areas of influence of different universities.

Similar to the distribution of dormitories, university demand (number of German students (!s_de), number of foreign students (!s_aus) and total number of students (!s_all)) is assigned to isochrones based on the principle of spatial intersection. Universities may also lie within several isochrones, especially when accessibility times are long (30–35 minutes). Therefore, the demand for each university is divided among all the zones within which it falls. This method eliminates artificial doubling of demand and creates a balanced distribution that reflects the real impact of universities on the housing market within their accessibility zones. The final demand values are aggregated at the market area level and can be compared with the distributed student accommodation capacity.

```
for time_min in isochrone_times:
    # Load isochrone data
    iso_file = f"{base_path}/iso_without_data_{time_min}_min.gpkg"
    iso_gdf = gpd.read_file(iso_file)

    # Reproject immediately
    if iso_gdf.crs != "EPSG:25833":
        iso_gdf = iso_gdf.to_crs("EPSG:25833")

    # Build multipolygons per tt_id
    iso_multi = (
        iso_gdf.groupby(["tt_id", "origin_lon", "origin_lat"])
        .geometry.apply(lambda g: MultiPolygon([geom for geom in g]))
        .reset_index()
    )
    iso_multi_gdf = gpd.GeoDataFrame(iso_multi, geometry="geometry", crs="EPSG:25833")

    # Prepare central places (no geometry needed for merge)
    central_places_no_geom = central_places.drop(columns="geometry")
    central_places_no_geom["lon"] = central_places_no_geom["lon"].astype(float)
```

```

2051
2052 # Merge on origin_lon/lon
2053 iso_multi_gdf["origin_lon"] = iso_multi_gdf["origin_lon"].astype(float)
2054 iso_dem = iso_multi_gdf.merge(
2055     central_places_no_geom,
2056     left_on="origin_lon",
2057     right_on="lon"
2058 )
2059 iso_dem = gpd.GeoDataFrame(iso_dem, geometry="geometry", crs="EPSG:25833")
2060 iso_dem = iso_dem.drop(columns=["origin_lon", "url"])
2061 iso_dem["geometry"] = iso_dem["geometry"].buffer(0) # fix invalids
2062
2063 # From here everything is already in EPSG:25833
2064 pbsa_cap_ready = pbsa_cap.copy()
2065 if "pbsa_id" not in pbsa_cap_ready.columns:
2066     pbsa_cap_ready["pbsa_id"] = pbsa_cap_ready.index
2067
2068 iso_dem_ready = iso_dem.copy()
2069 if "iso_id" not in iso_dem_ready.columns:
2070     iso_dem_ready["iso_id"] = iso_dem_ready.index
2071
2072 uni_dem = uni_dem.copy()
2073 if 'uni_id' not in uni_dem.columns:
2074     uni_dem['uni_id'] = uni_dem.index
2075
2076 # PBSA allocation
2077 join_pbsa = gpd.sjoin(pbsa_cap_ready, iso_dem_ready, how="inner", predicate="intersects")
2078 n_iso_pbsa = join_pbsa.groupby("pbsa_id")["iso_id"].nunique().rename("n_iso_pbsa")
2079 join_pbsa = join_pbsa.merge(n_iso_pbsa, left_on="pbsa_id", right_index=True, how="left")
2080 cap_cols = ["!cap"]
2081 for c in cap_cols:
2082     join_pbsa[f"{c}_sum"] = join_pbsa[c] / join_pbsa["n_iso_pbsa"]

```

```

2083 iso_alloc_pbsa = (
2084     join_pbsa.groupby("iso_id")[[f"{c}_sum" for c in cap_cols]]
2085     .sum().reset_index()
2086 )
2087
2088 # - - - UNI allocation - - -
2089 join_uni = gpd.sjoin(uni_dem, iso_dem_ready, how='inner', predicate='intersects')
2090 n_iso_uni = join_uni.groupby('uni_id')['iso_id'].nunique().rename('n_iso_uni')
2091 join_uni = join_uni.merge(n_iso_uni, left_on='uni_id', right_index=True, how='left')
2092 dem_cols = ['!s_aus_left', '!s_de_left', '!s_all_left']
2093 for c in dem_cols:
2094     join_uni[f'{c}_sum'] = join_uni[c] / join_uni['n_iso_uni']
2095 iso_alloc_uni = (join_uni.groupby('iso_id')[[f'{c}_sum' for c in dem_cols]]
2096     .sum()
2097     .reset_index())
2098
2099 # Merge results
2100 market_area = iso_dem_ready.merge(iso_alloc_pbsa, on="iso_id", how="left")
2101 market_area = market_area.merge(iso_alloc_uni, on="iso_id", how="left")
2102
2103 market_area = market_area.rename(columns={
2104     "!s_aus_left_sum": "!s_aus_sum",
2105     "!s_de_left_sum": "!s_de_sum",
2106     "!s_all_left_sum": "!s_all_sum",
2107 })
2108
2109 # Ratios
2110 market_area["capacity_demand_ratio"] = (
2111     market_area["!cap_sum"] / market_area["!s_aus_sum"].replace(0, np.inf)
2112 )
2113 market_area["demand_capacity_pressure"] = (
2114     market_area["!s_aus_sum"] / market_area["!cap_sum"].replace(0, np.inf)

```

```

2115 )
2116 market_area["area"] = market_area.geometry.area / 1e6 # in sq km
2117
2118 # PBSA count
2119 pbsa_pts = pbsa_cap_ready.copy()
2120 wb = gpd.sjoin(pbsa_pts, market_area[["iso_id", "geometry"]], how="inner", predicate="within")
2121 pbsa_counts = wb.groupby("iso_id").size()
2122 market_area["pbsa_count"] = market_area["iso_id"].map(pbsa_counts).fillna(0).astype(int)
2123
2124 # Uni Count
2125 uni_pts = uni_dem.copy() # again, geometry is already active
2126 # Spatial join: UNI point within isochrone polygon
2127 wb = gpd.sjoin(uni_pts, market_area[['iso_id', 'geometry']], how='inner', predicate='within')
2128 # Count UNI per iso_id
2129 uni_counts = wb.groupby('iso_id').size()
2130 # Attach to market_area fill missing with 0
2131 market_area['uni_count'] = market_area['iso_id'].map(uni_counts).fillna(0).astype(int)
2132
2133 # Travel time
2134 market_area["travel_time_min"] = time_min
2135
2136 #Cleaning
2137 market_area['cluster'] = market_area['cluster'].map({
2138     'HH': 'High -High (HH)',
2139     'HL': 'High -Low (HL)',
2140     'LH': 'Low -High (LH)',
2141     'LL': 'Low -Low (LL)',
2142     'NS': 'Not Significant (NS)'
2143 })
2144 columns_to_drop = ['origin_lat', 'origin_lon', 'cluster_aus', 'iso_id', 'lon']
2145 market_area = market_area.drop(columns=columns_to_drop, errors='ignore')
2146

```

```

2147     # Save
2148     output_file = f"{base_path}/market_areas_{time_min}_min.gpkg"
2149     market_area.to_file(output_file, driver="GPKG")
2150
2151     # Summary stats
2152     total_capacity_inside = market_area["!cap_sum"].sum()
2153     total_demand_inside = market_area["!s_aus_sum"].sum()
2154
2155     uncovered_demand = total_demand_complete - total_demand_inside
2156     unused_capacity = total_capacity_complete - total_capacity_inside
2157
2158     results_summary.append({
2159         "travel_time_min": time_min,
2160         "total_capacity_inside": total_capacity_inside,
2161         "total_demand_inside": total_demand_inside,
2162         "uncovered_demand": uncovered_demand,
2163         "unused_capacity": unused_capacity,
2164     })

```

2165 By changing the time thresholds, it is possible to check the stability of market areas: whether the structure of supply
 2166 and demand zones changes or remains stable. To do this, at the final stage, a summary of key indicators is formed for
 2167 each tested time of availability, allowing us to assess how effectively the formed market areas reflect the real structure
 2168 of supply and demand. The table (Table 12) contains: the total capacity of dormitories that fall within the isochrones
 2169 (total_capacity_inside) and the total university demand covered by these same areas (total_demand_inside).

```

2170 # Create summary DataFrame
2171 summary_df = []
2172
2173 summary_df = pd.DataFrame(results_summary).round(1)
2174
2175 summary_df.sort_values("travel_time_min", inplace=True)
2176
2177 summary_df

```

	travel_time_min	total_capacity_inside	total_demand_inside	uncovered_demand	unused_capacity
0	10	3001.0	32358.5	16098.3	22816.0
1	15	6191.0	44461.9	3994.9	19626.0
2	20	9089.0	45344.2	3112.6	16728.0
3	25	15195.0	46592.4	1864.4	10622.0
4	30	17841.0	46976.1	1480.6	7976.0
5	35	20635.0	47224.8	1232.0	5182.0

Table 12. Travel Time Summary Statistics

```

output_path = r"../outputs/3.2_central_place_analysis/market_area_travel_time.xlsx"
with pd.ExcelWriter(output_path, engine='openpyxl') as writer:
    summary_df.to_excel(writer, sheet_name='base')

```

Comparing these indicators between different time thresholds (10–35 minutes) (Figure 32) allows us to determine the optimal radius of market influence of universities, as well as to identify structural imbalances between supply and demand associated with spatial constraints. This approach provides a quantitative basis for selecting the isochrone threshold and correctly forming market areas.

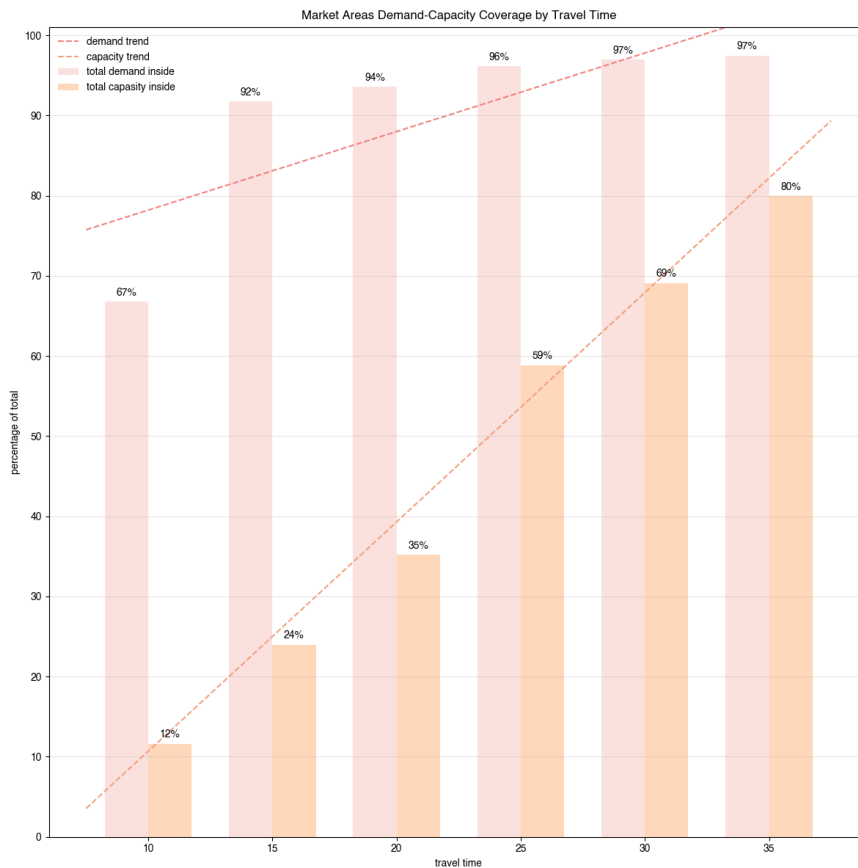


Figure 32. Market Areas Demand-Capacity Coverage by Travel Time

The results of the travel time test demonstrate that increasing the time threshold for accessibility significantly affects the degree of coverage of university demand and the capacity of student housing included in market areas. However, the effect grows unevenly, and it is precisely this dynamic that allows us to determine the optimal travel time value.

First, at thresholds of 10–25 minutes, there is a significant discrepancy between the share of demand covered and the share of capacity included. For example, at 25 minutes, the market already covers 96% of total demand, but only 59% of housing capacity. This imbalance indicates that with short isochrones, a significant portion of dormitories remain formally ‘outside the market,’ which leads to artificially inflated estimates of the deficit and local market pressure.

Secondly, starting at 30 minutes, the volume of included capacity increases sharply, reaching 69%, and by 35 minutes, 80%, which reflects a significant improvement in the spatial correspondence between university demand and student accommodation capacity. At the same time, there is no further noticeable growth in demand coverage: even at 20 minutes, more than 94% of student demand is covered, and at 30–35 minutes, 97%. This means that an increase in travel time in this range mainly affects the housing supply rather than changing the demand pattern.

Thus, the 35-minute threshold represents an optimal compromise.

Moreover, according to a study by Menz & Mandl (2024), students studying mostly in person spend 5 hours more

2203 per week on study-related activities compared to students studying mostly online. Thus, it can be concluded that full-
2204 time students spend an additional 5 hours per week commuting, which is equivalent to approximately 1 hour per day
2205 or 30 minutes each way.

2206 **4.6. Market Areas Analysis**

2207 Spatial economics and central place metrics were used to analyse market areas (Table 13).

	code	name	own	cluster	!cap_sum	!s_aus_sum	!s_de_sum	!s_all_sum
0	HS0180	Charite - Univer- sitätsmedi- zin Berlin	Public Univer- sity	Low- High (LH)	1455.4	3072.6	8126.0	11198.7
1	HS0180	Charite - Univer- sitätsmedi- zin Berlin	Public Univer- sity	High- Low (HL)	511.5	2563.9	10281.7	12845.5
2	HS5802	U of Eu- rope for Appl.Sc.Pots- dam i.Berlin	Privat Univer- sity	Low- High (LH)	1594.8	3520.2	9143.1	12663.4
3	HS7690	Hochschule für Tech- nik und Wirtschaft Berlin	Public Univer- sity	Low- High (LH)	3808.8	4446.4	19321.6	23768.1
4	HS7710	Berliner Hochschule für Tech- nik	Public Univer- sity	Low- High (LH)	1409.9	3023.1	7929.0	10952.1
5	HS7721	HWR Berlin, FB Wirtschaft	Public Univer- sity	High- High (HH)	884.2	5449.5	16643.3	22092.8
6	HS7971	SRH Berlin U of Appl. Sciences in Berlin	Privat Univer- sity	Low- High (LH)	1687.0	1244.6	1773.3	3017.9
7	HS0200	Humboldt- Universität Berlin	Public Univer- sity	Low- High (LH)	1245.5	4352.1	18945.9	23297.9

2210 **4.6.1. Spatial Economics Metrics**

2211 Spatial Economics indicators are important for further calculation of capacity–demand balance and central place
2212 metrics.

2213 **Capacity–Demand Ratio**

2214 This indicator reflects the ratio between the supply of student housing and university demand within each market area.
2215 It is calculated as the share of the total capacity of all student accommodation in the market area relative to the total
2216 demand from international students (`market_area['capacity_demand_ratio'] = !cap_sum / !s_aus_sum`).
2217 High ratio values indicate areas with a relative oversupply, while low values indicate a shortage of housing relative
2218 to demand. This indicator helps to identify areas where the market is potentially balanced or, conversely, under the
2219 pressure.

2220 **Demand–Capacity Pressure**

2221 This indicator is the inverse ratio and is calculated as the share of total university demand relative to student housing
2222 capacity (`market_area['demand_capacity_pressure'] = !s_aus_sum / !cap_sum`). It is interpreted as the
2223 degree of market pressure: the higher the value, the greater the load on the available supply and the more acute the
2224 shortage.

2225 **Number of Student Accommodations**

2226 This indicator determines how many student accommodation facilities are located within a specific market area (`pbsa_counts`
2227 `= wb.groupby('iso_id').size()`).

2228 **Number of Higher Education Universities**

2229 This indicator reflects how many higher education institutions fall within the boundaries of the market area (`uni_counts`
2230 `= wb.groupby('iso_id').size()`).

2231 **Market Area Size**

2232 The area of each market area is calculated as the area of its geometry in square kilometres (`market_area['area']`
2233 `= market_area.geometry.area / 1e6`).

2234 The main result of the analysis of spatial economic metrics is the capacity–demand balance (Figure 33), which
2235 allows for a direct comparison of the aggregate demand of international students within each market area with the
2236 available supply of places in student accommodation.

2237 As can be seen in the graph, in the vast majority of university market areas, university international demand sig-
2238 nificantly exceeds available student accommodation capacity. Almost all points are below the ideal equilibrium line
2239 (45°), indicating a structural supply shortage and systemic pressure on the student housing market. The size of the

points reflects the degree of pressure on a particular university. The smallest pressure is on the market area formed by SRH Berlin University of Applied Sciences, where capacity exceeds demand. The greatest pressure is on the market area formed by the University of Business and Law, where demand exceeds capacity by 6.2 times.

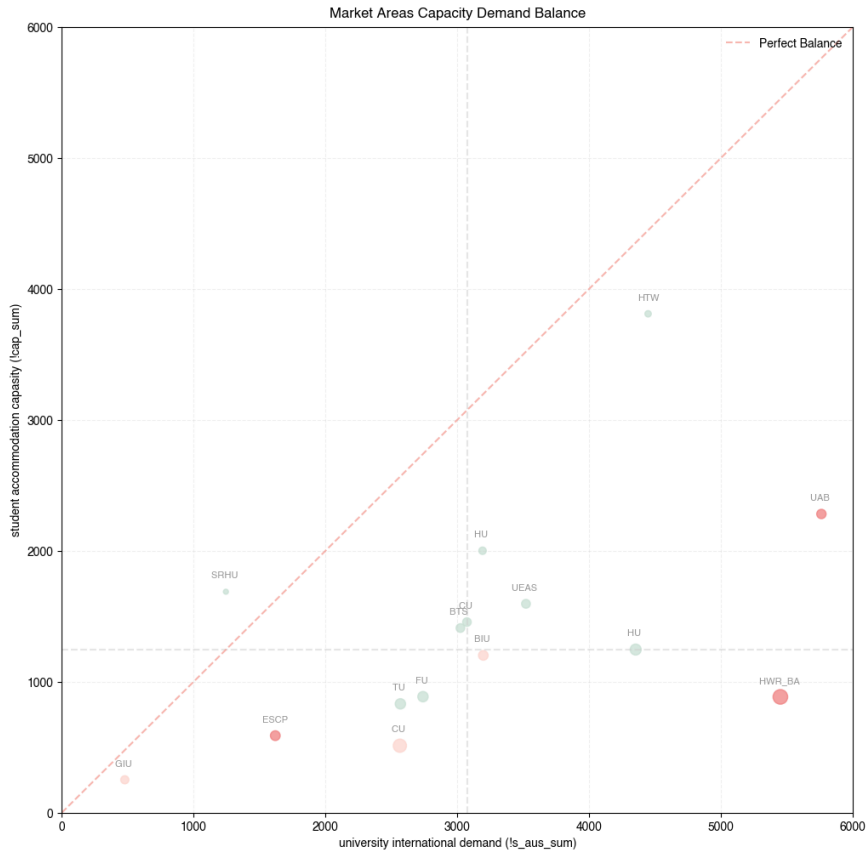


Figure 33. Market Areas Capacity Demand Balance

4.6.2. Central Place Metrics

Centrality

The first challenge was to find some way of measuring the centrality of an urban place (King, 1986). To calculate centrality Godlund (1956) used the following formula:

$$C = \sqrt{(B)(m) - (P)(k)} \quad (5)$$

where:

B — a count of the number of shops in the place m — a weighting factor that took into account the comparative average size of the shops

2252 P — the population of the town
 2253 k — was an index of the average accessibility to retail trade services enjoyed by any one inhabitant of the larger region
 2254 in which the central place was located

2255 In a study of the student housing market, centrality can be determined using a modified version of Godlund's
 2256 formula.

$$C = \sqrt{(Csa + Cuni)(m) - (D)(Ci/Di)} \quad (6)$$

2257 where:

2258 Csa +Cuni — a sum of the number of student accommodations and higher education universities in the market
 2259 area

2260 m — is a constant calculated during manual tests (m=50)

2261 D — the total international demand

2262 Ci/Di — a capacity-demand ratio

2263 m = 50

2264

```
2265 market_area["centrality"] = np.sqrt(((market_area["pbsa_count"] + market_area["uni_count"]) * m)
2266                                     (market_area["!s_aus_sum"] * market_area["capacity_demand_rat
```

2267

```
2268 market_area[["fname", "pbsa_count", "uni_count", "capacity_demand_ratio",
2269               "!s_aus_sum", "centrality"]].head(5).sort_values("centrality", ascending=False)
```

	fname	pbsa_count	uni_count	capacity_demandsratio_sum	centrality	
2	University of Europe for Applied Sciences	42	44	0.453028	3520.246	52.012
13	University of the Arts Berlin	50	48	0.395898	5760.527	51.180
5	Hochschule für Wirtschaft und Recht (Berlin, F...	32	37	0.162263	5449.490	50.653
0	Charité - Univer-sitätsmedizin Berlin	37	38	0.473682	3072.597	47.902
8	Humboldt University of Berlin	44	41	0.626782	3190.825	47.435
12	Berlin International University of Applied Sci...	33	35	0.375696	3196.726	46.894
4	Berlin School of Tech-nology (Technical Univers...	36	34	0.466377	3023.068	45.718
10	Technical University of Berlin	25	18	0.323625	2567.892	36.318
11	ESCP Busi-ness School	17	13	0.363232	1619.219	30.197

11

Nataliia, Design M.Sc., University of Berlin: *Preprint submitted to Elsevier*

Page 113 of 191

Table 14. Centrality Central Place Metrics

The market areas formed by the Humboldt University of Berlin and the University of Applied Sciences for Technology and Economics Berlin have zero centrality (Table 14), which can be explained by their spatial location in Treptow-Köpenick and Lichtenberg, respectively.

Service Range Adequacy

It is based on the key principle of central place theory: service range determines the size of the market territory necessary for the sustainable operation of the facility. As King (1986) notes, the higher the ‘service range’ of a product or service, the larger the area that must provide a sufficient flow of users, and the higher the probability that such a facility will be located in a larger urban centre. Service Range Adequacy (Table 15) allows us to assess how well each territory meets the expected level of service and contributes to the sustainable functioning of the student housing market. It is calculated as the relative share of demand attributable to each market area ($\text{market_area}[\text{“cntr_service_range_adequacy”}] = \text{market_area}[\text{“!s_aus_sum”}] / \text{total_demand_complete}$)

```
total_demand_complete = uni_dem['!s_aus'].sum()
market_area['cntr_service_range'] = (
    market_area['!s_aus_sum'] / total_demand_complete).round(3)

market_area[["fname", "!s_aus_sum", "centrality", "cntr_service_range"]].head(5).sort_values("cntr_service_range")
```

	fname	!s_aus_sum	centrality	cntr_service_range
13	University of the Arts Berlin	5760.527	51.180	0.119
5	Hochschule für Wirtschaft und Recht (Berlin, F...	5449.490	50.653	0.112
3	University of Applied Sciences for Technology ...	4446.418	0.000	0.092
7	Humboldt University of Berlin	4352.098	0.000	0.090
2	University of Europe for Applied Sciences	3520.246	52.012	0.073
8	Humboldt University of Berlin	3190.825	47.435	0.066
12	Berlin International University of Applied Sci...	3196.726	46.894	0.066
0	Charité - Universitätsmedizin Berlin	3072.597	47.902	0.063
4	Berlin School of Technology (Technical Univers...	3023.068	45.718	0.062
9	Free University of Berlin	2739.128	29.397	0.057
1	Charité - Universitätsmedizin Berlin	2563.897	25.269	0.053
10	Technical University of Berlin	2567.892	36.318	0.053
11	ESCP Business School Berlin	1619.219	30.197	0.033

Table 15. Service Range Adequacy Central Place Metrics

Hierarchy Rank

According to King's approach, the hierarchy of urban places is defined not only by their size, but primarily by the number and order of the functions they perform (King, 1986, p. 22).

Following this logic, the ranking of market areas in this study is based on their functional capacity, expressed through two key indicators: centrality and service range adequacy. These values are combined into a composite index ($\text{market_area}[\text{'cntr_hierarchy'}] = ((\text{market_area}[\text{'centrality'}]/100) + (\text{market_ara}[\text{'cntr_service_ran}])$). The hierarchical ranking (Table 16) allows us to see not only the current configuration of supply and demand, but also the structural role of each market area in the urban system, identifying higher-order centres and lower-level areas.

```
market_area['cntr_hierarchy'] = (  
    (market_area['centrality']/100) + (market_area['cntr_service_range'])).round(3)  
  
market_area['cntr_hierarchy_rank'] = pd.qcut(  
    market_area['cntr_hierarchy'],  
    q=5, labels=range(1, 6)    # 5 classes  
)  
  
market_area[["fname", "!s_aus_sum", "centrality", "cntr_service_range",  
    'cntr_hierarchy', 'cntr_hierarchy_rank']].head(5).sort_values("cntr_hierarchy", asce
```

	fname	!s_aus_sum	centrality	cntr_service_rang	cntr_hierarchy	cntr_hierarchy_rank
13	University of the Arts Berlin	5760.527	51.180	0.119	0.631	5
5	Hochschule für Wirtschaft und Recht (Berlin, F...	5449.490	50.653	0.112	0.619	5
2	University of Europe for Applied Sciences	3520.246	52.012	0.073	0.593	5
0	Charité - Univer- sitätsmedizin Berlin	3072.597	47.902	0.063	0.542	4
8	Humboldt University of Berlin	3190.825	47.435	0.066	0.540	4
12	Berlin In- ternational University of Applied Sci...	3196.726	46.894	0.066	0.535	4
4	Berlin School of Tech- nology (Technical Univers...	3023.068	45.718	0.062	0.519	3
10	Technical University of Berlin	2567.892	36.318	0.053	0.416	3
9	Free Univer- sity of Berlin	2739.128	29.397	0.057	0.351	3

Table 16. *Hierarchy Rank Central Place Metrics*

Market Areas Hierarchy Rank demonstrates the distribution of central places by hierarchical classes in accordance with Central Place Theory.

Hierarchy Rank - 5. Primary Central Places

Primary Central Places (Figure 34, Table 17) are formed around the University of the Arts Berlin (UAB), the University of Europe for Applied Sciences (UEAS) and the School of Economics and Law (HWR_BA). Although according to King (1986), centrality is not determined by the physical size of the territory, these market areas have some of the largest areas, more than 90 km² each. This is because top-level zones must have a sufficiently wide coverage to serve a large number of functions and significant volumes of demand.

Geographically, these market areas are located mainly in the districts of Mitte, Charlottenburg-Wilmersdorf and Friedrichshain-Kreuzberg, as well as partly in Tempelhof-Schöneberg, Steglitz-Zehlendorf and Neukölln, forming a distinct axis in a south-westerly direction. Together, they account for about 32% of total international student demand, which highlights their systemic role in the city's university infrastructure.

It is worth noting that two of the three primary centres belong to the High-High cluster, meaning they are stable magnets for international students, around which areas of high demand and growing market pressure are forming. However, the greatest pressure is observed in the School of Economics and Law market zone, where demand exceeds available capacity by 6.2 times, making this zone the most vulnerable in the hierarchy of central locations. This imbalance highlights the critical shortage of infrastructure and the need to prioritise the expansion of the housing stock near this educational cluster.

Together, the primary central places form the core of Berlin's student urbanisation, where key educational functions, large volumes of demand and the most significant infrastructure imbalances are concentrated. Their high hierarchical position confirms the central role of these areas in the urban structure, but at the same time demonstrates the vulnerability of the system: even higher-order centres face a significant housing shortage and growing market pressure.

Hierarchy Rank - 5. Primary Central Places

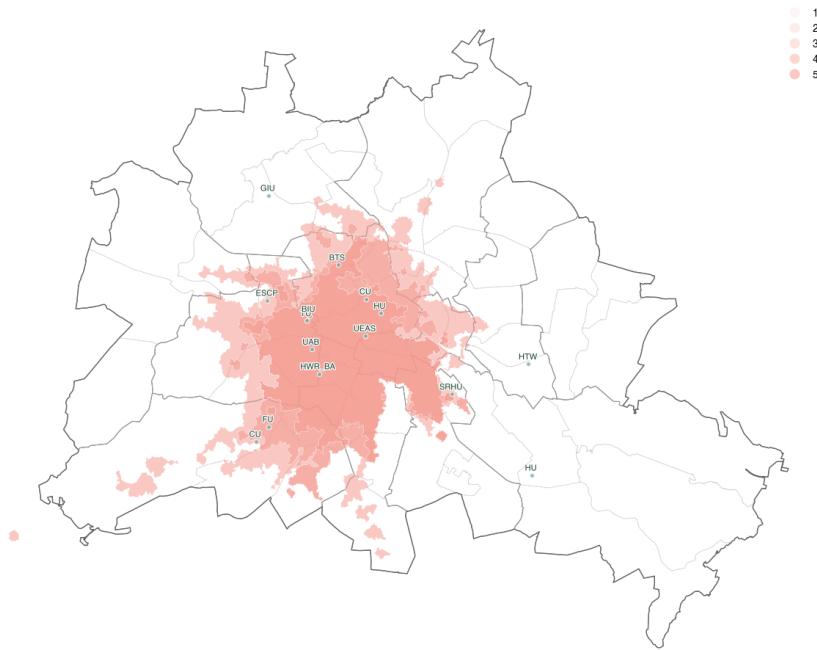


Figure 34. Primary Central Places

	fname	!s_aus_sum	centrality	cntr_service_range	cntr_hierarchy	cntr_hierarchy_rank
13	University of the Arts Berlin	5760.527	51.180	0.119	0.631	5
5	Hochschule für Wirtschaft und Recht (Berlin, F...	5449.490	50.653	0.112	0.619	5
2	University of Europe for Applied Sciences	3520.246	52.012	0.073	0.593	5

Table 17. Primary Central Places

Hierarchy Rank - 4. Secondary Central Places

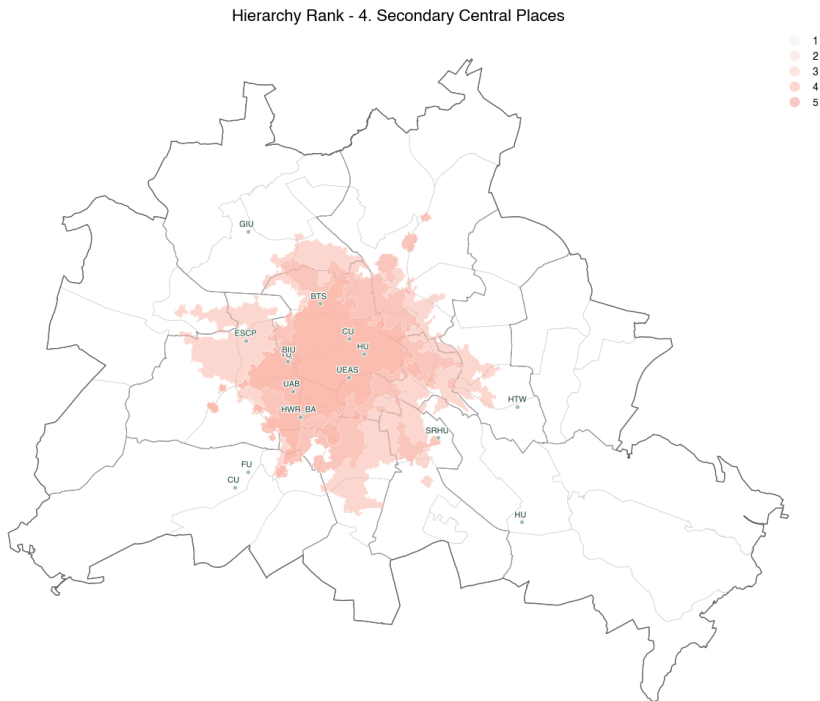
Secondary Central Places Centres (Figure 35, Table 18) are formed by Humboldt University of (HU), International

2338 University of Applied Sciences (BIU) and Charité University Hospital (CU). These market areas are secondary central
2339 places, i.e., high-order centres that have significant functional saturation but are lower than primary centres in terms
2340 of scale of attraction and infrastructure concentration.

2341 Geographically, secondary central places are located east of primary central places, covering parts of the Lichten-
2342 berg and Pankow districts and forming a distinct west-east axis in the spatial structure of the student housing market.
2343 This configuration reflects the expansion of the city's education system in an easterly direction, which is historically
2344 less saturated with higher education institutions. Together, these areas account for about 20% of the total demand from
2345 international students, confirming their significant but not dominant role in the urban hierarchy.

2346 The average demand–capacity pressure value of 2.1 confirms that demand exceeds current supply by more than
2347 double. This creates moderate but stable market pressure, indicating the need for strategic expansion of student infras-
2348 tructure and further development of public-private initiatives in the field of student accommodation.

2349 Thus, secondary central places function as sub-regional centres in the urban hierarchy, playing a key role in re-
2350 balancing demand, ensuring access to higher education and shaping future growth trends in Berlin's student housing
2351 market.



2352

2353 *Figure 35. Secondary Central Places*

	fname	!	s_aus_sum	centrality	cntr_service_rang	cntr_hierarchy	cntr_hierarchy_rank
0	Charité Univer- sitätsmedizin Berlin	-	3072.597	47.902	0.063	0.542	4
8	Humboldt University of Berlin		3190.825	47.435	0.066	0.540	4
12	Berlin In- ternational University of Applied Sci...		3196.726	46.894	0.066	0.535	4

Table 18. Secondary Central Places

Hierarchy Rank - 3. Tertiary Central Places

Tertiary Central Places (Figure 36, Table 19) are formed around the Technical University of Berlin (TUB), Free University (FU) and Berlin School of Technology (BTS). These universities represent significant but less dominant centres of demand, located on the periphery of the primary and secondary central places.

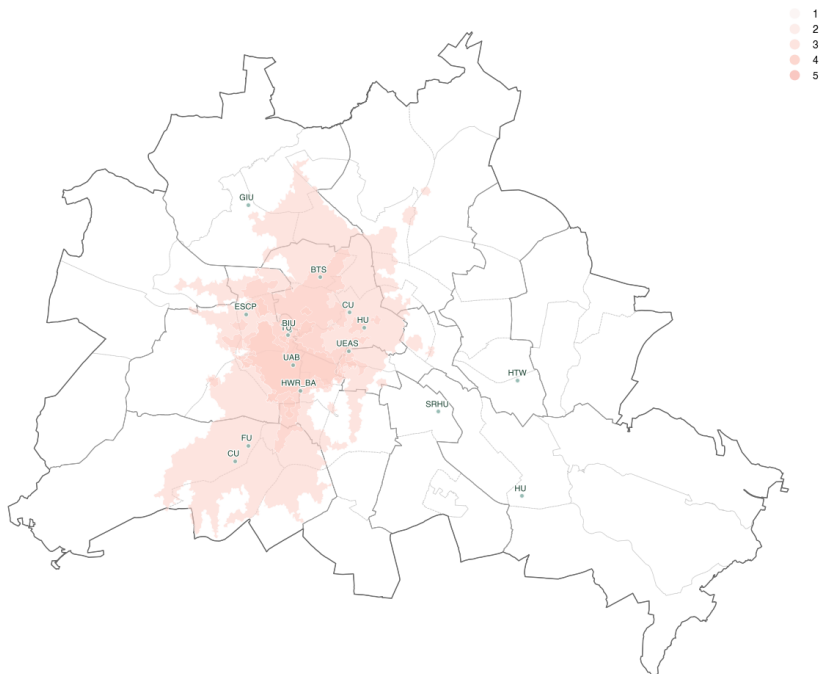
Geographically, tertiary central places occupy areas to the north and south of secondary central places, covering a significant part of Steglitz-Zehlendorf and Reinickendorf. In this way, they support the established south-western educational axis extending from the primary central places and, at the same time, form a new north-south axis in the spatial structure of the student housing market. This configuration reflects the complex multipolar organisation of demand characteristic of a large polycentric city.

Together, these market areas account for about 17% of total international student demand, making them significant but not dominant elements in the structure of central places.

It is noteworthy that all three universities forming tertiary central places belong to the Low–High (LH) cluster and are public. This combination indicates the high educational potential of these areas and their ability to be included in existing clusters, but at the same time demonstrates the lack of full strategic integration into established centres of student urbanisation.

The average demand–capacity pressure value is 2.8, which is higher than that of secondary central places. This indicates that demand is almost three times higher than the available supply. Consequently, tertiary central places are critically important areas of hidden deficit, where a growing gap is forming between the educational potential of

2374 universities and the lack of adequate housing infrastructure.
Hierarchy Rank - 3. Tertiary Central Places



2375

2376 *Figure 36. Tertiary Central Places*

2377

	fname	!s_aus_sum	centrality	cntr_service_range	cntr_hierarchy	cntr_hierarchy_rank
4	Berlin School of Technology (Technical Univers...	3023.068	45.718	0.062	0.519	3
10	Technical University of Berlin	2567.892	36.318	0.053	0.416	3
9	Free University of Berlin	2739.128	29.397	0.057	0.351	3

2378

2379

2380

2381

Table 19. Tertiary Central Places

Hierarchy Rank - 2. Local Central Places

Local Central Places (Figure 37, Table 20) are formed around SRH Berlin University of Applied Sciences (SRHU), Charité University Hospital (CU) and ESCP Business School Berlin (ESCP). These universities act as local centres,

2382 primarily serving the surrounding areas and acting as a link between larger secondary and tertiary central places.

2383 Geographically, these market areas are mainly located in the districts of Friedrichshain-Kreuzberg and Neukölln,
2384 as well as partly in Mitte, Charlottenburg-Wilmersdorf, Tempelhof-Schöneberg, Treptow-Köpenick and Lichtenberg.
2385 Spatially, they occupy positions south of secondary central places and simultaneously support two key axes of the
2386 urban structure: the west-east axis formed by secondary central places and the north-south axis formed by tertiary
2387 central places. In this way, local central places reinforce the polycentric organisation of the student housing market,
2388 ensuring connectivity between areas of different hierarchical levels and serving 11% of student demand.

2389 What is particularly interesting is that the three universities that form this hierarchical level belong to different types
2390 of spatial clusters: ESCP Business School Berlin is characterised by a High-High (HH) pattern, Charité University
2391 Hospital by a High-Low (HL) pattern, and SRH Berlin University of Applied Sciences by a Low-High (LH) pattern.
2392 This highlights the heterogeneity of these areas and their varying degrees of integration into existing educational
2393 clusters.

2394 The intensity of market pressure within local central places also varies. Market areas formed by private universities
2395 ESCP and SRHU show relatively moderate indicators: 2.7 and 0.7, respectively. At the same time, the area formed
2396 by the state-owned Charité University Hospital experiences a pronounced imbalance between supply and demand:
2397 demand exceeds available capacity by five times. This indicates a structural shortage of student housing in one of the
2398 city's most important medical and scientific centres.

2399 Thus, local central places reflect an important intermediate level of the city's educational and housing structure.
2400 They combine areas with stable patterns of demand and supply, adjacent to local shortages, and form a significant
2401 component of Berlin's polycentric student space.

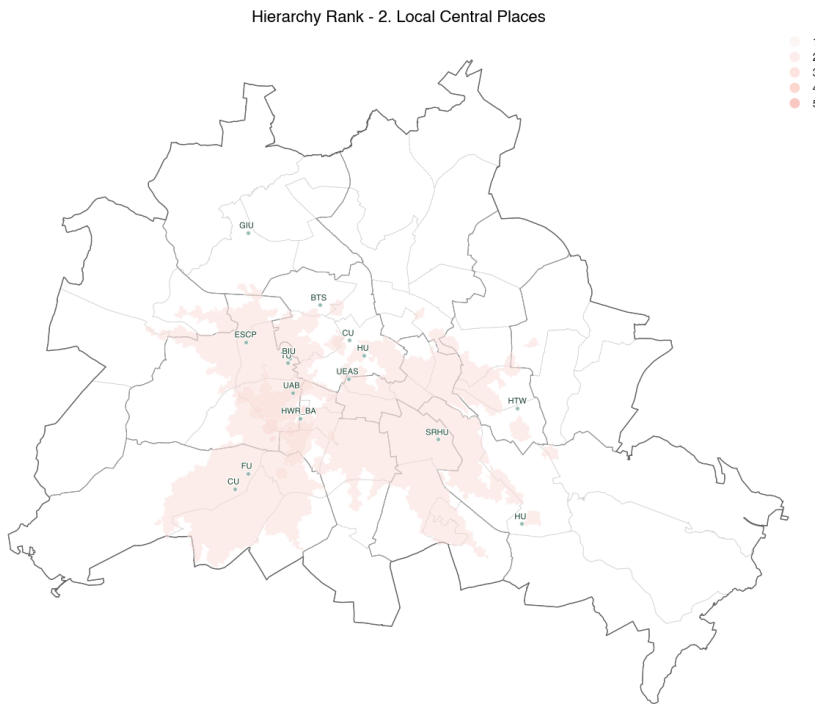


Figure 37. Local Central Places

Table 20. Local Central Places

Hierarchy Rank - 1. Basic Service Areas

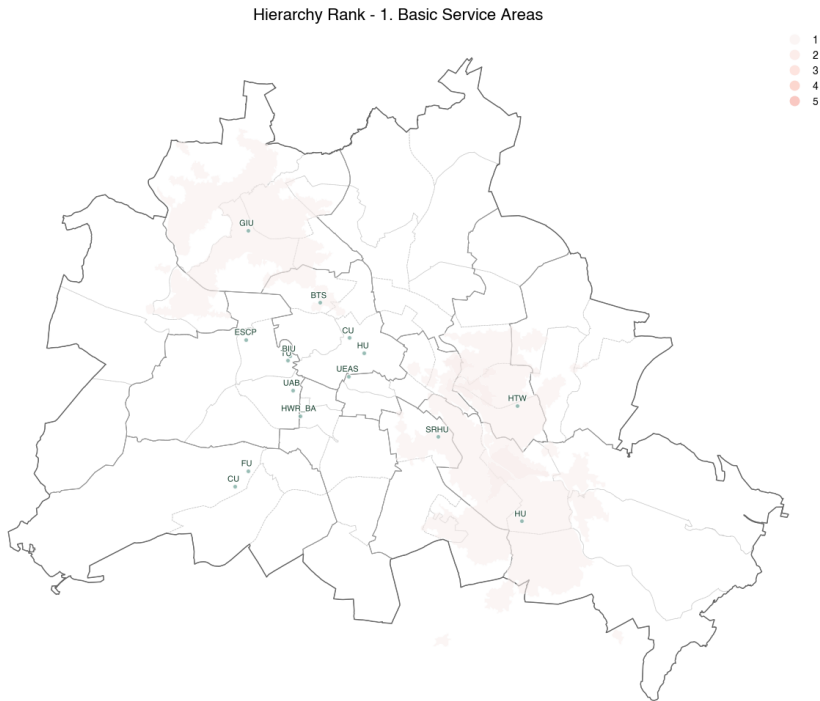
Basic Service Areas (Figure 3, Table 21) are formed around the German International University Berlin (GIU), University of Applied Sciences for Technology, Economics Berlin (HTW) and Humboldt University of Berlin (HU). These universities are basic service centres whose role in the structure of central locations is primarily to provide fundamental educational functions and support peripheral parts of the urban system.

Geographically, these market areas are located mainly in the districts of Reinickendorf, Treptow-Köpenick and Lichtenberg. They are situated to the south-east and north-west of the secondary central places. This location emphasises their functional role as areas serving peripheral or poorly integrated parts of the city's educational space, acting as 'edge' centres in Berlin's polycentric structure.

Collectively, Basic Service Areas account for about 19% of international student demand — more than tertiary central places. This indicates that even the most basic centres in the hierarchy play a significant role in the distribution of student mobility and the formation of market pressure, serving significant shares of both educational and housing demand in the periphery.

It is worth noting that two of the three universities forming this hierarchical level belong to the Low-High (LH) cluster, demonstrating low supply with higher regional demand. The third university is private and belongs to the High-

2420 Low (HL) cluster, reflecting the opposite dynamic: a high value at the facility level in an environment with relatively
2421 low values. This distribution of clusters shows that the basic market areas are characterised by strong heterogeneity in
2422 demand and supply, as well as varying levels of integration into existing educational clusters.



2423
2424 *Figure 38. Basic Service Areas*

2425 *Table 21. Basic Service Areas*

2426 The constructed hierarchy of market areas revealed a pronounced polycentricity of the city’s educational space:
2427 central locations at the highest levels concentrate the greatest student demand and form the main clusters of attraction,
2428 while lower-level zones perform supporting functions. At the same time, demand significantly exceeds supply at all
2429 levels.

2430 Thus, the analysis shows that the student housing crisis in Berlin has a clearly defined spatial structure. The uneven
2431 distribution of both universities and accommodation infrastructure exacerbates market pressure, creating persistent
2432 areas of overload and local shortages, which require further strategic intervention and planning.

2433 `market_area.to_crs(epsg=25833)`

2434

2435 `ef = r"../outputs/3.2_central_place_analysis/market_area_central_place_analysis.gpkg"`

2436 `market_area.to_file(ef, driver="GPKG")`

4.7. 3.3 Market Pressure Analysis

In the chapter Central Place Analysis, spatial economics metrics were calculated, including demand–capacity pressure for each market area. The task of the chapter Market Pressure Analysis is to move from numerical characteristics to spatial ones and to calculate the local pressure on the student housing market and particular student accommodations.

4.7.1. Set Up

Import Libraries

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import geopandas as gpd
from matplotlib.colors import LinearSegmentedColormap
import seaborn as sns
import matplotlib.path as path_effects
from matplotlib.colors import ListedColormap
from shapely.ops import unary_union
import rasterio
from rasterio import features
from affine import Affine
from scipy.ndimage import gaussian_filter

Berlin Spatial Boundaries

pf = r"../inputs/3.3_market_pressure_analysis/lor_pgr.gpkg"
pgr = gpd.read_file(pf)
pgr.replace({'-': np.nan, 0: np.nan, '' : np.nan}, inplace=True)

bf = r"../inputs/3.3_market_pressure_analysis/adm_bz.gpkg"
bz = gpd.read_file(bf)
bz.replace({'-': np.nan, 0: np.nan, '' : np.nan}, inplace=True)

of = r"../inputs/3.3_market_pressure_analysis/adm_out.gpkg"
out = gpd.read_file(of)
```

```
2467 out.replace({'-': np.nan, 0: np.nan, '' : np.nan}, inplace=True)
```

```
2468     Data Import
```

```
2469 ff = r"../inputs/3.3_market_pressure_analysis/market_area_central_place_analysis.gpkg"
```

```
2470 market_area = gpd.read_file(ff)
```

```
2471
```

```
2472 cf = r"../inputs/3.3_market_pressure_analysis/central_places.gpkg"
```

```
2473 central_places = gpd.read_file(cf)
```

```
2474
```

```
2475 pf = r"../inputs/3.3_market_pressure_analysis/pbsa_capacity_analysis.gpkg"
```

```
2476 pbsa_cap = gpd.read_file(pf)
```

2477 **4.8. Overlap Problem**

2478 Whereas in the original Central Place, theory market areas do not overlap, in order to analyse the aggregate market
2479 pressure on the student housing market, it is necessary to move from the overlapping market areas obtained as a result
2480 of the Central Place Analysis (Figure 39) to a more accurate local structure that reflects the aggregate pressure on a
2481 particular part of the city.

```
2482 market_area['demand_capacity_pressure_rank'] = pd.qcut(  
2483     market_area['demand_capacity_pressure'],  
2484     q=5, labels=range(1, 6)    # 15 classes  
2485 )
```

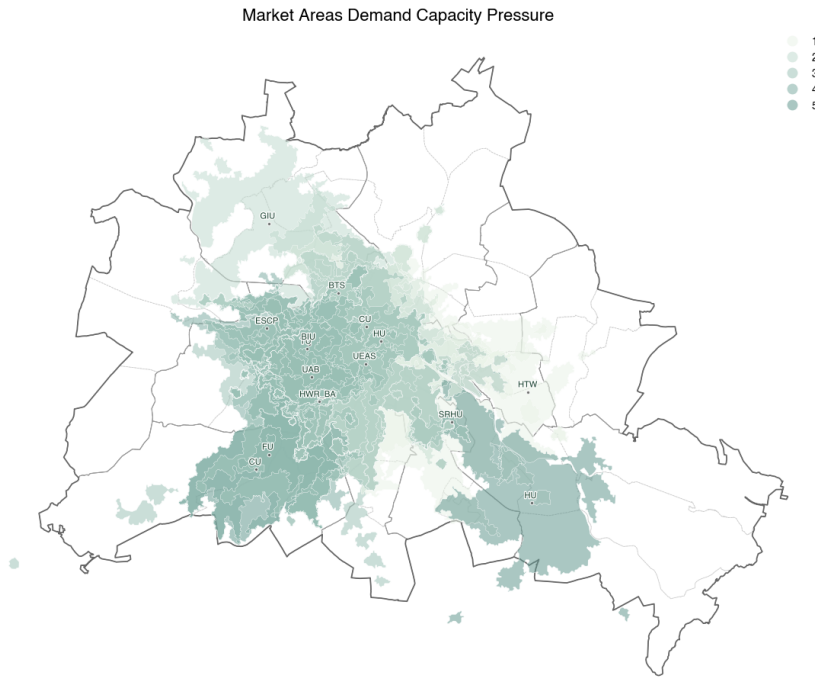


Figure 39. Market Areas Demand Capacity Pressure Rank

Market Areas Demand Capacity Pressure Rank (Figure XX) illustrates that each university forms its own accessibility zone, and these zones naturally overlap. On the one hand, this overlap reflects the real competition for student flows for the same housing stock. On the other hand, it creates analytical complexity: some territories are part of several market areas at the same time, which complicates the interpretation of local pressure.

This is partly due to the fact that, initially, isochrones were calculated using multipolygons, which form complex geometries and consist of several polygon parts. These fragments overlap with each other and with the polygons of other universities.

According to the results of spatial analysis, individual market areas of overlap and are part of at least five different market areas, while particularly central areas, such as parts of the Mitte district, are part of 14 market areas simultaneously. In order to simplify such a dense network of intersecting zones of influence, it is necessary to eliminate overlaps.

```
market_area['er_overlap'] = 0
```

```
# Calculate overlaps
```

```
for i, current_poly in market_area.iterrows():
```

```
    overlap_count = 0
```

```
    for j, other_poly in market_area.iterrows():
```

```

2505         if i != j: # Don't compare with itself
2506             if current_poly.geometry.intersects(other_poly.geometry):
2507                 overlap_count += 1
2508         market_area.at[i, 'er_overlap'] = overlap_count
2509
2510 print(f"Overlap Counts Range for Market Areas: {market_area['er_overlap'].min()} to {market_area[
2511
2512 ff = r'../outputs/3.3_market_pressure_analysis/market_areas_not_explode_overlap.gpkg'
2513 market_area.to_file(ff, driver="GPKG")
2514
2515 Overlap Counts Range for Market Areas: 5 to 14
2516
2517 One way to simplify geometry is to divide complex multipolygon geometry into simple polygons market_area.explode(ig
2518 In this case, polygons intersect between 0 and 51 times.
2519
2520 market_area_exp = market_area.explode(ignore_index=True)
2521
2522 market_area_exp['er_overlap'] = 0
2523
2524 for i, current_poly in market_area_exp.iterrows():
2525     overlap_count = 0
2526     for j, other_poly in market_area_exp.iterrows():
2527         if i != j:
2528             if current_poly.geometry.intersects(other_poly.geometry):
2529                 overlap_count += 1
2530         market_area_exp.at[i, 'er_overlap'] = overlap_count
2531
2532 print(f"Overlap Counts Range for Exploded Market Areas: {market_area_exp['er_overlap'].min()} to
2533
2534 Overlap Counts Range for Exploded Market Areas: 0 to 51
2535
2536 This means that in order to switch to simple geometry, it will be necessary to apply the division 51 more times.
2537 Such a task is not only difficult in terms of computing power, but will also result in the resulting non-intersecting
2538 polygons being extremely small.

```

4.9. Rasterisation

Another way to eliminate the problem of multiple market area overlaps is to switch from vector geometry to raster representation, i.e., to a regular grid of fixed size. Rasterisation allows complex, overlapping polygons to be replaced with a set of uniform pixel cells. Unlike a vector model, where a single territory can belong to several market areas at once, a raster grid forces a unique spatial correspondence: one cell equals one pressure value.

The use of a regular 25-meter grid provides a sufficient level of spatial detail to identify micro-zones of local congestion, especially in areas where the density of universities and student accommodation is highest.

```
value_col = "demand_capacity_pressure"

# Define raster resolution (in meters)
resolution = 25

# Bounding box
minx, miny, maxx, maxy = market_area.total_bounds

# Create transform (affine)
transform = Affine.translation(minx, maxy) * Affine.scale(resolution, -resolution)

# Calculate raster shape
width = int((maxx - minx) / resolution)
height = int((maxy - miny) / resolution)

raster = np.zeros((height, width), dtype=np.float32)

# For each polygon, add pressure values
for _, row in market_area.iterrows():
    shapes = [(row.geometry, row[value_col])]
    layer = features.rasterize(
        shapes=shapes,
        out_shape=(height, width),
        transform=transform,
        fill=0,
```

```
2565         all_touched=True,
2566         dtype=np.float32
2567     )
2568     raster += layer
```

2569 **4.10. Demand Capacity Pressure**

2570 Thus, rasterisation can be used to assess Market Areas Local Demand Capacity Pressure.

2571 The Local Demand–Capacity Pressure (Figure 40) shows the spatial distribution of the total market pressure arising
2572 from the overlap of the catchment areas of all universities. Unlike capacity–demand pressure, which is calculated at the
2573 level of individual market areas, this visualisation shows the ‘accumulated’ pressure, reflecting the cumulative effect
2574 of student competition for housing resources at each point in the urban area.

2575 The most intense pressure zones are marked in saturated red shades and are concentrated in the central part of the
2576 city — primarily in the districts of Mitte, Charlottenburg-Wilmersdorf, and Friedrichshain-Kreuzberg. This core coin-
2577 cides with the spatial location of the largest educational clusters: the University of the Arts Berlin (UAB), University
2578 of Europe for Applied Sciences (UEAS), Berlin International University (BIU), Berlin School of Technology (BTS),
2579 and School of Economics and Law (HWR_BA). The catchment areas of these universities overlap most densely, which
2580 greatly increases the pressure on the limited housing stock in the central districts. It is here that local pressure reaches
2581 its maximum values — over 30, which indicates extremely high competition among international students for available
2582 places.

2583 As move away from the central core, the intensity of the pressure gradually decreases, transitioning into moderate
2584 greenish hues. However, the decrease is asymmetrical: in the south-west (Steglitz-Zehlendorf), south-east (Treptow-
2585 Köpenick), and partly in the north-east (Pankow), there are still local areas of increased local pressure caused by the
2586 influence of the large universities FU, HU, and HTW. At the same time, the districts of Reinickendorf, Spandau, and
2587 Marzahn-Hellersdorf show the lowest local pressure values, indicating weak involvement in competition for housing
2588 stock and relatively low attractiveness for students.

2589 Thus, the local demand–capacity pressure map clearly shows that market tension in the student housing market in
2590 Berlin is concentrated and polycentric, reaching its peak in the central ‘education corridor’ and gradually weakening
2591 towards the periphery.

2592 Importantly, this visualisation emphasises that the pressure is not generated by individual market areas, but by the
2593 overlap of multiple accessibility zones, which explains why it is the concentration of university functions — and not
2594 just the local shortage of student accommodation — that is the key driver of market pressure in the central areas of the
2595 city.

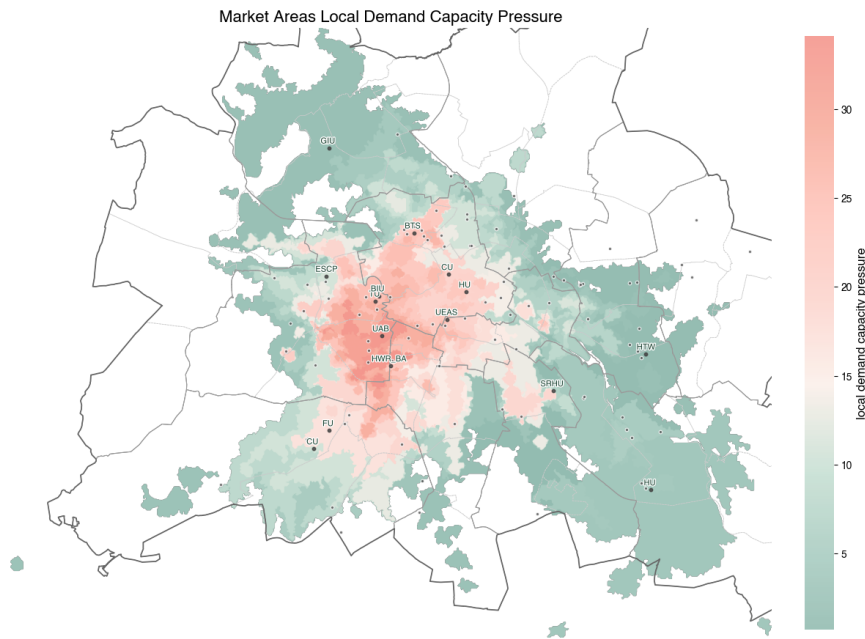


Figure 40. Market Areas Local Demand Capacity Pressure

```
out_raw = r'../outputs/3.3_market_pressure_analysis/market_pressure_sum_raw.tif'
```

4.11. Student Accommodation Pressure

Data Import

```
out_raw = r'../outputs/3.3_market_pressure_analysis/market_pressure_sum_raw.tif'
```

```
with rasterio.open(out_raw) as src:
```

```
    raster = src.read(1) # read first band as numpy array
```

```
    transform = src.transform
```

```
    crs = src.crs
```

```
ef = r"../inputs/3.3_market_pressure_analysis/pbsa_capacity_analysis.gpkg"
```

```
pbsa_cap = gpd.read_file(ef)
```

```
pbsa_cap.drop(columns=['link', '!cap_slag', '!cap_std', '!cap_slag_std', 'cluster', 'model', 'col
```

```
coords = [(geom.x, geom.y) for geom in pbsa_cap.geometry]
```

```
with rasterio.open(out_raw) as src:
```

```
    pbsa_cap["local_demand_capacity_pressure"] = [val[0] for val in src.sample(coords)]
```

2615

```
2616 pbsa_cap["local_demand_capacity_pressure"] = pbsa_cap["local_demand_capacity_pressure"].round(3)
```

2617 Student accommodations with the lowest level of local market pressure (Table 22) form a separate and segment
2618 of the Berlin student housing market. This group primarily includes properties located in the eastern districts of the
2619 city — Pankow, Lichtenberg and Friedrichsfelde — and are mainly state-owned (Berlinovo and Studierendenwerk
2620 Berlin). The lowest local demand–capacity pressure values (≈ 0.74 – 1.17) are recorded, for example, at the Walter-
2621 Friedländer-Straße (Pankow) and Storckower Straße (Lichtenberg) halls of residence, which means that supply here is
2622 either comparable to demand or even exceeds it.

```
2623 pbsa_cap[pbsa_cap["local_demand_capacity_pressure"] > 0][[  
2624     "name",  
2625     "plz",  
2626     "ownership",  
2627     "!cap",  
2628     "ownership_model",  
2629     "cluster_cap",  
2630     "local_demand_capacity_pressure",  
2631 ]].sort_values("local_demand_capacity_pressure").head(10)
```

	name	plz	ownership	!cap	ownership_model	cluster_cap	local_demand_capacity
86	Walter-Friedländer-Straße Pankow	10249	Berlinovo	285.0	State-Owned	HL	0.738
37	Storckower Straße Lichten- enberg I	10369	Berlinovo	141.0	State-Owned	HL	0.738
31	Rhinstraße Lichten- berg	10315	Berlinovo	800.0	State-Owned	HH	1.167
55	WH Aris- totelessteig	10318	Studierendenwerk Berlin	913.0	State-Owned	HH	1.167
54	WH Allee der Kos- monauten	10315	Studierendenwerk Berlin	298.0	State-Owned	HH	1.167
42	Studentenwohn- Alfred- Jung- Straße	10319	Home in Berlin GmbH	800.0	Private Owned	HH	1.167
8	Elfriede- Tygör- Straße	10319	Berlinovo	202.0	State-Owned	HL	1.167
11	Friedenhorster Straße Friedrichs- felde	10319	Berlinovo	138.0	State-Owned	HL	1.167
80	WH Sewanstraße	10319	Studierendenwerk Berlin	760.0	State-Owned	HH	1.167
83	WH Storckower Straße	10367	Studierendenwerk Berlin	598.0	State-Owned	HH	1.595

Table 22. *Less Pressured Student Accommodations*

Low market pressure in these locations is explained by a combination of several factors. Firstly, they are located on the periphery or semi-periphery of key university centres and educational clusters, which reduces competition from students focused on central areas. Secondly, these dormitories tend to have a relatively large capacity (up to 800 places), which helps to smooth out local peaks in demand. Thirdly, the state management model is focused not on maximising profits, but on providing stable and affordable housing, which reduces selectivity in occupancy and lessens structural pressure.

It is important to emphasise that low demand–capacity pressure in this case does not mean that there are no problems, but rather indicates a spatial imbalance in demand: students are willing to accept less central and less prestigious locations in exchange for affordability and relative stability. Thus, halls of residence with minimal pressure perform a compensatory function in Berlin’s student housing system, redistributing demand from congested central areas to the eastern and peripheral areas of the city.

Student accommodation with the highest level of local market pressure (Table 23) forms the opposite pole of the Berlin student housing market and clearly demonstrates the depth of the structural deficit in the central areas of the city. Extremely high local demand–capacity pressure values (from ~27 to over 34) were recorded for properties located mainly in Charlottenburg-Wilmersdorf, Schöneberg, Tiergarten and Mitte — areas with a high concentration of universities, private educational institutions and international student demand.

```
pbsa_cap[pbsa_cap["local_demand_capacity_pressure"] > 0][[  
    "name",  
    "plz",  
    "ownership",  
    "!cap",  
    "ownership_model",  
    "cluster_cap",  
    "local_demand_capacity_pressure",  
]].sort_values("local_demand_capacity_pressure", ascending=False).head(10)
```

	name	plz	ownership	!cap	ownership_model	cluster_cap	local_demand_capacity
2	Am Mühlenberg	10825	Gewobag	158.0	State-Owned	LL	34.094002
15	Insider Apartments Berliner Straße	10715	Insider Apartment Community Living GmbH	200.0	Private-Owned	LL	33.355999
63	WH Seldorfer Straße	10719	Studierendenwerk Berlin	72.0	State-Owned	LL	31.245001
78	WH Pfalzburgerstraße	10717	Studierendenwerk Berlin	28.0	State-Owned	LL	31.245001
62	WH Derfflingerstraße (Haus Unger)	10785	Studierendenwerk Berlin	143.0	State-Owned	LL	28.344000
81	WH Siegmunds Hof	10555	Studierendenwerk Berlin	59.0	State-Owned	LH	28.344000
17	Insider Apartments Leibniz Straße	10625	Insider Apartment Community Living GmbH	200.0	Private-Owned	LL	28.344000
76	WH Nollendorfstraße	10777	Studierendenwerk Berlin	183.0	State-Owned	LL	28.344000
70	WH Hardenbergstraße	10623	Studierendenwerk Berlin	46.0	State-Owned	LL	28.344000
47	Studentenwohnheim Walter-May-Haus	10315	Home in Berlin GmbH	276.0	Private-Owned	NS	27.153999

Table 23. Most Pressured Student Accommodations

The greatest pressure is observed (Figure 41) in the Am Mühlenberg (Schöneberg) and Insider Apartments Berliner Straße halls of residence, where demand exceeds available capacity by more than 30 times. It is noteworthy that most of the properties in this group belong to the Low–Low (LL) cluster, i.e. they are characterised by both low local capacity and extremely high demand. In some cases, the capacity of such halls of residence is only 28–70 places (e.g. WH Pfalzburgerstraße or WH Hardenbergstraße), which makes them particularly vulnerable to overcrowding in conditions of high spatial competition.

It is important to note that high market pressure is observed in both private and public halls of residence, including those run by Studierendenwerk Berlin. This indicates that the reason for overcrowding lies not so much in the form of ownership as in the spatial location and integration into dense university clusters. Even relatively large halls of residence, such as WH Siegmunds Hof (590 places), are under extreme pressure due to their overlap with several university catchment areas and the concentration of international demand.

Thus, student residences with the highest level of demand–capacity pressure serve as indicators of bottlenecks in Berlin’s student housing market. They are located in the most prestigious and functionally rich areas of the city, where educational centrality is combined with limited opportunities for expanding supply. These areas form the core of the crisis, where the housing shortage is becoming chronic and cannot be compensated for by local increases in capacity without systemic urban planning and housing policy interventions.

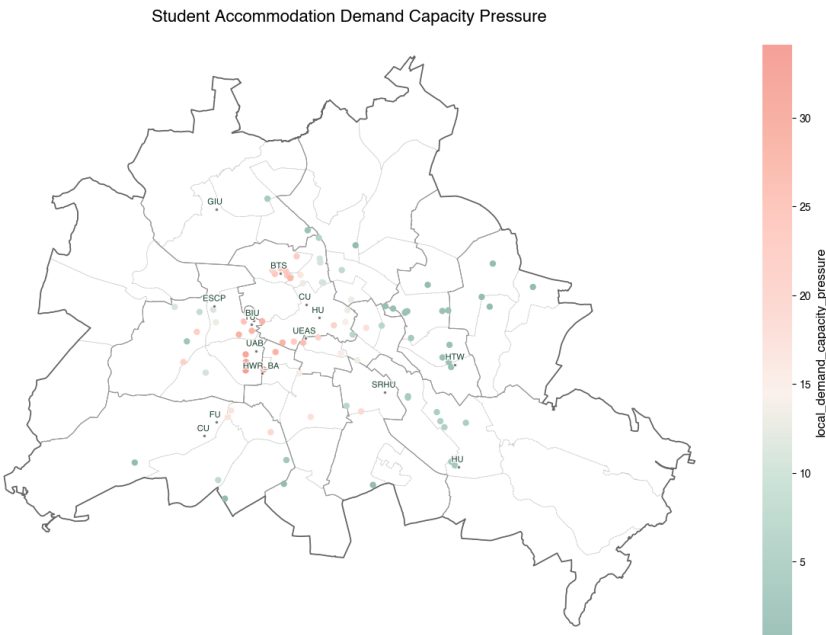


Figure 41. Student Accommodation Demand Capacity Pressure

5.1. 4.1 Overview

While the previous chapters described and examined the components of student housing and the market relationships between them, this chapter focuses on the users of student accommodations.

5.1.1. *What? Students Behind the Numbers*

Behind the student housing crisis numbers there are actual users. In order to rethink the housing crisis, it is necessary not only to assess the student housing market, but also to address users—customers—directly.

The first attempt to understand users was made in the chapter on defining demand. It is worth remembering that for this research, the target group is international students enrolled in Berlin's higher education institutions, at any level, regardless of current residency. However, international students themselves do not represent a homogeneous group, but rather a multitude of disparate qualitative groups that differ from each other in terms of housing needs and preferences.

This chapter helps to get to know international students as users in more detail.

5.1.2. *Why? Bridge Demand and Supply*

The aim of this chapter is to define users and assess user experience by answering the following research questions:

Who are the users of student accommodations?

What experience they have in ongoing housing crisis? What are common needs, preferences, difficulties and problems?

The answers to these questions can be used in developing strategies to improve the situation on the student housing market and bridge demand and supply.

5.1.3. *How? Behavioral Studies*

This chapter was written in collaboration with Joseph Kampfmann as part of the preparations for the ENHANCE workshop Perspectives on the European Student Accommodation Crisis, which was organised by Hannah Berner and Jesco Lippke.

Surveys, Questionnaires, Interviews

Surveys, questionnaires, and interviews are traditional behavioral study methods widely used to collect data on people's opinions, attitudes, and behavior. To improve the accuracy and relevance of the results, it is important not to generalise students but to divide them into qualitative groups. This takes into account the diversity of their needs and preferences, which contributes to a deeper understanding of the phenomena under study.

Storytelling & Narrative Analysis

Storytelling & Narrative Analysis can be seen as an alternative method of behavioral studies that allows the explo-

2710 ration of participants' personal stories and experiences, revealing the emotional and social aspects of their behavior
2711 (Riessman, 2008). Although this method differs from an overall data-driven approach, it makes it difficult to automate
2712 data analysis. Nevertheless, individual experiences are important for understanding hidden barriers to finding student
2713 housing, cultural sensitivity, and adaptation in a new city.

2714 Future-proof urban solutions require collaborative planning, where city authorities, public and private investors,
2715 and citizens act together.

5.2. 4.2 Student Needs and Preferences

As part of behavioural studies, an online survey of student dormitory users was conducted, in which more than 100 people took part.

The survey of Berlin students partially reproduces and refines the study by Von Wittenhorst Zu Sonsfeld and Beusker conducted in 2024 in the German city of Aachen.

As part of the study, the survey was published in thematic groups on housing search, and was also distributed in the form of QR codes on the campus of the Technical University of Berlin (TU) and Studierendenwerk Berlin accommodations.

Affordability Theory was chosen as the theoretical basis for behavioural studies, which assumes that affordability is not limited to financial indicators, but also includes education, transport and social indicators.

5.2.1. Set Up

Import Libraries

```
import pandas as pd
import geopandas as gpd
import chardet
import gspread
import seaborn as sns
import matplotlib.pyplot as plt
import textwrap
from pywaffle import Waffle
import numpy as np
from matplotlib.patches import Patch
from math import pi
```

5.2.2. Data Mining

The input data for the analysis was tabular data downloaded from Google Forms responses.

```
fp = r"../inputs/4.2_students_needs/student_experience_090126.xlsx"

df = pd.read_excel(fp)
```

The survey consists of 34 questions, designed to take a total of 10-15 minutes to complete, and consists of three thematic sections:

Personal Data

The first section is aimed at collecting basic socio-demographic and contextual information necessary to understand the profile of respondents and correctly interpret behavioural patterns in the student housing market.

Socio-demographic characteristics

- gender
- age group
- origin (origin)
- study status (stat)

These questions allow us to analyse whether housing problems vary between different social groups and help to determine whether international students face a higher degree of market pressure than local students.

Financial and housing information

- residential address (adr)
- postcode (plz)
- current rent (price),
- sources of funding (fin)
- housing model (model)

These questions provide insight into budget constraints and types of accommodation.

This block forms the basis for analysis and helps answer the question:

Who are the users of student accommodation?

Needs and Preferences

The second section is devoted to identifying the factors that determine the choice of housing, subjective assessment of needs and living conditions. The structure of the section reflects the key parameters of housing affordability, location choice, mobility behaviour and social environment.

Affordability The questions assess the financial component of current housing:

- monthly rent within budget (a_price) — whether the rent is in proportion to personal budget.
- price matches apartment condition (a_cond) — subjective perception of the fairness of the cost.

- hidden costs (a_hidd) — the presence of additional financial barriers (deposits, commissions).

This block allows us to assess the degree of financial stress and identify the gap between expectations and market reality.

Location & Proximity

This measures the significance of spatial factors:

- proximity to the university (l_uni);
- accessibility of the ring-bahn as a key transport axis (l_ring);
- availability of inexpensive food (Mensa) (l_mensa);
- availability of green areas (l_green);
- proximity to shops and essential services (l_grocer).

This data allows us to assess which amenities have the greatest influence on the choice of place of residence.

Transportation

The questions are aimed at assessing the quality of the transport infrastructure:

- quality and frequency of public transport (t_connect);
- cycling infrastructure (t_bike);
- availability of night transport (t_night).

This block is critical for understanding how students commute to and from university and how important this is to them.

Social & Living Conditions

The questions clarify the quality of the social environment:

- neighbourhood safety (s_safe);
- access to cultural/student events (s_cultur);
- proximity to student communities (s_common);
- friendliness towards international students (s_env);

- inclusiveness (`s_incl`).

This section identifies how the social atmosphere affects satisfaction with accommodation and choice of neighbourhood.

Difficulties and Satisfaction

The third section explores the subjective experience of finding accommodation, stress levels and potential barriers, including discrimination.

The housing search process

- method of finding housing (`way`);
- duration of the search (`term`);
- existence of a long-term contract (`longtermagree`);
- number of applications submitted (`applic_count`).

These questions provide insight into the time and effort required to integrate into the rental market.

Emotional and social impact

- level of stress when searching for housing (`stress`);
- level of satisfaction with current housing (`satisfied`).

This helps to identify the degree of individual market pressure.

Problems and discrimination

- the existence of difficulties (`diff`);
- types of discrimination (`discrimination`).

These questions allow us to analyse the vulnerability of different groups of students and assess the social justice of the market.

The interpretation of the third section is beyond the scope of this chapter and will be discussed in more detail in the next chapter, Students' Experiences.

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 103 entries, 0 to 102
```

2822 Data columns (total 34 columns):

2823	#	Column	Non -Null Count	Dty
2824	- - - - -		- - - - -	
2825	0	Gender	103 non -null	obj
2826	1	Age Group	103 non -null	obj
2827	2	Origin	103 non -null	obj
2828	3	University	103 non -null	obj
2829	4	Study Status	103 non -null	obj
2830	5	Address (street & house number)	59 non -null	obj
2831	6	Postal Code	103 non -null	obj
2832	7	Current Rental Price	103 non -null	obj
2833	8	Sources of Financing	103 non -null	obj
2834	9	Housing Model	103 non -null	obj
2835	10	Monthly Rent Within Budget	103 non -null	int
2836	11	Price Matches Apartment Condition	103 non -null	int
2837	12	Hidden Costs (deposit, agency fees)	103 non -null	int
2838	13	Proximity to the University	103 non -null	int
2839	14	Proximity to the Ring -Bahn	103 non -null	int
2840	15	Proximity to the Mensa (or affordable food options)	103 non -null	int
2841	16	Proximity to Green Spaces (parks, canals)	103 non -null	int
2842	17	Proximity to Grocery Stores/Essential Services	103 non -null	int
2843	18	Quality of public transport connections (frequency, reliability)	103 non -null	int
2844	19	Bike -friendly infrastructure (storage, parking places, bicycle paths)	103 non -null	int
2845	20	Night transport options (for late classes/part -time work)	103 non -null	int
2846	21	Safety in the Neighborhood	103 non -null	int
2847	22	Access to Cultural/Student events	103 non -null	int
2848	23	Proximity to Student Communities	103 non -null	int
2849	24	International student -friendly environment	103 non -null	int
2850	25	Inclusivity	103 non -null	int
2851	26	How did you find your accomodation in Berlin?	103 non -null	obj
2852	27	How long have you searched for your long term accommodation in Berlin?	103 non -null	obj
2853	28	Do you have a Long Term Rental Agreement?	103 non -null	obj

```

2854 29 How stressful would you describe your housing search? 103 non -null int
2855 30 How many accommodations did you apply for? 103 non -null obj
2856 31 How satisfied are you with your current housing situation? 103 non -null int
2857 32 Did you experience any kinds of difficulties? 103 non -null obj
2858 33 What kind of discrimination did you have to face with? 103 non -null obj
2859 dtypes: int64(18), object(16)
2860 memory usage: 27.5+ KB

```

When working with survey responses, standard data preparation operations used in data science were performed. First, the columns and response categories were renamed to shorter and more uniform designations. This simplified data access made the code more compact and ensured uniformity in the structure of the dataset.

```

2864 # Clean up any hidden characters
2865 df.columns = df.columns.str.strip().str.replace('\xa0', ' ', regex=False)
2866
2867 # Renaming colums for data clarity
2868 df = df.rename(columns={
2869     'Gender': 'gender',
2870     'Age Group': 'age',
2871     'Origin': 'origin',
2872     'University': 'uni',
2873     'Study Status': 'stat',
2874     'Address (street & house number)': 'adr',
2875     'Postal Code': 'plz',
2876     'Current Rental Price': 'price',
2877     'Sources of Financing': 'fin',
2878     'Housing Model': 'model',
2879     'Monthly Rent Within Budget': 'a_price',
2880     'Price Matches Apartment Condition': 'a_cond',
2881     'Hidden Costs (deposit, agency fees)': 'a_hidd',
2882     'Proximity to the University': 'l_uni',
2883     'Proximity to the Ring -Bahn': 'l_ring',
2884     'Proximity to the Mensa (or affordable food options)': 'l_mensa',

```

```

2885 'Proximity to Green Spaces (parks, canals)': 'l_green',
2886 'Proximity to Grocery Stores/Essential Services': 'l_grocer',
2887 'Quality of public transport connections (frequency, reliability)': 't_connect',
2888 'Bike -friendly infrastructure (storage, parking places, bicycle paths)': 't_bike',
2889 'Night transport options (for late classes/part -time work)': 't_night',
2890 'Safety in the Neighborhood': 's_safe',
2891 'Access to Cultural/Student events': 's_cultur',
2892 'Proximity to Student Communities': 's_common',
2893 'International student -friendly environment': 's_env',
2894 'Inclusivity': 's_incl',
2895 'How did you find your accomodation in Berlin?': 'way',
2896 'How long have you searched for your long term accommodation in Berlin?': 'term',
2897 'Do you have a Long Term Rental Agreement?': 'longtermagree',
2898 'How stressful would you describe your housing search?': 'stress',
2899 'How many accommodations did you apply for?': 'applic_count',
2900 'How satisfied are you with your current housing situation?': 'satisfied',
2901 'Did you experience any kinds of difficulties?': 'diff',
2902 'What kind of discrimination did you have to face with?': 'discrimination'
2903 })

2904 #Renaming long answers
2905 model_mapping = {
2906     "Studierendenwerk Dorms (State -subsidised student housing)": "PBSA",
2907     "PBSA (e.g., Neon Wood private investor housing with high amenities)": "PBSA",
2908     "Wohngemeinschaften (WGs shared flats, informal sublets)": "Wohngemeinschaften (WG)",
2909     "Living at Home (with family, commuting)": "Living at Home",
2910     "Commons / Co -ops (e.g., STUK - self -managed, solidarity -based housing)": "Co -ops / Commons",
2911     "PPPs (e.g., Camplus public -private university housing models)": "PPP Housing",
2912     "Private Apartment / Studio": "Private Apartment"
2913 }

2914
2915 #Apply the renaming
2916 df["model_short"] = df["model"].replace(model_mapping)

```


2917 Secondly, since the correct type of variables is critical for further analysis, numerical indicators were converted
2918 from text format to numerical format. This made it possible to perform statistical calculations, construct distributions,
2919 and apply quantitative analysis methods.

```
2920 int_columns = [  
2921     'a_price', 'a_cond', 'a_hidd', 'l_uni', 'l_ring', 'l_mensa', 'l_green',  
2922     'l_grocer', 't_connect', 't_bike', 't_night', 's_safe', 's_cultur',  
2923     's_common', 's_env', 's_incl', 'stress', 'satisfied'  
2924 ]  
  
2925  
2926  
2927 for col in int_columns:  
2928     df[col] = pd.to_numeric(df[col], errors='coerce').astype('Int64')
```

2929 Thirdly, multiple-choice questions were standardised: answers were broken down into binary indicators (dummy
2930 variables), which made it possible to correctly account for each selected option when modelling and analysing frequen-
2931 cies. This approach ensures data interpretability and prevents the loss of information that is characteristic of aggregated
2932 text fields. In addition, gaps were eliminated, and categories were brought into a uniform format.

```
2933 fin_categories = ['Public Funding', 'Private Support', 'Part -Time Job', 'Scholarship']  
  
2934  
2935 # Step 1: Clean and split the values  
2936 df['fin_e'] = (  
2937     df['fin']  
2938     .str.split(',') # Split by comma  
2939     .apply(lambda x: [item.strip() for item in x] if isinstance(x, list) else [])  
2940     .apply(lambda x: [item for item in x if item]) # Remove empty strings  
2941 )
```

2942 Together, these steps provided a clean, consistent, and analysable dataset that meets the methodological require-
2943 ments of quantitative behavioral research.

2944 **5.3. Users Personal Data**

2945 **5.3.1. Student Status**

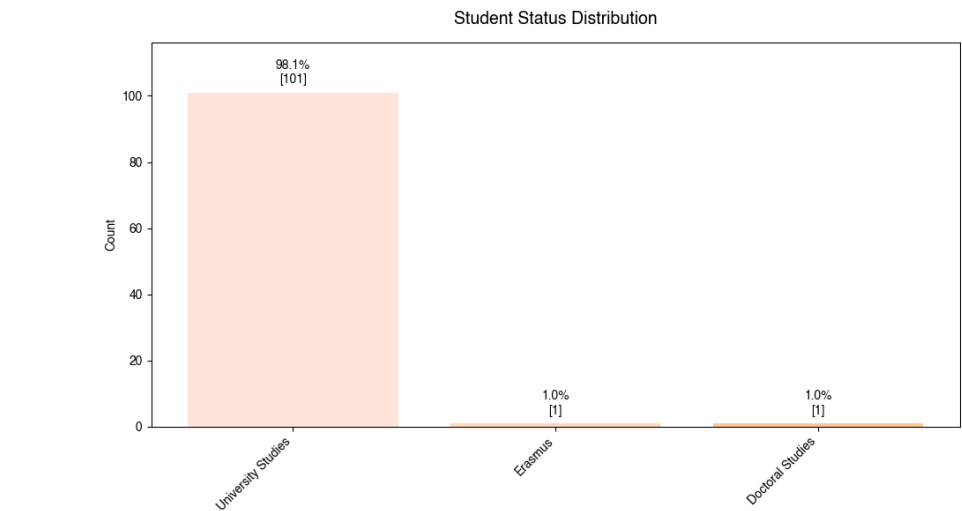
2946 The majority of survey participants (98.1%) were university students, including undergraduates, postgraduates and
2947 college students (Figure 42). This group constitutes the core user base of student accommodation and most accurately
2948 reflects the typical demand for long-term, affordable housing near universities.

2949 However, it is important to emphasise that student accommodation in Berlin serves a much wider range of educa-
2950 tional tracks. A significant proportion of users are applicants from other countries who are studying at a preparatory
2951 college (Studienkolleg) and preparing to enrol at universities in the German capital. These students have specific
2952 needs related to their short stay and high dependence on the availability of housing when taking their first steps in a
2953 new country.

2954 Exchange students, including participants in Erasmus and Double-Degree programs, form a separate category.
2955 Unlike regular university students, they are mainly interested in short-term rentals of 3 to 12 months, which creates
2956 additional seasonal and structural demand for flexible, temporary forms of accommodation.

2957 In addition, the urban student ecosystem includes students at higher levels of education — doctoral students, gradu-
2958 ates seeking additional qualifications, interns, and young research and teaching staff. Their housing needs vary greatly,
2959 ranging from short-term stays during internships to long-term accommodation near research institutes.

2960 All of these groups represent a significant research gap, as they fall outside the scope of both the original study
2961 by Von Wittenhorst Zu Sonsfeld and Beusker (2024) and the current work. Their inclusion requires separate analysis,
2962 as they create additional hidden demand, which in practice increases the burden on the student housing market but is
2963 rarely taken into account in regulatory and statistical assessments.



2964
2965 *Figure 42. Student Status Distribution*

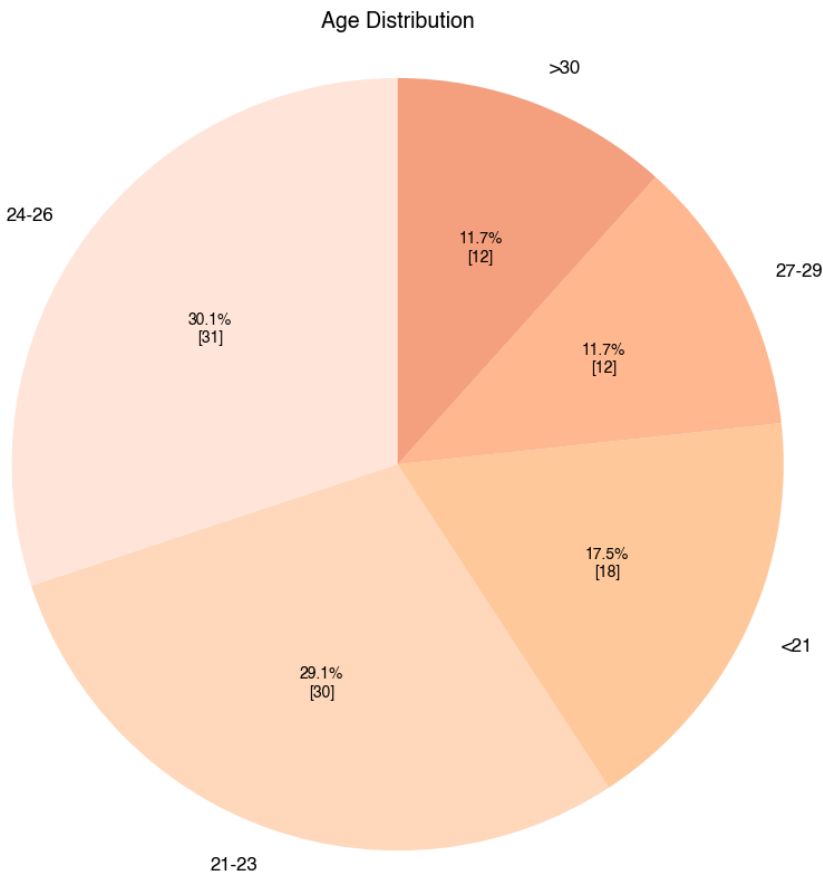
2966 **5.3.2. Age**

2967 According to the survey results, the largest groups were students aged 24–26 (30.1%) and 21–23 (29.1%), who
2968 together make up almost two-thirds of the entire sample. These age categories reflect the typical structure of Berlin’s
2969 student population, which is dominated by senior undergraduate and master’s students.

2970 The younger group — students under the age of 21 — accounts for 17.5%, which corresponds to the proportion
2971 of first-year students and students in preparatory programs. The remaining categories — 27–29 years old and over 30
2972 years old — include late starters, doctoral students, participants in international programs, and young professionals
2973 who combine work with study; each of these groups accounts for 11.7%.

2974 The data on student age obtained from the survey is consistent with the official statistics from Statistik Berlin
2975 Brandenburg (2025), where 57.8% of students are over 25 years old.

2976 Thus, the age distribution (Figure 43) shows a predominance of mature students in the later stages of their studies,
2977 which influences their housing preferences, level of independence, and requirements for the quality and affordability
2978 of housing.



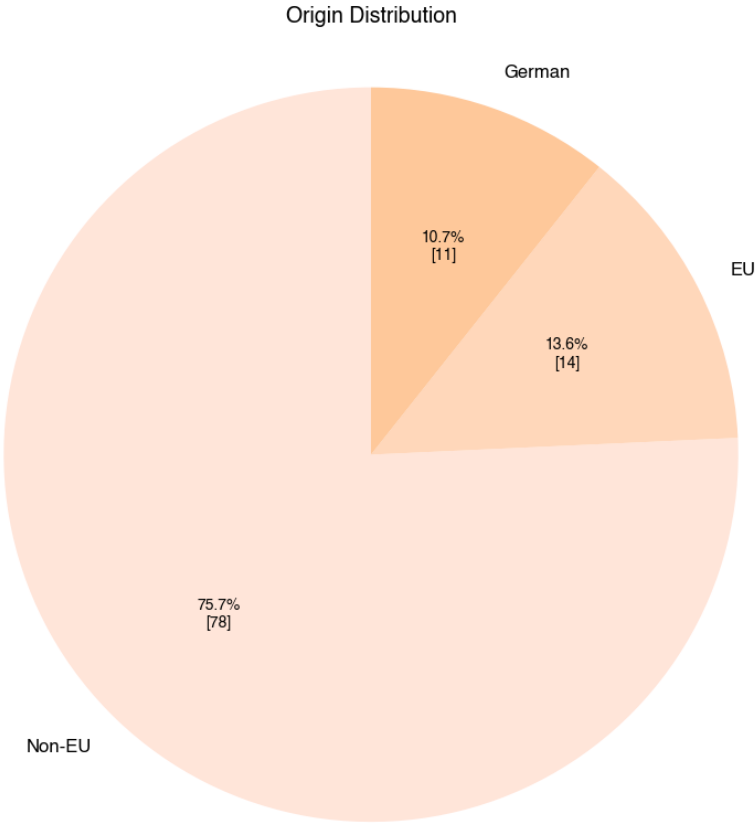
2979
2980 *Figure 43. Age Distribution*

2981 **5.3.3. Origin**

2982 Most of the survey participants are international students, with the majority coming from non-European countries
2983 (Non-EU) — 75.7%, while students from European Union countries make up another 13.6% of the sample. Only 10.7%
2984 of participants are German citizens.

2985 This distribution (Figure 44) is fully consistent with the target audience of the study, which focuses on analysing the
2986 living conditions and barriers faced by international students — the group experiencing the most significant difficulties
2987 in the Berlin rental market.

2988 The high proportion of non-EU students further reflects Berlin’s growing international appeal as an educational
2989 center and highlights the relevance of studying the pressure on the student housing market.



2990
2991 *Figure 44. Origin Distribution*

2992 **5.3.4. Housing Model**

2993 Students in Berlin use a wide range of accommodation models (Figure 45), each with its own structural features,
2994 advantages, risks and local context. These models form a heterogeneous student housing market, where public mecha-
2995 nisms coexist with the private sector, cooperatives and informal solutions. Let us consider three of them that are most
2996 frequently mentioned in the literature.

2997 **Purpose-Built Student Accommodation (PBSA)** Purpose-Built Student Accommodation can be divided into two
2998 categories: public and private.

2999 State-owned purpose-built student accommodation is funded and managed by public agencies and is the most
3000 affordable form of student housing. Its key advantages are low cost and social diversity among residents. However,
3001 limited capacity and long waiting lists significantly reduce their availability. This makes them a scarce resource and
3002 increases pressure on other types of housing.

3003 Private purpose-built student accommodation is an investment-oriented segment of student housing, offering mod-
3004 ern infrastructure, additional services and the possibility of immediate occupancy. At the same time, such properties
3005 are very expensive, which makes them inaccessible to many students and contributes to spatial segregation.

3006 Although 53.4% of users indicated that they live in PBSAs, in reality, only 5-6% of students in Berlin live in PBSAs.
3007 The majority of students live in shared flats or other corporate models due to the predominantly rental housing situation
3008 in Germany (Janet Merkel, 2025).

3009 **Wohngemeinschaften (WGs)** One of the most common forms of accommodation for students in Berlin is shared
3010 rented flats with an informal system of renting and subletting. The most important advantages of WGs are flexibility,
3011 affordability and a high level of social interaction. However, such conditions are often accompanied by instability,
3012 short-term contracts and discrimination when moving in, which particularly affects foreign students.

3013 **Living at Home**

3014 This option involves living in the parental home with daily commutes to the place of study. It ensures no or minimal
3015 rental costs, but increases the risk of social isolation and imposes time costs associated with long commutes. Unlike
3016 in Eastern European countries, this model is much less common in Berlin (Janet Merkel, 2025).

3017 With regard to Housing Model Distribution (Figure XX), the survey results are heavily skewed towards users of
3018 purpose-built student accommodation, as the main target group of the study.

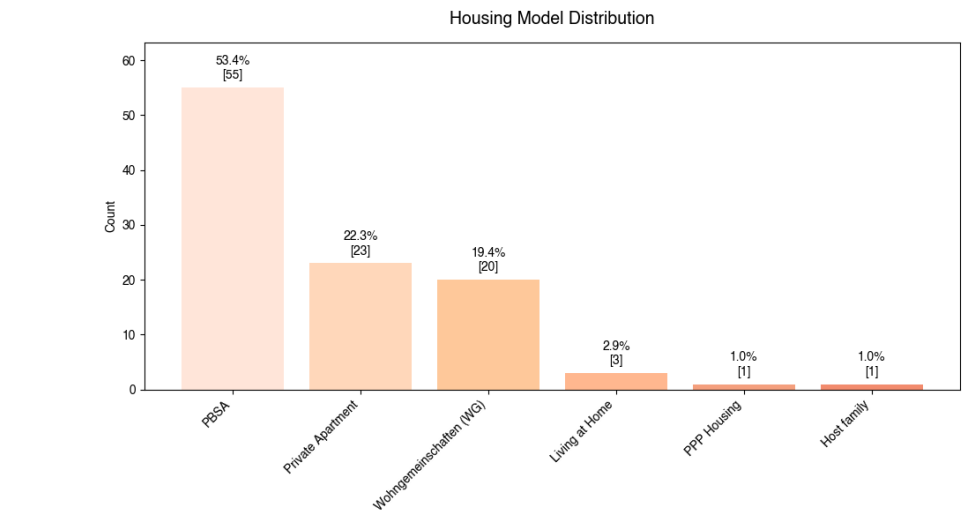


Figure 45. Housing Model Distribution

5.3.5. Sources of Financing

According to the survey results (Figure 46), the most common source of financing for students' educational and personal expenses is part-time employment: almost half of the respondents (47.3%) work alongside their studies to cover their daily expenses and living costs.

A significant proportion of respondents (36.6%) receive financial support from private individuals, most often from parents or relatives. However, formal support mechanisms, such as government funds (5.3%) and academic scholarships (6.9%), are used relatively rarely, which highlights the limited official channels of assistance for foreign students.

It is particularly telling that 81.3% of students identify rent as their largest expense. This confirms the key finding of the study: the cost of living in Berlin has a decisive influence on students' economic behavior and constitutes the main financial pressure affecting both their choice of place of residence and the need for additional employment.

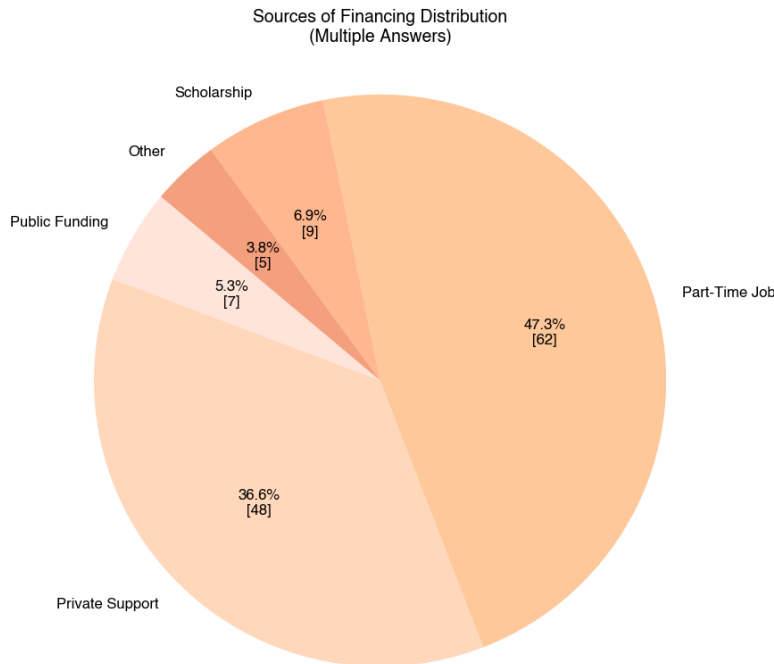


Figure 46. Sources of Financing Distribution

5.4. Needs and Preferences

In the second part of the survey, the target audience was asked to answer the question: what is most important when looking for housing? This helps to identify the needs and preferences of international students across the city.

The **Affordability** category covers issues such as whether the rent fits the budget, whether the price is commen-

surate with the quality of the accommodation, and whether there are any hidden costs. The **Location & Proximity** block records students' spatial preferences, including proximity to the university, everyday infrastructure and leisure areas. **Transportation** reflects accessibility and quality of mobility, while **Social & Living** Conditions describes the social environment, sense of security, inclusiveness, and access to student communities.

To ensure comparability of results, all responses were converted to a numerical format and aggregated within the relevant groups. For each question, respondents were asked to rate the importance of a particular factor on a scale from 1 (not important at all) to 10 (extremely important). In addition, a uniform order of display of indicators was established, which made it possible to maintain the logical structure of the analysis and ensure consistent interpretation of the results both at the level of individual indicators and at the level of thematic blocks.

Thus, this stage of the analysis serves as a diagnostic cross-section, allowing us to identify key areas of tension and satisfaction in students' perceptions of the Berlin housing market and to create a basis for further research of subjective assessments.

5.4.1. General Statistics

As a preliminary analysis of the data, similar to the Descriptive Market Statistics chapter, a generalised summary statistics assessment of respondents' answers was conducted across four key thematic topics reflecting the main dimensions of the student housing experience: affordability, location and proximity, transportation, and social and living conditions.

```
group_columns = {  
    'Affordability': ['a_price', 'a_cond', 'a_hidd'],  
    'Location & Proximity': ['l_uni', 'l_ring', 'l_mensa', 'l_green', 'l_grocer'],  
    'Transportation': ['t_connect', 't_bike', 't_night'],  
    'Social & Living Conditions': ['s_safe', 's_cultur', 's_common', 's_env']  
}  
  
order = [  
    'a_price', 'a_cond', 'a_hidd', 'Affordability',  
    'l_uni', 'l_ring', 'l_mensa', 'l_green', 'l_grocer', 'Location & Proximity',  
    't_connect', 't_bike', 't_night', 'Transportation',  
    's_safe', 's_cultur', 's_common', 's_env', 's_incl', 'Social & Living Conditions'  
]  
  
# 'stress', 'satisfied'
```

3069

```
3070 item_rename = {
3071     'a_price': 'Monthly Rent Within Budget',
3072     'a_cond': 'Price Matches Apartment Condition',
3073     'a_hidd': 'Hidden Costs',
3074     'l_uni': 'Proximity to the University',
3075     'l_ring': 'Proximity to the Ring -Bahn',
3076     'l_mensa': 'Proximity to the Mensa',
3077     'l_green': 'Proximity to Green Spaces (parks, canals)',
3078     'l_grocer': 'Proximity to Essential Services',
3079     't_connect': 'Quality of Public Transport Connections',
3080     't_bike': 'Bike -Friendly Infrastructure',
3081     't_night': 'Night Transport Options',
3082     's_safe': 'Safety in the Neighborhood',
3083     's_cultur': 'Access to Cultural/Student Events',
3084     's_common': 'Proximity to Student Communities',
3085     's_env': 'International Student -Friendly Environment',
3086     's_incl': 'Inclusivity',
3087     'stress': 'Level of Stress',
3088     'satisfied': 'Level of Satisfaction',
3089     'Affordability': 'Affordability',
3090     'Location & Proximity': 'Location & Proximity',
3091     'Transportation': 'Transportation',
3092     'Social & Living Conditions': 'Social & Living Conditions'
3093 }
```

3094

```
3095 int_columns = [
3096     'a_price', 'a_cond', 'a_hidd', 'l_uni', 'l_ring', 'l_mensa', 'l_green',
3097     'l_grocer', 't_connect', 't_bike', 't_night', 's_safe', 's_cultur',
3098     's_common', 's_env', 's_incl'
3099 ]
3100 # 'stress', 'satisfied'
```



```

3101 group_columns = {
3102     'Affordability': ['a_price', 'a_cond', 'a_hidd'],
3103     'Location & Proximity': ['l_uni', 'l_ring', 'l_mensa', 'l_green', 'l_grocer'],
3104     'Transportation': ['t_connect', 't_bike', 't_night'],
3105     'Social & Living Conditions': ['s_safe', 's_cultur', 's_common', 's_env']
3106 }

```

3107 Based on the descriptive statistics presented (Table XX), several key patterns can be identified in students' prefer-
3108 ences for living conditions.

3109 The **Affordability** category shows the highest and most consistent ratings among all the blocks analysed. The
3110 average value for the group is 8.29 with a low standard deviation, which indicates relative uniformity of perception.
3111 The affordability of rent is particularly highly rated (mean = 9.12, median = 10), reflecting students' adaptation to
3112 financial constraints or forced compromises when choosing accommodation. At the same time, issues of hidden costs
3113 and value for money have a higher variance, indicating heterogeneous experiences and the presence of problematic
3114 cases.

3115 The **Location & Proximity** block is characterised by a significantly lower average value (5.74) and greater variabil-
3116 ity in ratings. The most important aspect is proximity to canteens and affordable places to eat (mean = 3.52), as well
3117 as to universities and the Ring-Bahn, reflecting the spatial gap between places of study and actual areas of residence
3118 for students. At the same time, access to everyday services is rated relatively highly, indicating the compensatory role
3119 of urban infrastructure outside university clusters.

3120 In the **Transportation** category, there is a marked contrast between individual indicators. The quality of public
3121 transport receives one of the highest ratings in the sample (mean = 8.53), confirming the important role of Berlin's
3122 transport system as a compensatory mechanism for long-distance commuting. At the same time, cycling infrastructure
3123 is rated significantly lower (mean = 4.42) and shows high dispersion, indicating a spatially uneven distribution of
3124 bicycle-friendly conditions.

3125 The most vulnerable group of indicators is **Social & Living Conditions**, with an average score of 5.19. Despite a
3126 relatively high sense of security in the neighbourhood, access to student communities, cultural events and a sense of an
3127 inclusive environment receive low ratings and are characterised by high variability. This indicates social fragmentation
3128 in student housing and weak integration of some students, especially international students, into local social structures.

3129 Overall, the results of the initial analysis show that economic affordability and transport connectivity are priority
3130 factors in a stressed market.

	Item	mean	std	min	median	max
0	Monthly Rent Within Budget	9.12	1.13	5.0	10.0	10.0
1	Price Matches Apartment Condition	8.06	1.82	3.0	8.0	10.0
2	Hidden Costs	7.7	2.06	2.0	8.0	10.0
3	Affordability	8.29	1.17	<NA>	8.33	<NA>
4	Proximity to the University	6.1	2.18	1.0	6.0	10.0
5	Proximity to the Ring- Bahn	5.67	2.47	1.0	6.0	10.0
6	Proximity to the Mensa	3.52	2.4	1.0	3.0	10.0
7	Proximity to Green Spaces (parks, canals)	5.67	2.39	1.0	6.0	10.0
8	Proximity to Essential Ser- vices	7.72	1.91	1.0	8.0	10.0
9	Location & Proximity	5.74	1.6	<NA>	5.8	<NA>
10	Quality of Public Transport Connections	8.53	1.6	4.0	9.0	10.0
11	Bike- Friendly Infrastructure	4.42	2.84	1.0	4.0	10.0

Table XX. General Student Needs and Preferences

Correlation analysis of student needs and preferences (Figure 47) reveals stable relationships between individual aspects of housing choice, thereby deepening the interpretation of descriptive statistics results. Overall, the correlation matrix shows moderate but structurally meaningful relationships, indicating the multidimensional nature of student housing capacity, where no single factor is completely isolated.

Affordability indicators show relatively weak correlations with spatial and social characteristics. Rent affordability and hidden costs are more closely related to each other than to other blocks, confirming their autonomous role in the decision-making process. This indicates that financial constraints are seen by students as a separate structural condition rather than an element of the integrated quality of the living environment.

Blocks associated with **Location & Proximity** form a dense cluster of interrelationships. Proximity to the university, the ring-bahn, the mensa, green areas, and basic services shows consistent positive correlations with each other (in the range of approximately 0.27–0.50). This indicates that students tend to view accessibility not in terms of individual objects, but as a comprehensive characteristic of the living environment. A particularly strong link is observed between proximity to the Mensa and student communities, as well as between green areas and cultural access, which emphasises the role of everyday routes and infrastructure in creating a comfortable environment.

Transportation indicators act as a link between spatial and social characteristics. The quality of public transport correlates moderately with proximity to key urban functions and night-time transport, reflecting the importance of the transport system as a mechanism for commuters. At the same time, cycling infrastructure shows more selective links, including a negative correlation with the sense of safety in the area, which may indicate local conflicts between different types of mobility or spatial heterogeneity of cycling conditions in the city.

Social & Living Conditions form a separate but interrelated cluster. Access to cultural events, proximity to student communities, an internationally friendly environment, and inclusiveness demonstrate moderate positive correlations. The strongest link is observed between student communities and an international-friendly environment, which highlights the importance of social networks and “social density” for international students. At the same time, the feeling of safety is weakly correlated with most social indicators, which may indicate a gap between the physical and social comfort of the environment.

Overall, correlation analysis confirms that student housing choices are shaped by the intersection of three factors: economic adaptation, spatial accessibility, and social integration. At the same time, it is precisely spatial and social connections that are the most structured, while financial parameters act more as a strict limitation than as an active factor in shaping preferences.

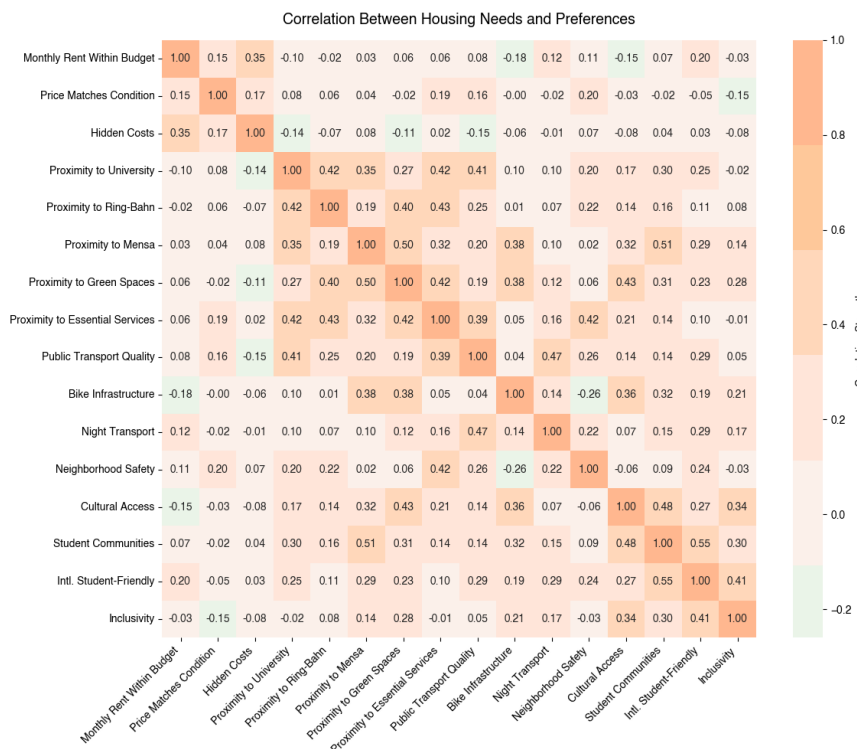


Figure 47. Correlation Between Housing Needs and Preferences

5.4.2. Statistics by Origin

In order to avoid generalising students, it is necessary to conduct a comparative analysis of students' housing needs and preferences (Table 24) depending on their country of origin (German, EU, Non-EU) and identify both common patterns and structural differences between groups.

origin_cat	German			EU			Non-EU		
stat	mean	median	std	mean	median	std	mean	median	std
Item									
Monthly Rent Within Budget	9.09	5.0	0.94	9.29	5.0	0.99	9.09	5.0	1.19
Price Matches Apartment Condition	7.18	5.0	1.33	7.50	5.0	1.87	8.28	5.0	1.83
Hidden Costs	7.36	5.0	1.50	8.57	5.0	1.83	7.59	5.0	2.15
Affordability	7.88	5.0	0.98	8.45	5.0	0.96	8.32	5.0	1.23
Proximity to the University	5.00	5.0	2.28	5.64	5.0	2.27	6.33	5.0	2.11
Proximity to the Ring-Bahn	4.55	5.0	2.77	5.21	5.0	2.67	5.91	5.0	2.36
Proximity to the Mensa	4.00	5.0	2.49	4.14	5.0	2.88	3.35	5.0	2.30
Proximity to Green Spaces (parks, canals)	6.00	5.0	2.14	5.50	5.0	2.79	5.65	5.0	2.37

Table 24. Student Needs and Preferences by Origin

In the **Affordability** section, the differences between the groups are relatively slight. For all categories of students, the affordability of rent is rated very highly (9), confirming the universal importance of the price factor. At the same time, students from the EU and especially non-EU countries rate the value for money of housing and the importance of hidden costs slightly higher, which may reflect their lower awareness of the local market and higher costs of searching for housing. In the aggregate indicator of affordability, it is foreign students who demonstrate the highest values, which emphasises their increased sensitivity to financial constraints.

The **Location & Proximity** section shows more noticeable differences. On average, non-EU students rate proximity to the university, the ring-bahn and, in particular, basic services higher than German students. This indicates that international students are more dependent on a compact and well-served urban environment where everyday functions are accessible without complicated navigation around the city. German students, on the other hand, give lower ratings for proximity, which may be due to their better knowledge of the city and greater willingness to make spatial compromises.

The differences are most noticeable in the Transportation category. Foreign students (EU and non-EU) rate the quality of public transport and night-time transport options significantly higher, while German students tend to value cycling infrastructure more. This reflects differences in mobility practices: for local students, bicycles are a familiar mode of transport, while international students rely more on public transport, especially at night.

The Social & Living Conditions section reveals the most contrasting results. Non-EU students rate the importance of neighbourhood safety significantly higher, indicating increased vulnerability and a need for a subjective sense of security in an unfamiliar urban environment. At the same time, foreign students rate access to cultural events, student communities and an international-friendly environment lower than German students, which may indicate structural barriers to integration and limited inclusion in local social networks. German students, on the other hand, rate their proximity to student communities and cultural life higher, reflecting their more stable position in the urban and academic environment.

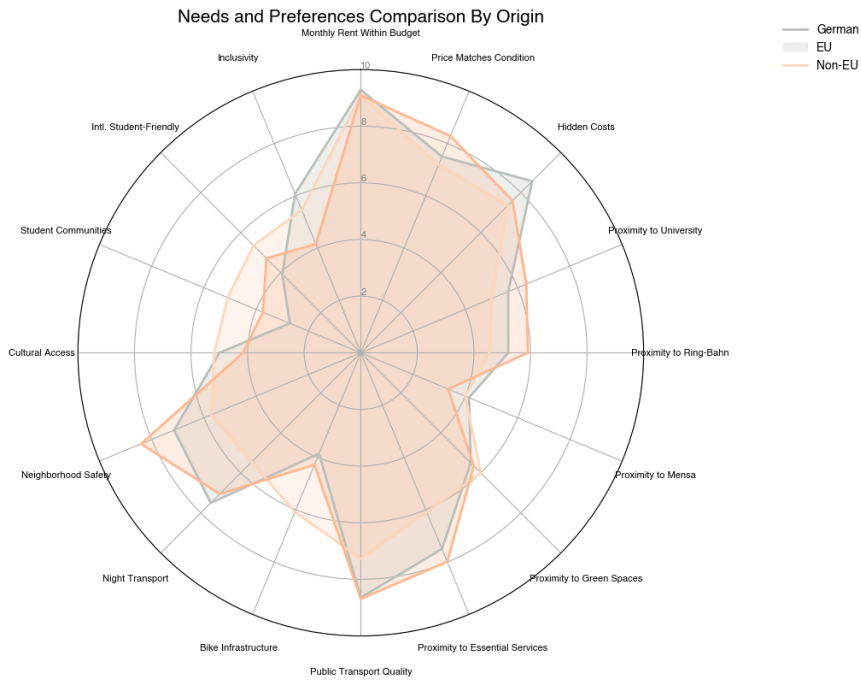


Figure 48. Student Needs and Preferences by Origin

Overall, the results (Figure 48) show that students' origins have a significant impact on the structure of their housing priorities. For international students, accessibility, transport and safety are key, while for German students, social and cultural aspects as well as alternative forms of mobility play a greater role. These differences highlight the need for a differentiated approach to analysing student housing capacity and confirm that a single 'student' model does not reflect the real diversity of user groups in the student housing market in Berlin.

```
df.to_excel(os.path.join(hf, "user_data.xlsx"), index=False)
```

3201 **5.5. 4.3 Student Experiences**

3202 The student experience of finding accommodation clearly demonstrates the difficulties and problems faced by
3203 the most vulnerable groups during the ongoing housing crisis. In order to better understand the students, individual
3204 interviews were conducted, where open-ended questions were asked to guide the storytelling.

3205 These problems and difficulties need to be analysed before proposing ways to resolve the student housing crisis, as
3206 they serve as barriers between student accommodation and users.

3207 **5.5.1. Set Up**

3208 Import Libraries

```
3209 import pandas as pd
3210 import geopandas as gpd
3211 import chardet
3212 import gspread
3213 import seaborn as sns
3214 import matplotlib.pyplot as plt
3215 import textwrap
3216 from pywaffle import Waffle
3217 import numpy as np
3218 from matplotlib.patches import Patch
3219 from math import pi
```

3220 Data Import

```
3221 fp = r"../inputs/4.3_students_experiences/user_data.xlsx"
3222
3223 df = pd.read_excel(fp)
```

3224 **5.6. Difficulties and Problems**

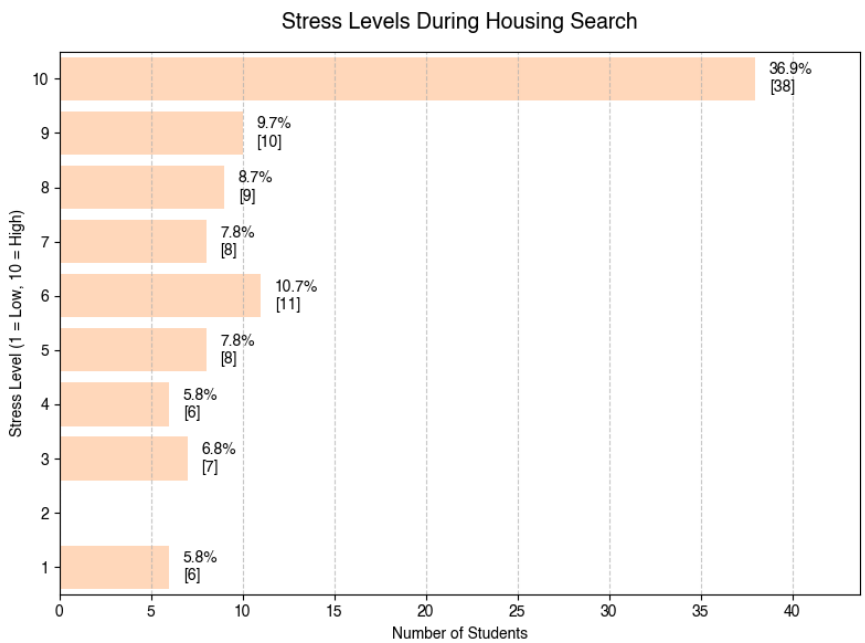
3225 **5.6.1. Too Much Competition**

3226 One of the most frequently mentioned problems when looking for accommodation is excessive competition, which
3227 permeates all segments of the student rental market. Virtually all respondents describe a situation in which the
3228 number of applicants significantly exceeds the available supply, especially in the budget segment. Even when ac-
3229 tively submitting applications, students rarely receive responses, and scheduled viewings are often cancelled at the
3230 last minute. As one participant noted: The worse problem is competition. I wrote people and even

3231 arranged meetings, but they did not respond and cancelled meetings because they rented the
3232 place already to another person. In such a competitive environment, time becomes a critical factor — a de-
3233 lay of even a few hours can mean losing an opportunity: You had to apply very quickly, or you'd be too
3234 late.

3235 **5.6.2. Stress**

3236 Due to excessive competition, 55.3% of student housing users experience stress levels (Figure 49) above 8 points.
3237 Similar to the high demand–capacity pressure indicators recorded in the spatial analysis, high stress levels reflect not
3238 only a structural shortage of student housing, but also a combination of related difficulties, including uncertainty,
3239 temporary solutions, and emotional costs associated with the housing search process.



3240
3241 *Figure 49. Stress Levels During Housing Search*

3242 The stress levels diagram (Figure 50) demonstrates differences and similarities in stress levels among students with
3243 different backgrounds when searching for accommodation. For German students, the median stress level is around 6
3244 points, with a relatively wide range of responses, indicating a heterogeneous experience — from a relatively comfort-
3245 able search to an extremely stressful one. At the same time, for students from EU and non-EU countries, the median
3246 values are significantly higher (around 8–9 points), and the upper quartiles in both cases reach the maximum value on
3247 the scale, indicating a systematically high level of stress.

3248 It is particularly telling that the distributions for the EU and non-EU groups largely overlap: despite differences in
3249 legal status and access to resources, both groups face comparable levels of emotional stress.

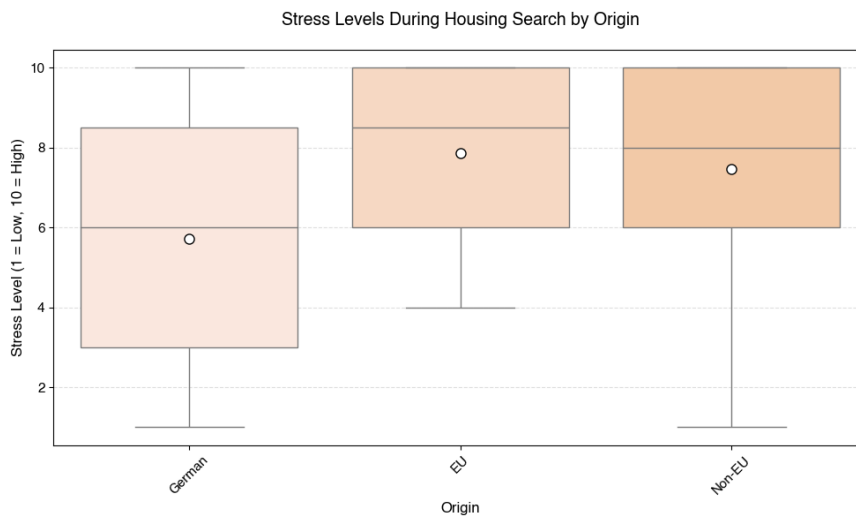


Figure 50. Stress Levels During Housing Search by Origin

Uncertainty, constant rejections and fear of being left homeless directly affect mental health and academic adaptation. One participant directly links the housing crisis to depression: It was kind of difficult and, to be honest, scary because of the new country.

Another student emphasises that the stress reached such a level that professional help was required: I was really very depressed, but I took therapy at the TU. Thus, housing instability becomes not a temporary difficulty, but a factor that directly affects the well-being and ability of students to study and integrate.

5.6.3. Searching Time

The distribution of searching time (Figure 51) demonstrates the high heterogeneity of student experiences and highlights the structural tension in the market. For the largest share of respondents, the search took between 1 and 3 months (35.0%), which indicates a relatively ‘standard’ scenario, requiring, nevertheless, significant time and emotional resources. Another 27.2% of students were able to find accommodation in less than a month, which is probably due to a combination of favorable market conditions and an increased budget.

At the same time, almost one in five students (21.4%) searched for accommodation for more than six months, and another 16.5% searched for 3 to 6 months. Taken together, this means that more than a third of respondents faced a protracted and problematic search that went beyond one semester. Such long search periods directly correlate with increased stress and housing instability and point to a systemic shortage of affordable and long-term student housing in Berlin, as well as a general lack of transparency in the search process.

Searching Time Distribution

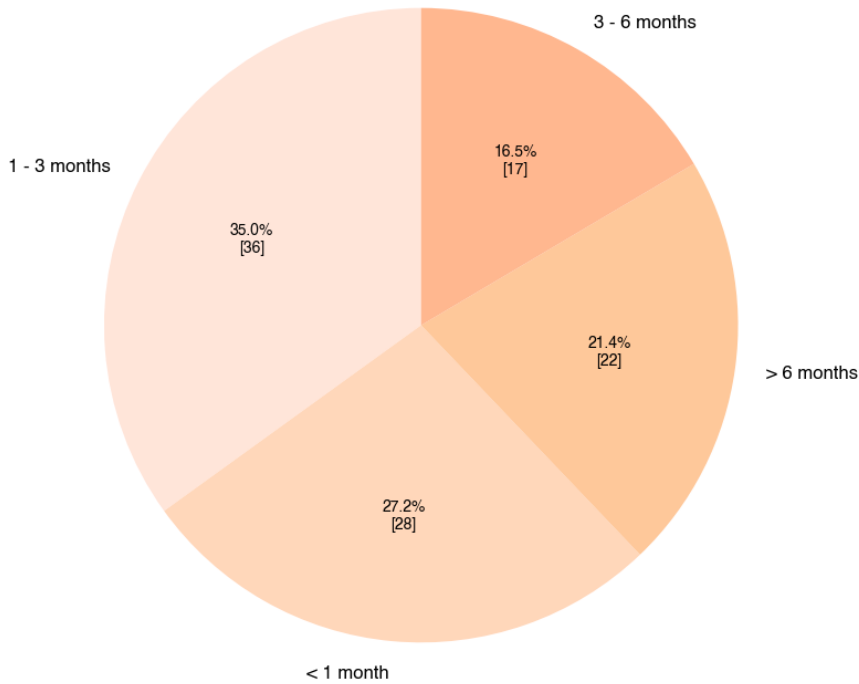


Figure 51. Searching Time Distribution

5.6.4. Applications Count

The application count distribution diagram (Figure 52) clearly reflects the scale of competition in the student housing market in Berlin and the efforts that students have to make in their search.

The largest group of respondents (35.9%) submitted between 1 and 5 applications, which is typically characteristic of students who managed to find accommodation relatively quickly — most often through acquaintances, university channels or state-run student halls of residence. Another 16.5% submitted between 6 and 15 applications, which may also indicate relatively favorable search conditions or a lucky coincidence in terms of timing and budget.

At the same time, almost half of the survey participants reported a significantly higher level of search effort. About 14.6% of students submitted 16–30 applications, and the same number submitted 31–50 applications. It is particularly noteworthy that 18.5% of respondents submitted more than 50 applications, including 10.7% who submitted more than 100 applications. These figures indicate a high degree of frustration, a prolonged search and a systemic shortage of available housing options.

Thus, the distribution of applications highlights the polarisation of the search experience: some students find accommodation relatively quickly, while for a significant proportion, the search becomes a protracted and time-consuming

process. Combined with high levels of stress, this confirms the existence of a structural imbalance between supply and demand in the student housing market in Berlin.

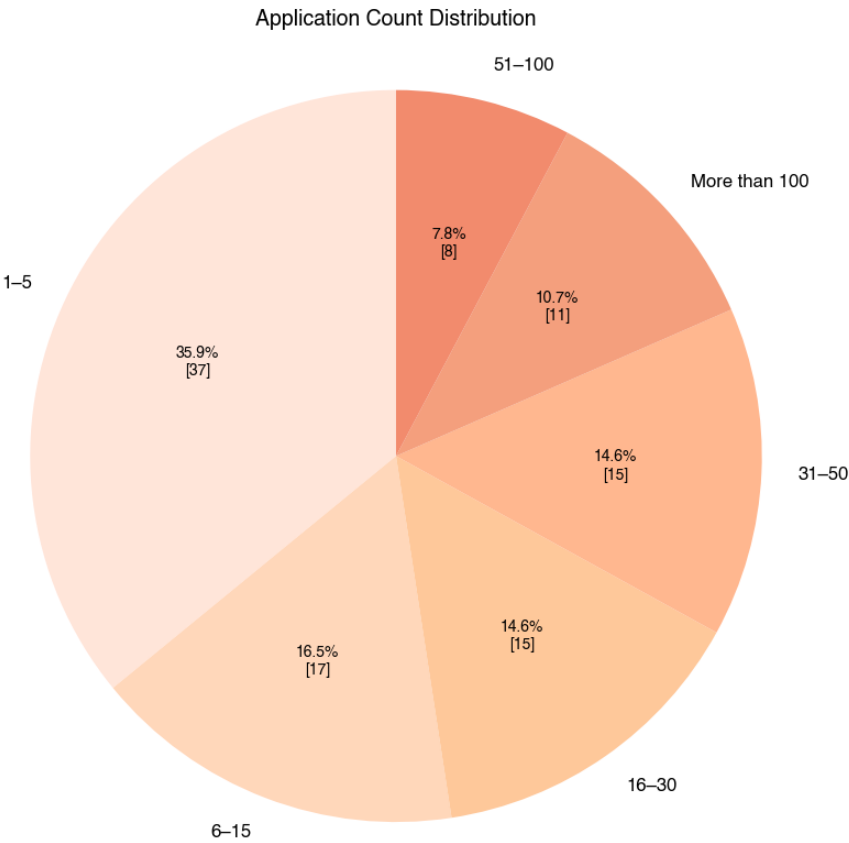


Figure 52. Application Count Distribution

5.6.5. Short Term Agreement

Almost a quarter of students (24.3%) do not have a long-term rental contract for a period of one year or more throughout their studies (Figure 53). Given that the duration of studies can range from 2 to 6 years, this means that a significant proportion of students have to repeatedly search for accommodation, facing recurring stress, uncertainty, and additional financial costs.

Against this backdrop, state-run student halls of residence remain the most stable and reliable landlords. They usually offer one-year contracts with the possibility of renewal, providing basic housing security and predictability of living conditions. However, the limited capacity of such accommodations significantly reduces their systemic role and does not compensate for the instability characteristic of the private segment of the student housing market.

Long Term Agreement Distribution

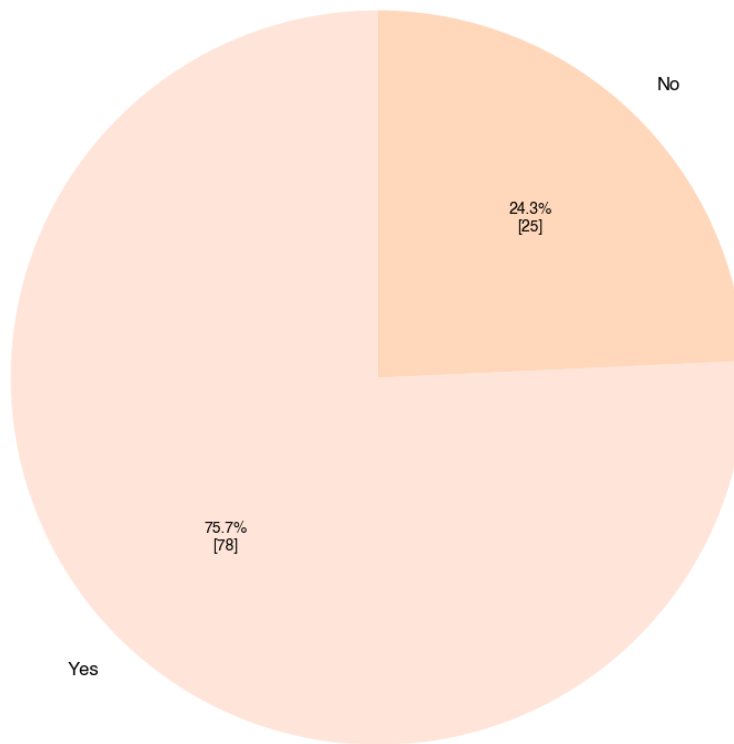


Figure 53. Long Term Agreement Distribution

5.6.6. No Anmeldung

The lack of a long-term rental agreement is directly linked to the issue of registering at the place of residence (Anmeldung), which, in turn, is a mandatory condition for extending visas and residence permits in Germany. Several survey participants who did not have a long-term contract indicated that they had to pay extra to register their residence or register at the address of acquaintances, which indicates the prevalence of informal and legally vulnerable practices in the student housing market.

In addition, the absence of a long-term contract often indicates subletting or re-renting from previous tenants. This practice is usually accompanied by higher rents and, at the same time, deprives students of the right to official registration. This creates a vicious circle of housing instability, in which short-term rentals increase financial pressure, legal insecurity and administrative risks, especially for international students.

Several participants indicated that the possibility of registration was promised before signing the contract, but later turned out to be impossible due to illegal subletting. This creates serious legal and administrative risks: Before signing the contract, I was told that Anmeldung would be possible, and afterwards the landlord

told us that it isn't. In one interview, the situation is described as a vicious circle: You need papers to get an Anmeldung, and you need an Anmeldung before you can apply for your residence permit.

Many students are forced to live without official registration or use old addresses, which creates a chain of further problems — from renewing their residence permits to accessing public services. Anmeldung is transformed from a basic administrative procedure into a tool of pressure and control by landlords.

5.6.7. *Hidden Costs*

In addition to high rents, students regularly encounter hidden and non-transparent costs that significantly increase the actual cost of living and make housing less affordable than advertised. These costs include high deposits, additional payments for utilities, furniture, and internet, as well as administrative fees and agency services.

The search for housing takes place in parallel with the start of studies, adaptation to a new country and, in some cases, a lack of social and financial support. One respondent sums up the general mood as follows: The entire system is screwed. Students who don't have financial support from their families have it especially rough.

5.6.8. *Scams*

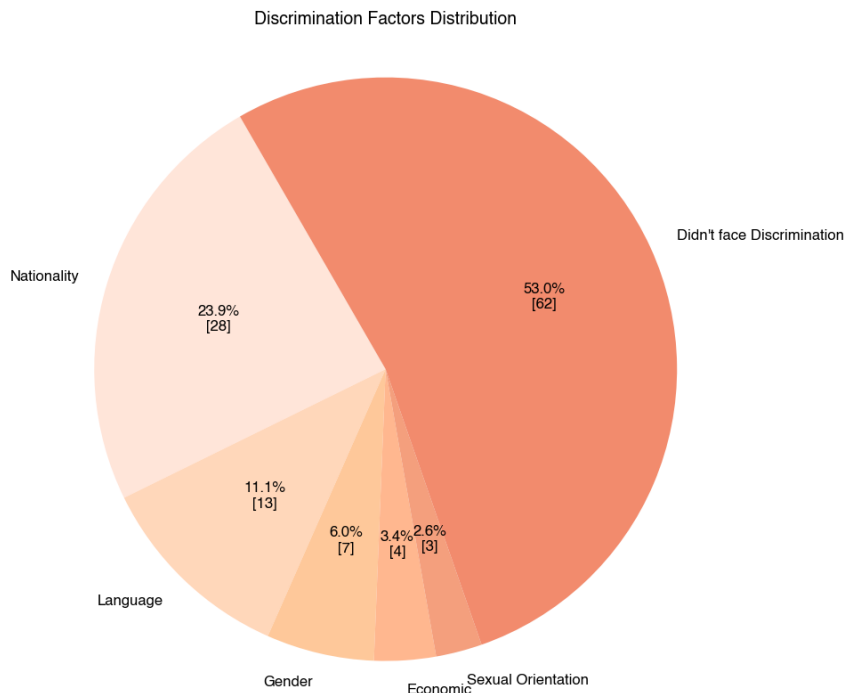
Housing shortages and high competition make students particularly vulnerable to fraud in the rental market. Interviews show that some respondents lost significant amounts of money due to fake advertisements, fictitious landlords, or requests to transfer a deposit before viewing the property.

One of the most dangerous and financially painful types of fraud in the rental market involves fake offers and advance deposit payments, where the victim is told that a flat has been found and that a deposit (Kaution) must be transferred before meeting or signing the contract. This happened to one participant in the study, who noted: That's how 300 other people lost 2000 euros of deposit... The scheme involves the prospective tenant transferring a significant amount of money as a deposit or advance payment, waiting for the keys and confirmation of the transaction, but after the transfer, contact with the 'landlord' is lost and they disappear. This is a typical sign of a rental scam — the scammer creates the appearance of legitimacy, but has no right to rent out the property, and the transferred funds become non-refundable.

When time is short and there aren't many options, students often have to make quick decisions, which makes it harder for them to think critically about the risks. International students who are outside Germany or aren't familiar with local laws and market conditions are especially vulnerable. The fear of being left without accommodation leads to unsafe decisions, including temporary accommodation without a contract or in illegal conditions. In this context, fraud is not a separate problem, but a direct consequence of the structural crisis in student housing, in which the shortage and asymmetry of information systematically work against students.

3344 5.6.9. Discrimination

3345 Although more than half of respondents formally stated that they had not encountered direct discrimination (Figure
 3346 54), qualitative interviews show that discrimination often takes hidden and institutionalised forms. The most common
 3347 practices are ignoring applications, failing to respond, and rejecting applications without explanation. One student em-
 3348 phasises: Haven't encountered direct discrimination, rather landlords ignore applications. Such
 3349 "invisible discrimination" is particularly characteristic of an overheated market, where refusal does not require justifi-
 3350 cation but systematically affects students, foreigners and people with diverse social profiles.



3351
3352 *Figure 54. Discrimination Factors Distribution*

3353 Discrimination is most evident on the basis of nationality and origin, as confirmed by both quantitative data and
 3354 interviews. Some respondents encountered outright refusals related to their citizenship or name: I don't get a lot
 3355 of replies to be honest, I'm assuming that the reason behind it may be me being a student and
 3356 my non-German name, while another participant notes: Actually, you can really see how important a
 3357 German surname can sometimes be when competing for accommodation. In some cases, discrimination is
 3358 explicit, such as refusing to accept students of a certain nationality: Due to today's situation, the landlord
 3359 decided not take Russian students. These examples highlight the vulnerability of international students in
 3360 the absence of transparent control mechanisms.

3361 Financial discrimination is another key barrier and is closely linked to student status. Many landlords and hous-
 3362 ing companies require proof of income at a level that is unaffordable for most students: Many housing companies
 3363 require proof of income and require you to make at least three times the monthly rent, which

is entirely unattainable for most students. As a result, housing is effectively reserved for working professionals, while students are systematically excluded from the market: Apartments only for people with work contracts. This is particularly painful for foreign students who cannot provide German guarantors or credit history (Schufa).

A separate and particularly vulnerable group are students who face discrimination on the basis of gender, family status and queer identity. One interviewee describes being denied a rental because of their 'non-traditional' family configuration: Another agent denied our application because we are an alternative family configuration and not a straight couple. Others mention advertisements with strict restrictions: Women only, no LGBT, which directly excludes certain groups from the housing market. The inevitable consequence of such discrimination is a retreat into the grey area of renting, accompanied by risks and exploitation: I had to live in flats without contracts, hiding from the Hausverwaltung, creating a very stressful power struggle.

Taken together, this evidence shows that discrimination in the student housing sector is rarely an isolated or purely interpersonal phenomenon. Rather, it is embedded in the structure of a market where competition, scarcity and excessive requirements legitimise the exclusion of students, particularly international and socially vulnerable groups.

5.7. Key Findings

The housing crisis in Berlin has a systemic and multi-level impact on students' lives, significantly limiting their choices and undermining their sense of stability. High prices and intense competition dramatically narrow the range of available options, making even minimally acceptable housing difficult to obtain. Many students note that there are virtually no affordable options on the market that are specifically tailored to their needs: nothing affordable for students or fewer flats, more competition. In this situation, housing ceases to be a matter of preference and becomes a matter of survival.

As a result, students are forced to lower their expectations and settle for inadequate or temporary solutions. Some remain in dormitories with outdated facilities and a lack of privacy, as alternatives are financially unattainable: I'm stuck with my affordable dormitory option with older amenities and a lack of privacy. Others consider moving to neighbouring cities or the suburbs, which increases the time and cost of daily commutes: Flats are expensive and/or in districts not suitable for my daily living. Thus, the crisis directly affects students' spatial mobility and their daily organisation of life.

The crisis has a particularly acute impact on students' psychological and emotional state. The search for housing is described as an exhausting and demanding process that takes priority over studying and social life. Respondents emphasise that they had to spend several hours a day exclusively on searching, often without any results: It was very stressful and time-consuming. I couldn't do much else besides search for an apartment. For

3395 some, this situation becomes almost unbearable: Not yet, but soon it will break me, which indicates the risk
3396 of long-term emotional exhaustion.

3397 Overall, the results show that the housing crisis creates an atmosphere of constant instability and forced gratitude
3398 for minimally acceptable conditions. Even those students who have managed to find accommodation realise that they
3399 were simply lucky, rather than that the system works effectively: I was lucky, but just a couple of months
3400 later it was almost impossible to find a room. Awareness of the shortage and lack of alternatives forces
3401 students to give up actively searching in advance and rely entirely on student dormitories as the only realistic option.
3402 This highlights the structural nature of the crisis and its long-term consequences for the student population, especially
3403 those who do not have financial support or social connections.

6.1. 5.1 Conclusion

I attempted to combine spatial, quantitative, and qualitative aspects to rethink the student housing crisis in Berlin and came up with the following conclusions:

1. Public student accommodations are more evenly distributed across the periphery, reflecting their role as affordable, welfare-oriented infrastructure. Private facilities are more concentrated in central areas and high-demand areas, particularly in Mitte, Friedrichshain-Kreuzberg and Charlottenburg-Wilmersdorf, reflecting market logic that prioritises profitable locations with high rents over equal access.
2. East Berlin shows a high concentration of large student complexes and compact clusters. Western Berlin, on the other hand, is characterised by a predominance of stable areas with low capacity and a shortage of supply.
3. Private universities are mainly concentrated in Mitte, Charlottenburg-Wilmersdorf and Friedrichshain-Kreuzberg – areas that are already characterised by high demand for housing – while public universities occupy both central areas and more peripheral areas such as Lichtenberg, Steglitz-Zehlendorf and Treptow-Köpenick.
4. Universities with high numbers of international students are not grouped geographically, but rather distributed in a fragmented manner and often separated from similar neighbors. The West Berlin traditionally concentrates international academic demand, while the East remains unattractive in an international context.
5. In the vast majority of university market areas, university international demand significantly exceeds available student accommodation capacity.
6. Market pressure is not generated by individual market areas, but by the overlap of multiple accessibility zones, which explains why it is the concentration of university functions — and not just the local shortage of student accommodation.
7. Berlin have pronounced polycentricity of educational space: central locations at the highest levels concentrate the greatest student demand and form the main clusters of attraction, while lower-level zones perform supporting functions. At the same time, demand significantly exceeds supply at all levels. Thus, the analysis shows that the student housing crisis in Berlin has a clearly defined spatial structure.
8. Student accommodation in Berlin serves a wider range of educational tracks. A significant proportion of users university students, including preparatory undergraduates, college students, postgraduates, exchange students, interns, young research workers and teaching staff.
9. The affordability of rent is highly rated, reflecting students' adaptation to financial constraints or forced compromises when choosing accommodation. The location and proximity to universities reflecting the spatial gap between places of study and actual areas of residence for students. In other words students priorities affordability over proximity to university.

10. The housing crisis in Berlin has a systemic and multi-level impact on students' lives, significantly limiting their choices and undermining their sense of stability. High prices and intense competition dramatically narrow the range of available options, making even minimally acceptable housing difficult to obtain. The crisis has a particularly acute impact on students' psychological and emotional state. The search for housing is described as an exhausting and demanding process that takes priority over studying and social life. The housing crisis creates an atmosphere of constant instability and forced gratitude for minimally acceptable conditions.

In a broader context, these findings imply that the student housing crisis in Berlin is not a temporary market failure or the result of individual student decisions. Rather, it is the result of institutional and political priorities that have placed social student housing secondary to the investment logic of urban development. The spatial structure of the crisis points to a systemic mismatch between education policy, migration dynamics and urban housing planning.

The polycentric structure of Berlin's educational space, with demand concentrated in central areas and insufficient supply of affordable housing at all levels, demonstrates a lack of coordinated policy between universities, senates and housing operators. Universities continue to increase their international student enrolment without accompanying this growth with an equivalent expansion of public housing infrastructure. As a result, responsibility for student accommodation is effectively shifted to the private market, which operates on the logic of profit maximisation rather than equal access.

Particularly problematic is the fact that purpose-built student accommodation, despite being formally positioned as a solution to the crisis, reinforces social selection and spatial inequality. The concentration of private halls of residence in central and prestigious areas does not reduce overall market pressure but, on the contrary, contributes to rising rents and the displacement of socially vulnerable groups. Thus, private purpose-built student accommodation functions more as an investment asset than as an element of social housing policy, which contradicts Berlin's stated goals of providing affordable housing.

At the city policy level, this points to the need to rethink the role of student housing as critical social infrastructure rather than a niche segment of the real estate market. The research data shows that without active public sector involvement — through the expansion of state-owned student housing, regulation of private operators, and the integration of housing into educational and spatial planning — the crisis will continue and deepen. This particularly affects international students, for whom the lack of stable housing is directly linked to their legal status, access to registration and the ability to participate fully in city life.

In a political sense, the student housing crisis is becoming an indicator of a broader problem—the crisis of accessibility in the city as such. Limited choices, the normalisation of temporary and unstable forms of housing, increased psychological stress, and forced gratitude for minimally acceptable conditions all point to the gradual loss of the city as an inclusive space. This calls into question the implementation of the Right to the City principle and requires a shift

3468 from reactive measures to a long-term, structural strategy in which students are seen not as temporary users of the city,
3469 but as its rightful residents.

3470

6.2. 5.3 Future Research

3471

here will be future actions and collaborations that can take place

7.1. 6.1 References

7.1.1. Literature Research

1. Aalbers, M.B. and Christophers, B., 2014. Centring Housing in Political Economy. *Housing, Theory and Society*, 31(4), pp. 373–394. Aalbers and Christophers (2014)
2. Aaron, L., 2024. Digital Arrival Infrastructures: Housing Platforms and Residency Governance in Berlin's Rental Sector. *UP*, 9(8480). Aaron (2024)
3. Ahlfeldt, G.M., Maennig, W. and Richter, F.J., 2017. Urban renewal after the Berlin Wall: a place-based policy evaluation. *Journal of Economic Geography*, 17(1), pp. 129–156. Ahlfeldt et al. (2016)
4. Anna Granath Hansson, 2018. Combatting the housing shortage through institutional reform: The parallel cases of Germany and Sweden. *Zeitschrift für Geodäsie, Geoinformation und Landmanagement*, 2(1), pp. 93–99.
5. Attia, S., Alphonsine, P., Amer, M. and Ruellan, G., 2020. Towards a European rating system for sustainable student housing: Key performance indicators (KPIs) and a multi-criteria assessment approach. *Environmental and Sustainability Indicators*, 7(100052). Attia et al. (2020)
6. Avogo, F.A., Appau, W.M. and Attakora-Amaniampong, E., 2022. The effects of word-of-mouth and online review marketing strategies on students' satisfaction with their housing selection during COVID-19 season. *F*, 40(2), pp. 394–411. Avogo et al. (2022)
7. Bellini, O.E., Gullace, M.T. and Mocchi, M., 2022. Re-Start: Student Housing Polycrisis post-Coronavirus. *Techne*, pp. 94–103. Bellini et al. (2022)
8. Bernt, M., 2022. The Commodification Gap: Gentrification and Public Policy in London, Berlin and St Petersburg. 1st ed. Wiley.
9. Bernt, M., Grell, B. and Holm, A. (Eds.), 2013. The Berlin Reader: A Compendium on Urban Change and Activism. transcript Verlag.
10. CBRE GmbH, 2025. Housing Market Report Berlin 2024. Berlin.
11. Deutsches Studierendenwerk e. V., 2024. Wohnraum für Studierende: Statistische Übersicht 2024. Berlin.
12. Diehl, C., Andorfer, V.A., Khoudja, Y., Krause, K., 2013. Not In My Kitchen? Ethnic Discrimination and Discrimination Intentions in Shared Housing among University Students in Germany. *Journal of Ethnic and Migration Studies*, 39(10), pp. 1679–1697. Diehl et al. (2013)
13. Duso, T., Michelsen, C., Schaefer, M. and Tran, K.D., 2024. Airbnb and rental markets: Evidence from Berlin. *Regional Science and Urban Economics*, 106(104007). Duso et al. (2024)
14. ESN, ESU, Contributors: Sara Tagliabracci, Lina Mohoric, Pedro Gonzalez, Rita Dias, 2023. International Student Housing: How Are Exchange Students in Europe Navigating the Housing Crisis? European Students' Union (ESU), Erasmus Student Network (ESN).

15. Fields, D. and Uffer, S., 2016. The financialisation of rental housing: A comparative analysis of New York City and Berlin. *Urban Studies*, 53(10), pp. 1486–1502. Fields and Uffer (2014)
16. Franz, Y. and Gruber, E., 2022. The Changing Role of Student Housing as Social Infrastructure. *UP*, 7(4). Franz and Gruber (2022)
17. Frederike Gerdes and Martina Kroher, 2024. Studying and Living on Your Own: Possible But not Affordable? The Current State of Student Housing in Germany. *Autonomie locali e servizi sociali*, pp. 371–404. Frederike Gerdes and Martina Kroher (2024)
18. French, N., Bhat, G., Matharu, G., Guimarães, F.O. and Solomon, D., 2018. Investment opportunities for student housing in Europe. *Journal of Property Investment & Finance*, 36(5), pp. 578–584. French et al. (2018)
19. Granath Hansson, A., 2019. City strategies for affordable housing: the approaches of Berlin, Hamburg, Stockholm, and Gothenburg. *International Journal of Housing Policy*, 19(1), pp. 95–119. Granath Hansson (2017)
20. Gwosc, C., 2024. Students' housing situation, in: Social and Economic Conditions of Student Life in Europe: Eurostudent 8 Synopsis of Indicators 2021-2024. wbv Media GmbH & Co. KG.
21. Hahn, A.M., Kholodilin, K.A., Walzl, S.R. and Fongoni, M., 2024. Forward to the Past: Short-Term Effects of the Rent Freeze in Berlin. *Management Science*, 70(6), pp. 1901–1923. Hahn et al. (2024)
22. Hauschildt, K., 2024. Policy considerations, in: Social and Economic Conditions of Student Life in Europe: Eurostudent 8 Synopsis of Indicators 2021-2024. wbv Media GmbH & Co. KG.
23. Hauschildt, K., Gwosc, C., Schirmer, H., Mandl, S. and Menz, C., 2024. Social and Economic Conditions of Student Life in Europe: Eurostudent 8 Synopsis of Indicators 2021-2024. wbv Publikation, Bielefeld.
24. Heeg, S., 2022. Finanzinvestoren auf dem deutschen Wohnungsmarkt. *WSI*, 75(3), pp. 197–204. HEEG (2022)
25. Hofstede, H., 2024. “Making sense of rural identities in future horizons of Dutch and German students living in rural areas.” *Journal of Rural Studies*, 107(103219). Hofstede (2024)
26. Holm, Andrej, Gerhardt, Sebastian, Scheller, David and Gastaminza Vacas, Itziar, 2024. Keine Profite mit der Miete: Strategien für eine bestandssichernde, nachhaltige und soziale Bewirtschaftung großer Wohnbestände. 2nd ed. Studien. Rosa-Luxemburg-Stiftung, Berlin.
27. Iglesias, M., Huesca, L. and Lozada, K., 2024. Focusing on academic tourism: demographics, behaviours and impacts. *fg*, 23(1), pp. 117–131. Iglesias et al. (2024)
28. Investitionsbank Berlin, 2024. IBB Wohnungsmarktbericht 2024: Schwerpunktthema – Wohnbedarfe und Verteilung von Wohnraum auf dem Berliner Wohnungsmarkt, Berliner Wohnungsmarktbericht. Berlin.
29. Lehner, M., Richter, J.L., Kreinin, H., Mamut, P., Vadovics, E., Henman, J., Mont, O. and Fuchs, D., 2024. Living smaller: acceptance, effects and structural factors in the EU. *B&C*, 5(438). Lehner et al. (2024)

30. Lindenthal, T. and Eichholtz, P., 2012. Prolonged Crisis: Housing in Germany and Berlin, in: Bardhan, A., Edelstein, R.H., Kroll, C.A. (Eds.), *Global Housing Markets*. Wiley, pp. 69–99.
31. Lisa von Witten Sonsfeld and Elisabeth Beusker, 2023. The future development of student housing in Germany by the example of the city Aachen.
32. Miessner, M., 2021. Studentification in Germany: How investors generate profits from student tenants in Goettingen and the impacts on urban segregation. *European Urban and Regional Studies*, 28(1), pp. 133–154. Miessner (2020)
33. Mitze, T., Burgard, C. and Alecke, B., 2015. The tuition fee ‘shock’: Analysing the response of first-year students to a spatially discontinuous policy change in Germany. *Papers in Regional Science*, 94(3), pp. 385–420. Mitze et al. (2015)
34. MLP Finanzberatung SE, 2024. MLP Studentenwohnreport 2024. Institut der deutschen Wirtschaft, Wiesloch.
35. Özkan, D. and Büyüksaraç, G.B. (Eds.), 2020. Contradictions of housing commons Between middle-class and anarchist models in Berlin, in: *Commoning the City: Empirical Perspectives on Urban Ecology, Economics and Ethics, Space, Materiality and the Normative*. Routledge, Taylor & Francis Group, Abingdon, Oxon New York, NY.
36. Piekut, M., 2024. Housing conditions in European one-person households. *PLoS ONE*, 19(3), p. e0303295. Piekut (2024)
37. Preetz, R., Greifenberg, J., Hülsemann, J. and Filser, A., 2022. Moving Back to the Parental Home in Times of COVID-19: Consequences for Students’ Life Satisfaction. *International Journal of Environmental Research and Public Health*, 19(17), p. 10659. Preetz et al. (2022)
38. Sage, J., Smith, D. and Hubbard, P., 2013. New-build Studentification: A Panacea for Balanced Communities? *Urban Studies*, 50(12), pp. 2623–2641. Sage et al. (2013)
39. Schwittay, A., 2025. Students take over: prefiguring urban commons in student housing co-operatives. *Housing Studies*, 40(7), pp. 966–987. Schwittay (2024)
40. Smith, D., 2018. The Politics of Studentification and ‘(Un)balanced’ Urban Populations: Lessons for Gentrification and Sustainable Communities? *Urban Studies*, 45(12), pp. 2541–2564. Smith (2008)
41. Steinführer, A. and Haase, A., 2019. Flexible–inflexible: socio-demographic, spatial and temporal dimensions of flat sharing in Leipzig (Germany). *GeoJournal*, 74(7), pp. 567–587. Steinführer and Haase (2009)
42. Stoll, N., 2022. Vergesellschaftung als Transformationsstrategie: Deutsche Wohnen & Co. enteignen“ im diskursiven und politischen Kontext. *PROKLA*, 52(209), pp. 631–648. Stoll (2022)
43. Twan Dimmendaal, 2022. Afraid of becoming homeless; the effects of a housing crisis on international students.

44. Vollmer, L. and Gutiérrez, D., 2022. Organizing for expropriation: How a tenants campaign convinced Berliners to vote for expropriating big landlords. *RHJ*, 4(1), pp. 47–66. Vollmer and Gutiérrez (2022)
45. Von Wittenhorst Zu Sonsfeld, L. and Beusker, E., 2024. An examination of the needs and preferences for student housing in Germany in relation to different financial backgrounds using the example of the city of Aachen. *Journal of European Real Estate Research*, 17(2), pp. 286–308. von Wittenhorst zu Sonsfeld and Beusker (2024)
46. Wijburg, G. and Aalbers, M.B., 2017. The alternative financialization of the German housing market. *Housing Studies*, 32(6), pp. 968–989. Wijburg and Aalbers (2017)
47. Wijburg, G., Aalbers, M.B. and Heeg, S., 2018. The Financialisation of Rental Housing 2.0: Releasing Housing into the Privatised Mainstream of Capital Accumulation. *Antipode*, 50(4), pp. 1098–1119. Wijburg et al. (2018)
48. Wittowsky, D., Hoekveld, J., Welsch, J. and Steier, M., 2020. Residential housing prices: impact of housing characteristics, accessibility and neighbouring apartments – a case study of Dortmund, Germany. *Urban, Planning and Transport Research*, 8(1), pp. 44–70. Wittowsky et al. (2020)
49. Wolter, F., Raviv, O.C. and Mertens, M., 2023. Discriminatory Residential Preferences in Germany—A Vignette Study. *Köln Z Soziol*, 75(2), pp. 263–288. Wolter et al. (2023)
50. Zunder, T.H., Aditjandra, P.T., Schoemaker, J.T., Vaghi, C., Laparidou, K. and Österle, I., 2014. Engaging city stakeholders to achieve efficient and environmentally friendly urban freight movements. Zunder et al. (2014)

7.1.2. Master Thesis

1. CBRE GmbH, 2025. *Housing Market Report Berlin 2024*. Berlin.
2. Federal Foreign Office, 2025. *Ein Sperrkonto eröffnen und schließen*. Available at: auswaertiges-amt.de/en/sperrkonto-388600
3. Fields, D. and Uffer, S., 2016. The financialisation of rental housing: A comparative analysis of New York City and Berlin. *Urban Studies*, 53, pp. 1486–1502. doi:10.1177/0042098014543704
4. GIS Geography, 2025. *Rasterization and Vectorization Conversion*. Available at: gisgeography.com/rasterization-vectorization/
5. Giseke, U., Löw, M., Million, A., Misselwitz, P. and Stollmann, J. (eds.), 2021. *Urban Design Methods*. Berlin: De Gruyter. doi:10.1515/9783868599558
6. Godlund, S., 1956. The function and growth of bus traffic within the sphere of urban influence. *Lund Studies in Geography, Series B*.
7. Grekousis, G., 2020. *Spatial Analysis Methods and Practice: Describe – Explore – Explain through GIS*. 1st ed. Cambridge: Cambridge University Press. doi:10.1017/9781108614528

8. Heeg, S., 2022. Finanzinvestoren auf dem deutschen Wohnungsmarkt. *WSI*, 75, pp. 197–204. doi:10.5771/0342-300X-2022-3-197
9. Holm, A., 2013. Berlin's Gentrification Mainstream. In: Bernt, M., Grell, B. and Holm, A. (eds.) *The Berlin Reader*. Bielefeld: transcript Verlag, pp. 171–188. doi:10.1515/transcript.9783839424780.171
10. Iglesias, M., Huesca, L. and Lozada, K., 2024. Focusing on academic tourism: demographics, behaviours and impacts. *fg*, XXIII, pp. 117–131. doi:10.5775/fg.2024.1.3617
11. King, L.J., 1986. *Central Place Theory*. 3rd ed. Beverly Hills: Sage.
12. Kurowska, K., Kryszk, H. and Kietlinska, E., 2017. The use of gravity model in spatial planning. In: *Proceedings of the 10th International Conference "Environmental Engineering"*. Vilnius: Vilnius Gediminas Technical University. doi:10.3846/enviro.2017.109
13. Lindenthal, T. and Eichholtz, P., 2012. Prolonged crisis: Housing in Germany and Berlin. In: Bardhan, A., Edelstein, R.H. and Kroll, C.A. (eds.) *Global Housing Markets*. Hoboken: Wiley, pp. 69–99.
14. Louarn, A., Meur-Ferec, C. and Hervé-Fournereau, N., 2025. The concept of “nature-based solutions” applied to urban coastal risks: A bibliometric and content analysis review. *Ocean & Coastal Management*, 261, 107530. doi:10.1016/j.ocecoaman.2024.107530
15. Menz, C. and Mandl, S., 2024. Students' time budget. In: *Social and Economic Conditions of Student Life in Europe: Eurostudent 8 Synopsis of Indicators 2021–2024*. Bielefeld: wbv Media. doi:10.3278/6001920ew005
16. MLP Finanzberatung SE, 2024. *MLP Studentenwohnreport 2024*. Wiesloch: Institut der deutschen Wirtschaft.
17. Nijkamp, P. and Ratajczak, W., 2021. Gravitational analysis in regional science and spatial economics: A vector gradient approach to trade. *International Regional Science Review*, 44, pp. 400–431. doi:10.1177/0160017620980519
18. OpenStreetMap Wiki, 2025. *Isochrone Map*. Available at: wiki.openstreetmap.org/wiki/Isochrone
19. Parr, J.B., 2017. Central place theory: An evaluation. *Review of Urban & Regional Development Studies*, 29, pp. 151–164. doi:10.1111/rurd.12066
20. Rauch, S.L., 1996. A symptom provocation study of posttraumatic stress disorder using positron emission tomography and script-driven imagery. *Archives of General Psychiatry*, 53, p. 380. doi:10.1001/archpsyc.1996.0183005001
21. Rey, S.J., Arribas-Bel, D. and Wolf, L.J., 2023. *Geographic Data Science with Python*. Boca Raton: CRC Press. doi:10.1201/9780429292507
22. Riessman, C.K., 2008. *Narrative Methods for the Human Sciences*. Los Angeles: Sage.
23. Schwab, C., 2021. *Gentrification in Berlin: A Novel Conceptually Driven Quantitative Approach to Evaluating Anti-Displacement Legislation*. Chicago: University of Chicago.
24. Senate Department for Urban Development, Building and Housing, 2023. *Planning, Building, Development*. Berlin.

- 3630 25. Sevtsuk, A. and Kalvo, R., 2018. Patronage of urban commercial clusters: A network-based extension of the
3631 Huff model for balancing location and size. *Environment and Planning B: Urban Analytics and City Science*,
3632 45, pp. 508–528. doi:10.1177/2399808317721930
- 3633 26. Statistik Berlin Brandenburg, 2025a. *Education Universities Dashboard*. Available at: [statistik-berlin-brandenburg.de/ges](https://statistik-berlin-brandenburg.de/gesundheit)
- 3634 27. Statistik Berlin Brandenburg, 2025b. *Studierende an Hochschulen – Übersicht*. Available at: [statistik-berlin-](https://statistik-berlin-brandenburg.de/b-iii-2-j)
3635 [brandenburg.de/b-iii-2-j](https://statistik-berlin-brandenburg.de/b-iii-2-j)
- 3636 28. The Governing Mayor, Senate Chancellery, 1995. *The Constitution of Berlin*. Berlin.
- 3637 29. University of West Florida, Morgan, J., 2023. Spatial joins. *GIS&T BoK 2023*. doi:10.22224/gistbok/2023.1.1
- 3638 30. Urban, F., 2018. Large housing estates of Berlin, Germany. In: Hess, D.B., Tammaru, T. and Van Ham, M.
3639 (eds.) *Housing Estates in Europe*. Cham: Springer, pp. 99–120. doi:10.1007/978-3-319-92813-5_5
- 3640 31. Von Wittenhorst zu Sonsfeld, L. and Beusker, E., 2024. An examination of the needs and preferences for student
3641 housing in Germany in relation to different financial backgrounds using the example of the city of Aachen.
3642 *Journal of European Real Estate Research*, 17, pp. 286–308. doi:10.1108/JERER-10-2023-0042

3643 7.2. Colour Palettes Design

3644 , . :

- 3645 • sequential schemes - mapping orderable or numeric data that progress from low to high (or vice versa)
- 3646 • diverging schemes - mapping quantitative data above and below an interesting mid-point
- 3647 • qualitative schemes - mapping categorical differences in qualitative data

3648 , ColorBrewer, MapColPal and CartoColors , .

3649 7.2.1. Colour Definitions

- 3650 • sequential schemes - “Mint”, “Pi”, “Or”, “GrSc”
- 3651 • diverging schemes - “MintPi”, “MintOr”, “PiOr”
- 3652 • qualitative schemes - “MiPiOr”

```
3653 color_palettes = {
3654     "Mint": ["#EAF4E8", "#C8E0D4", "#A9CCC1", "#8FB9AE", "#76A79D", "#5F968E", "#498581"],
3655     "Pi": ["#F8EDEB", "#FAE1DD", "#FCD5CE", "#FEC5BB", "#F8AD9D", "#F4978E", "#F08080"],
3656     "Or": ["#FFE5D9", "#FFD7BA", "#FEC89A", "#FFB78F", "#F49F7D", "#F28B6D", "#E1755CFF"],
3657     "GrSc": ['#EFEFef', '#D9D9d9', '#CCCCcc', '#B7B7b7', '#999999', '#666666', '#434343'],
3658     "MintPi": ["#76A79D", "#8FB9AE", "#A9CCC1", "#C8E0D4", "#FBF0EA", "#FCD5CE", "#FEC5BB", "#F8A78F"],
3659     "MintOr": ["#76A79D", "#8FB9AE", "#A9CCC1", "#C8E0D4", "#FBF0EA", "#FFD7BA", "#FEC89A", "#FFB78F"],
3660     "PiOr": ["#F4978E", "#F8AD9D", "#FEC5BB", "#FCD5CE", "#FBF0EA", "#FFD7BA", "#FEC89A", "#FFB78F"],
3661     "MiPiOr": ['#EAF4E8', '#FAE1DD', '#C8E0D4', '#FFD7BA', '#F8EDEB', '#FCD5CE', '#FFE5D9']
3662 }
```

```
3663     "MiPiOr": ['#EAF4E8', '#FEC89A', '#FAE1DD', '#C8E0D4', '#FFD7BA', '#8FB9AE', '#F8EDEB', '#FCD5CE',
3664     '#A9CCC1', '#FFE5D9']["#e1755cff"]
3665     "GrSc": ["#434343", "#666666", "#999999", "#b7b7b7", "#cccccc", "#d9d9d9", "#efefef",
3666     colors = ['#FAE1DD', '#FEC89A', '#C8E0D4', '#FEC5BB', '#FFE5D9', '#EAF4E8', '#FBF0EA', '#FFB78F']
```

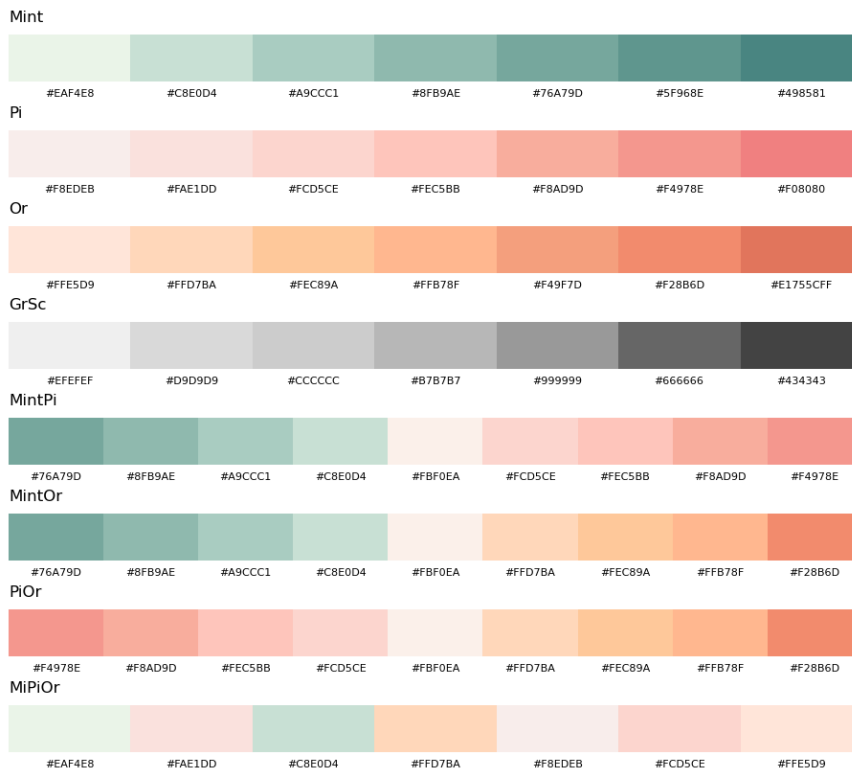


Figure XX. bla bla bla

7.2.2. Color Blindness Test

Differences in vision cover both blindness and color vision deficiency These can be described by three concepts: accessibility, usability, and universal design. (Iwarson and Ståhl, 2003)

To ensure graphical materials are universal and enjoyable for everyone theories about color blindness are investigated alongside with accessibility and usability theories.

Color Blindes can be seen as differences in vision cover both blindness and color vision deficiency. Color perception in the human eye is build up by three different types of cones. Each type is sensitive to a certain wavelength of light (red, green, and blue) and every perceived color is therefore a mixture of stimuli of those three cone types. (<https://www.color-blindness.com/what-is-color-blindness/>)

Coblis — Color Blindness Simulator



Figure XX. bla bla bla

! use Pi und Or together only with different color strenght !

! PiOr colour palette dont use !

TODO: MintPiOr check second time , do i really need it

```
color_palettes = {
    "Mint": ["#EAF4E8", "#C8E0D4", "#A9CCC1", "#8FB9AE", "#76A79D", "#5F968E", "#498581"],
    "Pi": ["#F8EDEB", "#FAE1DD", "#FCD5CE", "#FEC5BB", "#F8AD9D", "#F4978E", "#F08080"],
    "Or": ["#FFE5D9", "#FFD7BA", "#FEC89A", "#FFB78F", "#F49F7D", "#F28B6D", "#E1755CFF"],
    "GrSc": ['#EFEFEF', '#D9D9D9', '#CCCCCC', '#B7B7B7', '#999999', '#666666', '#434343'],
    "MintPi": ["#76A79D", "#8FB9AE", "#A9CCC1", "#C8E0D4", "#F8AD9D", "#F4978E", "#F08080", "#F8AD9D"],
    "MintOr": ["#76A79D", "#8FB9AE", "#A9CCC1", "#C8E0D4", "#F8AD9D", "#F4978E", "#F08080", "#F8AD9D"],
    "MiPiOr": ['#EAF4E8', '#FAE1DD', '#C8E0D4', '#FFD7BA', '#F8EDEB', '#FCD5CE', '#FFE5D9']
}
```

7.2.3. Colour Palettes Export

```
from matplotlib.colors import LinearSegmentedColormap
```



```

3696 cmap_mint = LinearSegmentedColormap.from_list("Mint", color_palettes["Mint"])
3697 cmap_pi = LinearSegmentedColormap.from_list("Pi", color_palettes["Pi"])
3698 cmap_or = LinearSegmentedColormap.from_list("Or", color_palettes["Or"])
3699 cmap_mintpi = LinearSegmentedColormap.from_list("MintPi", color_palettes["MintPi"])
3700 cmap_mintor = LinearSegmentedColormap.from_list("MintOr", color_palettes["MintOr"])
3701 cmap_grsc = LinearSegmentedColormap.from_list("GrSc", color_palettes["GrSc"])
3702 cmap_mipior = LinearSegmentedColormap.from_list("MiPiOr", color_palettes["MiPiOr"])

```

3703 **7.2.4. New color pallets**

3704 *here will be a color pallets fro new visualisations and for apendix content*

```

3705 import matplotlib.pyplot as plt
3706 import numpy as np
3707
3708 colors = ['#FAE1DD', '#FFD7BA', '#A9CCC1', '#FEC5BB', '#FFE5D9', '#C8E0D4', '#FFB78F', '#F08080']
3709
3710 fig, ax = plt.subplots(figsize=(8, 1))
3711 ax.set_xlim(0, len(colors))
3712 ax.set_ylim(0, 1)
3713 for i, c in enumerate(colors):
3714     rect = plt.Rectangle((i, 0), 1, 1, facecolor=c)
3715     ax.add_patch(rect)
3716     ax.text(i + 0.5, 0.5, c, ha='center', va='center', fontsize=8, color='black')
3717 ax.axis('off')
3718 plt.tight_layout()
3719 plt.savefig('colors.jpg', dpi=300, bbox_inches='tight')
3720 plt.close(fig)

```

References

- Aalbers, M.B., Christophers, B., 2014. Centring Housing in Political Economy. *Housing, Theory and Society* 31, 373–394. URL: <http://dx.doi.org/10.1080/14036096.2014.947082>, doi:10.1080/14036096.2014.947082.
- Aaron, L., 2024. Digital Arrival Infrastructures: Housing Platforms and Residency Governance in Berlin's Rental Sector. *Urban Planning* 9. URL: <http://dx.doi.org/10.17645/up.8480>, doi:10.17645/up.8480.
- Ahlfeldt, G.M., Maennig, W., Richter, F.J., 2016. Urban renewal after the Berlin Wall: a place-based policy evaluation. *Journal of Economic Geography* 17, 129–156. URL: <http://dx.doi.org/10.1093/jeg/lbw003>, doi:10.1093/jeg/lbw003.
- Attia, S., Alphonsine, P., Amer, M., Ruellan, G., 2020. Towards a European rating system for sustainable student housing: Key performance indicators (KPIs) and a multi-criteria assessment approach. *Environmental and Sustainability Indicators* 7, 100052. URL: <http://dx.doi.org/10.1016/j.indic.2020.100052>, doi:10.1016/j.indic.2020.100052.
- Avogo, F.A., Appau, W.M., Attakora-Amaniampong, E., 2022. The effects of word-of-mouth and online review marketing strategies on students' satisfaction with their housing selection during COVID-19 season. *Facilities* 40, 394–411. URL: <http://dx.doi.org/10.1108/F-09-2021-0085>, doi:10.1108/f-09-2021-0085.
- Bellini, O.E., Gullace, M.T., Mocchi, M., 2022. Re-Start: Student Housing Polycrisis post-Coronavirus. *TECHNE - Journal of Technology for Architecture and Environment*, 94–103 URL: <http://dx.doi.org/10.36253/techne-12155>, doi:10.36253/techne-12155.
- Diehl, C., Andorfer, V.A., Khoudja, Y., Krause, K., 2013. Not In My Kitchen? Ethnic Discrimination and Discrimination Intentions in Shared Housing among University Students in Germany. *Journal of Ethnic and Migration Studies* 39, 1679–1697. URL: <http://dx.doi.org/10.1080/1369183X.2013.833705>, doi:10.1080/1369183x.2013.833705.
- Duso, T., Michelsen, C., Schaefer, M., Tran, K.D., 2024. Airbnb and rental markets: Evidence from Berlin. *Regional Science and Urban Economics* 106, 104007. URL: <http://dx.doi.org/10.1016/j.regsciurbeco.2024.104007>, doi:10.1016/j.regsciurbeco.2024.104007.
- Fields, D., Uffer, S., 2014. The financialisation of rental housing: A comparative analysis of New York City and Berlin. *Urban Studies* 53, 1486–1502. URL: <http://dx.doi.org/10.1177/0042098014543704>, doi:10.1177/0042098014543704.
- Franz, Y., Gruber, E., 2022. The Changing Role of Student Housing as Social Infrastructure. *Urban Planning* 7. URL: <http://dx.doi.org/10.17645/up.v7i4.5661>, doi:10.17645/up.v7i4.5661.
- Frederike Gerdes, Martina Kroher, 2024. Studying and Living on Your Own: Possible But not Affordable? The Current State of Student Housing in Germany. *Autonomie locali e servizi sociali*, 371–404 URL: <https://doi.org/10.1447/115241>, doi:10.1447/115241.
- French, N., Bhat, G., Matharu, G., Guimarães, F.O., Solomon, D., 2018. Investment opportunities for student housing in Europe. *Journal of Property Investment & Finance* 36, 578–584. URL: <http://dx.doi.org/10.1108/JPIF-08-2018-0058>, doi:10.1108/jpif-08-2018-0058.
- Granath Hansson, A., 2017. City strategies for affordable housing: the approaches of Berlin, Hamburg, Stockholm, and Gothenburg. *International Journal of Housing Policy* 19, 95–119. URL: <http://dx.doi.org/10.1080/19491247.2017.1278581>, doi:10.1080/19491247.2017.1278581.
- Hahn, A.M., Kholodilin, K.A., Walzl, S.R., Fongoni, M., 2024. Forward to the Past: Short-Term Effects of the Rent Freeze in Berlin. *Management Science* 70, 1901–1923. URL: <http://dx.doi.org/10.1287/mnsc.2023.4775>, doi:10.1287/mnsc.2023.4775.
- HEEG, S., 2022. Finanzinvestoren auf dem deutschen Wohnungsmarkt. *WSI-Mitteilungen* 75, 197–204. URL: <http://dx.doi.org/10.5771/0342-300X-2022-3-197>, doi:10.5771/0342-300x-2022-3-197.
- Hofstede, H., 2024. “Making sense of rural identities in future horizons of Dutch and German students living in rural areas”. *Journal of Rural Studies* 107, 103219. URL: <http://dx.doi.org/10.1016/j.jrurstud.2024.103219>, doi:10.1016/j.jrurstud.2024.103219.

Iglesias, M., Huesca, L., Lozada, K., 2024. Focusing on academic tourism: demographics, behaviours and impacts. *Forum geografic XXIII*, 117–131. URL: <http://dx.doi.org/10.5775/fg.2024.1.3617>, doi:10.5775/fg.2024.1.3617.

Lehner, M., Richter, J.L., Kreinin, H., Mamut, P., Vadovics, E., Henman, J., Mont, O., Fuchs, D., 2024. Living smaller: acceptance, effects and structural factors in the EU. *Buildings & Cities 5*. URL: <http://dx.doi.org/10.5334/bc.438>, doi:10.5334/bc.438.

Miessner, M., 2020. Studentification in Germany: How investors generate profits from student tenants in Goettingen and the impacts on urban segregation. *European Urban and Regional Studies 28*, 133–154. URL: <http://dx.doi.org/10.1177/0969776420934850>, doi:10.1177/0969776420934850.

Mitze, T., Burgard, C., Alecke, B., 2015. The tuition fee 'shock': Analysing the response of firstyear students to a spatially discontinuous policy change in Germany. *Papers in Regional Science 94*, 385–420. URL: <http://dx.doi.org/10.1111/pirs.12067>, doi:10.1111/pirs.12067.

Piekut, M., 2024. Housing conditions in European one-person households. *PLOS ONE 19*, e0303295. URL: <http://dx.doi.org/10.1371/journal.pone.0303295>, doi:10.1371/journal.pone.0303295.

Preetz, R., Greifenberg, J., Hülsemann, J., Filser, A., 2022. Moving Back to the Parental Home in Times of COVID-19: Consequences for Students' Life Satisfaction. *International Journal of Environmental Research and Public Health 19*, 10659. URL: <http://dx.doi.org/10.3390/ijerph191710659>, doi:10.3390/ijerph191710659.

Sage, J., Smith, D., Hubbard, P., 2013. New-build Studentification: A Panacea for Balanced Communities? *Urban Studies 50*, 2623–2641. URL: <http://dx.doi.org/10.1177/0042098013477694>, doi:10.1177/0042098013477694.

Schwittay, A., 2024. Students take over: prefiguring urban commons in student housing co-operatives. *Housing Studies 40*, 966–987. URL: <http://dx.doi.org/10.1080/02673037.2024.2334816>, doi:10.1080/02673037.2024.2334816.

Smith, D., 2008. The Politics of Studentification and '(Un)balanced' Urban Populations: Lessons for Gentrification and Sustainable Communities? *Urban Studies 45*, 2541–2564. URL: <http://dx.doi.org/10.1177/0042098008097108>, doi:10.1177/0042098008097108.

von Wittenhorst zu Sonsfeld, L., Beusker, E., 2024. An examination of the needs and preferences for student housing in Germany in relation to different financial backgrounds using the example of the city of Aachen. *Journal of European Real Estate Research 17*, 286–308. URL: <http://dx.doi.org/10.1108/JERER-10-2023-0042>, doi:10.1108/jerer-10-2023-0042.

Steinführer, A., Haase, A., 2009. Flexible–inflexible: socio-demographic, spatial and temporal dimensions of flat sharing in Leipzig (Germany). *GeoJournal 74*, 567–587. URL: <http://dx.doi.org/10.1007/s10708-008-9248-3>, doi:10.1007/s10708-008-9248-3.

Stoll, N., 2022. Vergesellschaftung als Transformationsstrategie: »Deutsche Wohnen & Co. enteignen« im diskursiven und politischen Kontext. *PROKLA. Zeitschrift für kritische Sozialwissenschaft 52*, 631–648. URL: <http://dx.doi.org/10.32387/prokla.v52i209.2024>, doi:10.32387/prokla.v52i209.2024.

Vollmer, L., Gutiérrez, D., 2022. Organizing for expropriation: How a tenants campaign convinced Berliners to vote for expropriating big landlords. *Radical Housing Journal 4*, 47–66. URL: <http://dx.doi.org/10.54825/QHHD8116>, doi:10.54825/qhhd8116.

Wijburg, G., Aalbers, M.B., 2017. The alternative financialization of the German housing market. *Housing Studies 32*, 968–989. URL: <http://dx.doi.org/10.1080/02673037.2017.1291917>, doi:10.1080/02673037.2017.1291917.

Wijburg, G., Aalbers, M.B., Heeg, S., 2018. The Financialisation of Rental Housing 2.0: Releasing Housing into the Privatised Mainstream of Capital Accumulation. *Antipode 50*, 1098–1119. URL: <http://dx.doi.org/10.1111/anti.12382>, doi:10.1111/anti.12382.

Wittowsky, D., Hoekveld, J., Welsch, J., Steier, M., 2020. Residential housing prices: impact of housing characteristics, accessibility and neighbouring apartments – a case study of Dortmund, Germany. *Urban, Planning and Transport Research 8*, 44–70. URL: <http://dx.doi.org/10.1080/21650020.2019.1704429>, doi:10.1080/21650020.2019.1704429.

- 3797 Wolter, F., Raviv, O.C., Mertens, M., 2023. Discriminatory Residential Preferences in Germany—A Vignette Study. KZfSS Kölner
3798 Zeitschrift für Soziologie und Sozialpsychologie 75, 263–288. URL: <http://dx.doi.org/10.1007/s11577-023-00906-2>, doi:10.1007/
3799 s11577-023-00906-2.
- 3800 Zunder, T.H., Aditjandra, P.T., Schoemaker, J.T., Vaghi, C., Laparidou, K., Österle, I., 2014. Engaging city stakeholders to achieve efficient
3801 and environmentally friendly urban freight movements URL: <https://www.researchgate.net/doi/10.13140/2.1.2601.7920>, doi:10.
3802 13140/2.1.2601.7920.